Senior School Certificate Examination-2020
Marking Scheme - MATHEMATICS
Subject Code: 041 Paper Code: 65 (B)

***General instructions*:-**

1. You are aware that evaluation is the most important process in the actual and correct assessment of the candidates. A small mistake in evaluation may lead to serious problems which may affect the future of the candidates, education system and teaching profession. To avoid mistakes, it is requested that before starting evaluation, you must read and understand the spot evaluation guidelines carefully. **Evaluation is a 10-12 days mission for all of us. Hence, it is necessary that you put in your best efforts in this process.**
2. Evaluation is to be done as per instructions provided in the Marking Scheme. It should not be done according to one's own interpretation or any other consideration. Marking Scheme should be strictly adhered to and religiously followed. **However, while evaluating, answers which are based on latest information or knowledge and/or are innovative, they may be assessed for their correctness otherwise and marks be awarded to them.**
3. The Head-Examiner must go through the first five answer books evaluated by each evaluator on the first day, to ensure that evaluation has been carried out as per the instructions given in the Marking Scheme. The remaining answer books meant for evaluation shall be given only after ensuring that there is no significant variation in the marking of individual evaluators.
4. Evaluators will mark(√) wherever answer is correct. For wrong answer 'X"be marked. Evaluators will not put right kind of mark while evaluating which gives an impression that answer is correct and no marks are awarded. This is most common mistake which evaluators are committing.
5. If a question has parts, please award marks on the right-hand side for each part. Marks awarded for different parts of the question should then be totaled up and written in the left-hand margin and encircled. This may be followed strictly.
6. If a question does not have any parts, marks must be awarded in the left-hand margin and encircled. This may also be followed strictly.
7. If a student has attempted an extra question, answer of the question deserving more marks should be retained and the other answer scored out.
8. No marks to be deducted for the cumulative effect of an error. It should be penalized only once.
9. A full scale of marks 0 - 80 has to be used. Please do not hesitate to award full marks if the answer deserves it.
10. Every examiner has to necessarily do evaluation work for full working hours i.e. 8 hours every day and evaluate 20 answer books per day in main subjects and 25 answer books per day in other subjects (Details are given in Spot Guidelines).
11. Ensure that you do not make the following common types of errors committed by the Examiner in the past:-
    - Leaving answer or part thereof unassessed in an answer book.
    - Giving more marks for an answer than assigned to it.
    - Wrong totaling of marks awarded on a reply
    - Wrong transfer of marks from the inside pages of the answer book to the title page.
    - Wrong question wise totaling on the title page.
    - Wrong totaling of marks of the two columns on the title page.
    - Wrong grand total.
    - Marks in words and figures not tallying.
    - Wrong transfer of marks from the answer book to online award list.
    - Answers marked as correct, but marks not awarded. (Ensure that the right tick mark is correctly and clearly indicated. It should merely be a line. Same is with the X for incorrect answer.)
    - Half or a part of answer marked correct and the rest as wrong, but no marks awarded.
12. While evaluating the answer books if the answer is found to be totally incorrect, it should be marked as cross (X) and awarded zero (0)Marks.
13. Any unassessed portion, non-carrying over of marks to the title page, or totaling error detected by the candidate shall damage the prestige of all the personnel engaged in the evaluation work as also of the Board. Hence, in order to uphold the prestige of all concerned, it is again reiterated that the instructions be followed meticulously and judiciously.
14. The Examiners should acquaint themselves with the guidelines given in the Guidelines for spot Evaluation before starting the actual evaluation.
15. Every Examiner shall also ensure that all the answers are evaluated, marks carried over to the title page, correctly totaled and written in figures and words.
16. The Board permits candidates to obtain photocopy of the Answer Book on request in an RTI application and also separately as a part of the re-evaluation process on payment of the processing charges.

QUESTION PAPER CODE 65(B)

EXPECTED ANSWER/VALUE POINTS

SECTION – A

Question Numbers 1 to 20 carry 1 mark each.

Question Numbers 1 to 10 are multiple choice type questions.
Select the correct option.

Q.No. **Marks**

1. Total number of possible matrices of order 2 × 2 with each entry 2 or 3 is

(A) 4 **(B)** 8 **(C)** 16 **(D)** 32

Ans: (C) 16 **1**

2. The area of a triangle with vertices (–2, 0), (2, 0) and (0, k) is 4 sq. units. The value of k is

(A) 4 **(B)** 2 **(C)** – 4 **(D)** 6

Ans: (B) 2 **1**

3. The value of the expression $2\,\text{cosec}^{-1}2 + \cos^{-1}\left(\frac{1}{2}\right)$ is

(A) $\frac{\pi}{3}$ **(B)** $-\frac{\pi}{3}$ **(C)** $-\frac{2\pi}{3}$ **(D)** $\frac{2\pi}{3}$

Ans: (D) $\frac{2\pi}{3}$ **1**

4. $\int \frac{dx}{16 + 9x^2}$ is equal to

(A) $\frac{1}{4}\tan^{-1}\frac{3x}{4} + c$ **(B)** $\frac{1}{12}\tan^{-1}\frac{3x}{4} + c$ **(C)** $\frac{1}{3}\tan^{-1}\frac{3x}{4} + c$ **(D)** $\frac{1}{12}\tan^{-1}\frac{9x}{16} + c$

Ans: (B) $\frac{1}{12}\tan^{-1}\frac{3x}{4} + c$ **1**

5. The area of the trinagle whose two sides are represented by the vectors $\hat{i} + \hat{k}$ and $2\hat{i} + \hat{j} + \hat{k}$ is

(A) $\frac{\sqrt{3}}{2}$ **(B)** $\sqrt{3}$ **(C)** 3 **(D)** $\frac{3}{2}$

Ans: (A) $\frac{\sqrt{3}}{2}$ **1**

6. If the direction cosines of a line are a, a, a, then

(A) $a > 0$ **(B)** $a = 1$ or $a = -1$ **(C)** $0 < a < 1$ **(D)** $a = \frac{1}{\sqrt{3}}$ or $a = -\frac{1}{\sqrt{3}}$

Ans: (D) $a = \frac{1}{\sqrt{3}}$ or $a = -\frac{1}{\sqrt{3}}$ **1**

7. The plane $3x - 2y + 6z + 11 = 0$, makes an angle $\sin^{-1}(\alpha)$ with x-axis. The value of α is

(A) $-\frac{3}{7}$ **(B)** $\frac{3}{7}$ **(C)** $\frac{2}{7}$ **(D)** $\frac{\sqrt{3}}{2}$

Ans: (B) $\frac{3}{7}$ **1**

8. Let $f : \mathbf{R} \to \mathbf{R}$ be defined by $f(x) = 5x - 3$. Then, $f^{-1}(x)$ is given by

(A) $\frac{x+3}{5}$ **(B)** $\frac{x}{3} - 5$ **(C)** $\frac{x}{5} + 3$ **(D)** $\frac{x}{5} - 3$

Ans: (A) $\frac{x+3}{5}$ **1**

9. If A and B be two events such that $P(A) = 0{\cdot}2$, $P(B) = 0{\cdot}4$ and $P(A \cap B) = 0{\cdot}08$, then $P(A \mid B)$ is

(A) 0·02 **(B)** 0.2 **(C)** 0·4 **(D)** 0·08

Ans: (B) 0.2 **1**

10. A bag contains 4 red and 3 black balls. If 2 balls are drawn from the bag at random without replacement, then the probability of getting exactly one red ball is

(A) $\frac{1}{7}$ **(B)** $\frac{2}{7}$ **(C)** $\frac{4}{7}$ **(D)** $\frac{3}{14}$

Ans: (C) $\frac{4}{7}$ **1**

Fill in the blanks in questions numbers 11 to 15

11. Matrices A and B will be inverse of each other only if _____.

Ans: $AB = BA = I$ **1**

12. The function $f(x) = |x - 3|$, $x \in \mathbf{R}$ is not differentiable at $x =$ _____.

Ans: 3 **1**

13. The function $f(x) = \log_e (\sin x)$ is strictly _____ on $\left(\frac{\pi}{2}, \pi\right)$.

Ans: decreasing **1**

OR

The approximate value of $\sqrt{26}$, using differentials, up to 2 places of decimal is _____.

Ans: 5.10 **1**

14. In an LPP, the linear function which has to be maximised or minimised is called a linear _____ function.

Ans: Objective **1**

15. A vector of magnitude 14 in the direction of the vector $\vec{a} = -2\hat{i} + 3\hat{j} + 6\hat{k}$ is _____.

Ans: $-4\hat{i} + 6\hat{j} + 12\hat{k}$ **1**

Question numbers 16 to 20 are very short answer type questions.

16. Find the value of the determinant $\begin{vmatrix} bc & 1 & a(b+c) \\ ca & 1 & b(c+a) \\ ab & 1 & c(a+b) \end{vmatrix}$.

Ans: Writing determinant as $(ab + bc + ca)\begin{vmatrix} bc & 1 & 1 \\ ca & 1 & 1 \\ ab & 1 & 1 \end{vmatrix}$ **1/2**

$= 0$ **1/2**

17. Evaluate :

$$\int_{-\pi/4}^{\pi/4} \left(x^3 + x\cos x + \tan^5 x\right) dx$$

Ans: $f(x) = x^3 + x \cos x + \tan^5 x$ is odd **1/2**

$\therefore$ Value of Integral = 0 **1/2**

18. Find :

$$\int \frac{6\cos x - 9\sin x}{6\cos x + 4\sin x}\, dx$$

Ans: Given integral $= \frac{3}{2}\int \frac{2\cos x - 3\sin x}{3\cos x + 2\sin x}\, dx$ **1/2**

$= \frac{3}{2}\log|3\cos x + 2\sin x| + C$ **1/2**

OR

Find : $\int \frac{(x-5)e^x}{(x-3)^3} dx$

Ans: Given integral $= \int \left[\frac{1}{(x-3)^2} - \frac{2}{(x-3)^3} \right] e^x dx$ **1/2**

$= \frac{e^x}{(x-3)^2} + C$ **1/2**

19. Find :

$\int \tan^2(3x+5)\, dx$ m

Ans: Given integral $= \int \left(\sec^2(3x+5) - 1 \right) dx$ **1/2**

$= \frac{\tan(3x+5)}{3} - x + C$ **1/2**

20. Form the differential equation representing the family of curves $y = a \cos(x + b)$, where a, b are arbitrary constants.

Ans: $\frac{dy}{dx} = -a \sin(x+b)$ **1/2**

$\frac{d^2y}{dx^2} = -a \cos(x+b) \Rightarrow \frac{d^2y}{dx^2} = -y$ **1/2**

OR

Find the integrating factor for the solution of the differential equation $y\,dx - (x + 2y^2)\,dy = 0$.

Ans: Given differential equation can be written as

$\frac{dx}{dy} - \frac{x}{y} = -y$ **1/2**

Integrating factor is $e^{\int -\frac{1}{y} dy} = \frac{1}{y}$ **1/2**

SECTION-B

Question numbers 21 to 26 carry 2 marks.

21. Show at the relation R in the set of all positive real numbers defined by $R = \{(a, b) : a \le b^3\}$ is neither symmetric nor transitive.

Ans: $(1, 2) \in R$ but $(2, 1) \notin R$ $\therefore$ R is not symmetriac **1**

$(9, 3) \in R$, $(3, 2) \in R$ but $(9, 2) \notin R$ $\therefore$ R is not transitive **1**

OR

Prove that :

$$\cos^{-1}\left(\frac{12}{13}\right)+\sin^{-1}\left(\frac{3}{5}\right)=\sin^{-1}\left(\frac{56}{65}\right)$$

Ans: LHS $= \tan^{-1}\frac{5}{12}+\tan^{-1}\frac{3}{4}$ **1/2+1/2**

$$=\tan^{-1}\frac{\frac{5}{12}+\frac{3}{4}}{1-\frac{5}{12}\cdot\frac{3}{4}}=\tan^{-1}\frac{56}{33}$$ **1/2**

$$=\sin^{-1}\frac{56}{65}=\text{RHS}$$ **1/2**

22. If $(x^2+y^2)^2 = xy$, then find $\frac{dy}{dx}$.

Ans: Differentiating both sides w.r.t. x

$$2(x^2+y^2)\left(2x+2y\frac{dy}{dx}\right)=x\frac{dy}{dx}+y$$ **1+1/2**

Getting $\frac{dy}{dx}=\frac{y-4x^3-4xy^2}{4x^2y+4y^3-x}$ **1/2**

23. The radius of a right circular cylinder is increasing at the rate of 2cm/s and its height is decreasing at the rate of 8cm/s. Find the rate of change of its volume, when the radius is 3 cm and height is 6cm.

Ans: Let r be radius & h be the height

$$\frac{dr}{dt}=2\text{ cm/s}\qquad \frac{dh}{dt}=-8\text{cm/s}$$ **1/2**

$$v=\pi r^2h \Rightarrow \frac{dv}{dt}=\pi\left[r^2\frac{dh}{dt}+2hr\frac{dr}{dt}\right]$$ **1**

$$\frac{dv}{dt}\ (\text{at } r=3, h=6)=0$$ **1/2**

24. Show that the points with position vectors $2\hat{i}-\hat{j}+\hat{k}$, $\hat{i}-3\hat{j}-5\hat{k}$ and $3\hat{i}-4\hat{j}-4\hat{k}$ are the vertices of a right angled triangle.

Ans: Let $A(2\hat{i}-\hat{j}+\hat{k})$, $B(\hat{i}-3\hat{j}-5\hat{k})$, $C(3\hat{i}-4\hat{j}-4\hat{k})$

$$\overrightarrow{AB}=-\hat{i}-2\hat{j}-6\hat{k},\ \left|\overrightarrow{AB}\right|=\sqrt{41}$$ **1/2**

$\overrightarrow{AC} = \hat{i} - 3\hat{j} - 5\hat{k}, \left|\overrightarrow{AC}\right| = \sqrt{35}$ **1/2**

$\overrightarrow{BC} = 2\hat{i} - \hat{j} + \hat{k}, \left|\overrightarrow{BC}\right| = \sqrt{6}$ **1/2**

Clearly $\left|\overrightarrow{AC}\right|^2 + \left|\overrightarrow{BC}\right|^2 = \left|\overrightarrow{AB}\right|^2$ **1/2**

$\therefore$ A, B, C are vertices of right angled triangle

OR

Using vectors, find the area of triangle ABC with vertices A(1, 1, 1) B(1, 2, 3) and C(2, 3, 1).

Ans: $\overrightarrow{AB} = \hat{j} + 2\hat{k}$ **1/2**

$\overrightarrow{AC} = \hat{i} + 2\hat{j}$ **1/2**

$\overrightarrow{AB} \times \overrightarrow{AC} = -4\hat{i} + 2\hat{j} - \hat{k}$ **1/2**

Required area $= \frac{1}{2}\sqrt{21}$ **1/2**

25. Find the angle between the lines.

$\vec{r} = \left(2\hat{j} - 3\hat{k}\right) + \lambda\left(\hat{i} + 2\hat{j} + 2\hat{k}\right)$ and

$\vec{r} = \left(2\hat{i} + 6\hat{j} + 3\hat{k}\right) + \mu\left(2\hat{i} + 3\hat{j} - 6\hat{k}\right)$.

Ans: Let θ be angle between lines

$$\cos\theta = \left|\frac{\left(\hat{i} + 2\hat{j} + 2\hat{k}\right)\cdot\left(2\hat{i} + 3\hat{j} - 6\hat{k}\right)}{\sqrt{3}\cdot\sqrt{7}}\right|$$ **1**

$\theta = \cos^{-1}\frac{4}{\sqrt{21}}$ **1**

26. Two dice are thrown once. Given that two numbers appearing on the dice are different, find the probability of the event 'the sum of numbers on the dice is 6'.

Ans: A : Sum of numbers is 6

B : Number appearing on dice are different

$A \cap B = (1, 5)\ (5, 1)\ (2, 4)\ (4, 2)$ **1**

$$P\left(\frac{A}{B}\right) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{4}{36}}{\frac{30}{36}} = \frac{2}{15}$$ **1/2+1/2**

SECTION-C

Question numbers 27 to 32 carry 4 marks each.

27. Consider $f : \mathbb{R}_+ \to [-5, \infty]$ given by $f(x) = 9x^2 + 6x - 5$. Show that f is invertible with $f^{-1}(y) = \frac{\sqrt{y+6}-1}{3}$, where $\mathbb{R}_+$ is the set of all non-negative real numbers.

Ans: One-One

Let $x_1, x_2 \in \mathbb{R}_+$ such that $f(x_1) = f(x_2)$

$\Rightarrow \quad 9x_1^2 + 6x_1 - 5 = 9x_2^2 + 6x_2 - 5$

$\Rightarrow \quad x_1 = x_2$ or $x_1 + x_2 = -\frac{2}{3}$ rejected

$\therefore$ f is one-one **2**

For range let $f(x) = y$

i.e. $9x^2 + 6x - 5 = y$

which gives $x = \frac{\sqrt{y+6}-1}{3} \in R_+$ **1**

if $\sqrt{y+6} - 1 \geq 0$ i.e. $y \geq -5$

$\therefore$ Range = $[-5, \infty)$ = codomain **1/2**

f is onto

As f is one-one & onto $\therefore$ f is invertible & $f^{-1}(y) = \frac{\sqrt{y+6}-1}{3}$ **1/2**

28. If $y = (\cos x)^x$, find $\frac{dy}{dx}$.

Ans: Taking log on both sides gives $\log y = x \log \cos x$ **1**

differentiating w.r.t. x to get

$\frac{1}{y}\frac{dy}{dx} = -x \tan x + \log \cos x$ **2**

$\therefore \quad \frac{dy}{dx} = (\cos x)^x (-x \tan x + \log \cos x)$ **1**

OR

If x = a cos θ and y = b sin θ, then prove that $\frac{d^2y}{dx^2} = -\frac{b^4}{a^2y^3}$.

Ans: $\frac{dx}{d\theta} = -a \sin\theta, \ \frac{dy}{d\theta} = b \cos\theta$ **1/2+1/2**

$\therefore \ \frac{dy}{dx} = -\frac{b}{a}\cot\theta$ **1**

$\frac{d^2y}{dx^2} = +\frac{b}{a}\text{cosec}^2\theta \cdot -\frac{1}{a\sin\theta}$ **1**

$= -\frac{b}{a^2\sin^3\theta} \quad \left(\text{putting } \sin\theta = \frac{y}{b}\right)$ **1/2**

$= -\frac{b^4}{a^2y^3}$ **1/2**

29. Evaluate :

$$\int_1^2 \frac{x\,dx}{(x+1)(x+2)}$$

Ans: Writing $\frac{x}{(x+1)(x+2)}$ as $\frac{2}{x+2} - \frac{1}{x+1}$

Given integral= $2\int_1^2 \frac{dx}{x+2} - \int_1^2 \frac{dx}{x+1}$ **1**

$= 2\left|\log(x+2)\right|_1^2 - \left|\log(x+1)\right|_1^2$ **2**

$= 2\log\frac{4}{3} - \log\frac{3}{2}$ or $\log\frac{32}{27}$ **1**

30. Find the particular solution of the differential equation $\frac{dy}{dx} + y \cot x = 2x + x^2 \cot x$, $(x \neq 0)$, given that y = 0, when $x = \frac{\pi}{2}$.

Ans: Integrating factor is $e^{\int \cot x\,dx} = \sin x$ **1**

Solution is $y \cdot \sin x = \int\left(x^2\cos x + 2x\sin x\right)dx + C$ **1**

$$y \sin x = x^2 \sin x - \int 2x \sin x \, dx + \int 2x \sin x \, dx + C$$

$$y \sin x = x^2 \sin x + C \quad \textbf{1}$$

Putting $y = 0, x = \frac{\pi}{2}$ to get $C = -\frac{\pi^2}{4}$ **1/2**

$\therefore$ Particular solution is $y \sin x = x^2 \sin x - \frac{\pi^2}{4}$ **1/2**

31. A furniture firm manufactures chairs and tables, each requiring the use of three machines A, B and C. Production of one chair requires 2 hours on machine A, 1 hour on machine B and 1 hour on machine C. Each table requires 1 hour each on machines A and B and 3 hours on machine C. The profit obtained by sellling one chair is ₹ 300; while by selling one table, the profit is ₹ 600. The total time available per week on machine A is 70 hours, on machine B is 40 hours and on machine C is 90 hours. Formulate an LPP to determine the number of chairs and tables the firm should make per week in order to get maximum profit.

Ans: Let number of chairs be x and table be y.

Maximize profit $Z = 300x + 600y$ **1**

Subject to constraints

$$\left.\begin{aligned} 2x + y &\le 70 \\ x + y &\le 40 \\ x + 3y &\le 90 \\ x, y &\ge 0 \end{aligned}\right\} \quad \textbf{3}$$

32. A bag contains 2 white, 3 red and 4 blue balls. Two balls are drawn one-by-one without replacement from the bag. Find the probability distribution of the number of red balls. Also, find the mean of the number of red balls.

Ans: Let X denotes number of Red balls

X	0	1	2	**1**
P(X)	$\frac{15}{36}$ or $\frac{5}{12}$	$\frac{18}{36}$ or $\frac{1}{2}$	$\frac{3}{36}$ or $\frac{1}{12}$	**2**
XP(X)	0	$\frac{18}{36}$	$\frac{6}{36}$	**1/2**

Mean $= \sum XP(X) = \frac{24}{36}$ or $\frac{2}{3}$ **1/2**

OR

A card from a pack of 52 playing cards is lost. From the remaining cards of the pack, two cards are drawn (one-by-one without replacement) and both are found to be diamonds. Find the probability of the lost card being a diamond card.

Ans: Let E_1 : Lost card is diamond

E_2 : Lost card is not a diamond

A : Two cards drawn are diamonds

$$P(E_1) = \frac{1}{4} \qquad P(E_2) = \frac{3}{4}$$ **1**

$$P(A/E_1) = \frac{{}^{12}C_2}{{}^{51}C_2} \qquad P(A/E_2) = \frac{{}^{13}C_2}{{}^{51}C_2}$$ **1**

$$P(E_1/A) = \frac{P(E_1)P(A/E_1)}{P(E_1)P(A/E_1)+P(E_2)P(A/E_2)}$$

$$= \frac{\frac{1}{4}\times\frac{{}^{12}C_2}{{}^{51}C_2}}{\frac{1}{4}\times\frac{{}^{12}C_2}{{}^{51}C_2}+\frac{3}{4}\times\frac{{}^{13}C_2}{{}^{51}C_2}}$$ **1**

$$= \frac{11}{50}$$ **1**

SECTION-D

Question numbers 33 to 36 carry 6 marks each.

33. If $A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$, show that $A^3 - 6A^2 + 9A - 4I = O$ and hence find A^{-1}.

Ans: $A^2 = \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix}$ **1**

$A^3 = \begin{bmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{bmatrix}$ **1**

$\therefore\ A^3 - 6A^2 + 9A - 4I$

$$= \begin{bmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{bmatrix} - \begin{bmatrix} 36 & -30 & 30 \\ -30 & 36 & -30 \\ 30 & -30 & 36 \end{bmatrix} + \begin{bmatrix} 18 & -9 & 9 \\ -9 & 18 & -9 \\ 9 & -9 & 18 \end{bmatrix} - \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$= 0$ **2**

$A^3 - 6A^2 + 9A - 4I = 0 \Rightarrow A^2 - 6A + 9I - 4A^{-1} = 0$

$$\Rightarrow A^{-1} = \frac{1}{4}\left[A^2 - 6A + 9I\right]$$ **1**

$$\therefore\ A^{-1} = \frac{1}{4}\begin{bmatrix} 3 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 3 \end{bmatrix}$$ **1**

OR

Using the properties of determinants,

prove that $\begin{vmatrix} 3a & -a+b & -a+c \\ -b+a & 3b & -b+c \\ -c+a & -c+b & 3c \end{vmatrix} = 3(a + b + c)\,(ab + bc + ca)$

Ans: Applying $C_1 \to C_1 + C_2 + C_3$ $\quad LHS = \begin{vmatrix} a+b+c & -a+b & -a+c \\ a+b+c & 3b & -b+c \\ a+b+c & -c+b & 3c \end{vmatrix}$ **2**

$$= (a+b+c)\begin{vmatrix} 1 & -a+b & -a+c \\ 1 & 3b & -b+c \\ 1 & -c+b & 3c \end{vmatrix}$$ **1**

$R_2 \to R_2 - R_1,\ R_3 \to R_3 - R_1$

$$= (a+b+c)\begin{vmatrix} 1 & -a+b & -a+c \\ 0 & 2b+a & a-b \\ 0 & a-c & 2c+a \end{vmatrix}$$ **2**

Expanding to get

$= (a + b + c)\,(3ab + 3bc + 3ca)$ **1**

$= 3(a + b + c)\,(ab + bc + ca) = RHS$

34. Find the absolute maximum and absolute minimum values of a fucntion f given by $f(x) = 12x^{4/3} - 6x^{1/3}$, $x \in [-1, 1]$.

Ans: $f'(x) = 16x^{1/3} - \dfrac{2}{x^{-2/3}}$ **2**

$f'(x) = 0 \Rightarrow x = \dfrac{1}{8}$ **1**

Critical points are $-1, 0, \dfrac{1}{8}, 1$ **1**

$f(-1) = 18,\ f(0) = 0,\ f\left(\dfrac{1}{8}\right) = -\dfrac{9}{4},\ f(1) = 6$ **1**

absolute maximum value is 18 **1/2**

absolute minimum value is $-\dfrac{9}{4}$ **1/2**

35. Using integration, find the area of the region bounded by the lines $2x + y = 4$, $3x - 2y = 6$ and $x - 3y + 5 = 0$.

Ans: Point of intersection of given lines are (2, 0), (1, 2) and (4, 3) **$1\frac{1}{2}$**

Required area $= \dfrac{1}{3}\int_1^4 (x+5)dx - 2\int_1^2 (2-x)dx - \dfrac{3}{2}\int_2^4 (x-2)dx$ **2**

$= \dfrac{(x+5)^2}{6}\Big|_1^4 + \dfrac{2(2-x)^2}{2}\Big|_1^2 - \dfrac{3}{4}(x-2)^2\Big|_2^4$ **$1\frac{1}{2}$**

$= \dfrac{15}{2} - 1 - 3 = \dfrac{7}{2}$ **1**

36. Prove that the line through A (0, –1, –1) and B(4, 5, 1) intersects the line throught C(3, 9, 4) and D(– 4, 4, 4).

Ans: dr's of AB : 4, 6, 2 **1**

dr's of CD : –7, –5, 0 **1**

Consider $\begin{vmatrix} x_2 - x_1 & y_2 - y_1 & z_2 - z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = \begin{vmatrix} 3-0 & 9+1 & 4+1 \\ 4 & 6 & 2 \\ -7 & -5 & 0 \end{vmatrix}$ **3**

$= 30 - 140 + 110 = 0$

$\therefore$ lines intersect each other **1**

OR

Find the coordinates of the foot of perpendicular and the perpendicular distance from the point P(4, 3, 2) to the plane x + 2y + 3z = 2. Also, find the image of P in the plane.

Ans: Let foot of perpendicular be A

Equation of PA is $\frac{x-4}{1}=\frac{y-3}{2}=\frac{z-2}{3}=\lambda$ **1**

Coordinate of A $(\lambda + 4, 2\lambda + 3, 3\lambda + 2)$ **1**

Putting in equation of plane we get $\lambda = -1$ **1**

Coordinate of A $(3, 1, -1)$ **1**

distance AP = $\sqrt{(4-3)^2+(3-1)^2+(2+1)^2}$

$= \sqrt{14}$ **1**

For coordinate of image P′(x, y, z)

$\frac{x+4}{2}=3, \frac{y+3}{2}=1, \frac{z+2}{2}=-1$

to get P′(2, –1, –4) **1**

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Senior School Certificate Examination-2020

Marking Scheme - Mathematics 65/1/1

***General instructions*:-**

1. You are aware that evaluation is the most important process in the actual and correct assessment of the candidates. A small mistake in evaluation may lead to serious problems which may affect the future of the candidates, education system and teaching profession. To avoid mistakes, it is requested that before starting evaluation, you must read and understand the spot evaluation guidelines carefully. **Evaluation is a 10-12 days mission for all of us. Hence, it is necessary that you put in your best efforts in this process.**
2. Evaluation is to be done as per instructions provided in the Marking Scheme. It should not be done according to one's own interpretation or any other consideration. Marking Scheme should be strictly adhered to and religiously followed. **However, while evaluating, answers which are based on latest information or knowledge and/or are innovative, they may be assessed for their correctness otherwise and marks be awarded to them.**
3. The Head-Examiner must go through the first five answer books evaluated by each evaluator on the first day, to ensure that evaluation has been carried out as per the instructions given in the Marking Scheme. The remaining answer books meant for evaluation shall be given only after ensuring that there is no significant variation in the marking of individual evaluators.
4. Evaluators will mark(√) wherever answer is correct. For wrong answer 'X"be marked. Evaluators will not put right kind of mark while evaluating which gives an impression that answer is correct and no marks are awarded. This is most common mistake which evaluators are committing.
5. If a question has parts, please award marks on the right-hand side for each part. Marks awarded for different parts of the question should then be totaled up and written in the left-hand margin and encircled. This may be followed strictly.
6. If a question does not have any parts, marks must be awarded in the left-hand margin and encircled. This may also be followed strictly.
7. If a student has attempted an extra question, answer of the question deserving more marks should be retained and the other answer scored out.
8. No marks to be deducted for the cumulative effect of an error. It should be penalized only once.
9. A full scale of marks _____(example0-100 marks as given in Question Paper) has to be used. Please do not hesitate to award full marks if the answer deserves it.
10. Every examiner has to necessarily do evaluation work for full working hours i.e. 8 hours every day and evaluate 20 answer books per day in main subjects and 25 answer books per day in other subjects (Details are given in Spot Guidelines).
11. Ensure that you do not make the following common types of errors committed by the Examiner in the past:-
    - Leaving answer or part thereof unassessed in an answer book.
    - Giving more marks for an answer than assigned to it.
    - Wrong totaling of marks awarded on a reply
    - Wrong transfer of marks from the inside pages of the answer book to the title page.
    - Wrong question wise totaling on the title page.
    - Wrong totaling of marks of the two columns on the title page.
    - Wrong grand total.
    - Marks in words and figures not tallying.
    - Wrong transfer of marks from the answer book to online award list.
    - Answers marked as correct, but marks not awarded. (Ensure that the right tick mark is correctly and clearly indicated. It should merely be a line. Same is with the X for incorrect answer.)
    - Half or a part of answer marked correct and the rest as wrong, but no marks awarded.
12. While evaluating the answer books if the answer is found to be totally incorrect, it should be marked as cross (X) and awarded zero (0)Marks.
13. Any unassessed portion, non-carrying over of marks to the title page, or totaling error detected by the candidate shall damage the prestige of all the personnel engaged in the evaluation work as also of the Board. Hence, in order to uphold the prestige of all concerned, it is again reiterated that the instructions be followed meticulously and judiciously.
14. The Examiners should acquaint themselves with the guidelines given in the Guidelines for spot Evaluation before starting the actual evaluation.
15. Every Examiner shall also ensure that all the answers are evaluated, marks carried over to the title page, correctly totaled and written in figures and words.
16. The Board permits candidates to obtain photocopy of the Answer Book on request in an RTI application and also separately as a part of the re-evaluation process on payment of the processing charges.

QUESTION PAPER CODE 65/1/1

EXPECTED ANSWER/VALUE POINTS

SECTION – A

Question Numbers 1 to 20 carry 1 mark each.

Question Numbers 1 to 10 are multiple choice type questions. Select the correct option.

Q.No. **Marks**

1. If A is a square matrix of order 3 and |A| = 5, then the value of |2A'| is

(A) –10 **(B)** 10 **(C)** –40 **(D)** 40

Ans: (D) 40 **1**

2. If A is a square matrix such that $A^2 = A$, then $(I - A)^3 + A$ is equal to

(A) I **(B)** 0 **(C)** I – A **(D)** I + A

Ans: (A) I **1**

3. The principal value of $\tan^{-1}\left(\tan\frac{3\pi}{5}\right)$

(A) $\frac{2\pi}{5}$ **(B)** $\frac{-2\pi}{5}$ **(C)** $\frac{3\pi}{5}$ **(D)** $\frac{-3\pi}{5}$

Ans: (B) $-\frac{2\pi}{5}$ **1**

4. If the projection of $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$ on $\vec{b} = 2\hat{i} + \lambda\hat{k}$, is zero, then the value of λ is

(A) 0 **(B)** 1 **(C)** $\frac{-2}{3}$ **(D)** $\frac{-3}{2}$

Ans: (C) $-\frac{2}{3}$ **1**

5. The vector equation of the line passing through the point (–1, 5, 4) and perpendicular to the plane z = 0 is

(A) $\vec{r} = -\hat{i} + 5\hat{j} + 4\hat{k} + \lambda(\hat{i} + \hat{j})$ **(B)** $\vec{r} = -\hat{i} + 5\hat{j} + (4 + \lambda)\hat{k}$

(C) $\vec{r} = \hat{i} - 5\hat{j} - 4\hat{k} + \lambda\hat{k}$ **(D)** $\vec{r} = \lambda\hat{k}$

Ans: (B) $\vec{r} = -\hat{i} + 5\hat{j} + (4 + \lambda)\hat{k}$ **1**

6. The number of arbitrary constants in the particular solution of a differential equation of second order is (are)

(A) 0 **(B)** 1 **(C)** 2 **(D)** 3

Ans: (A) 0 **1**

7. $\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sec^2 x\,dx$

(A) -1 **(B)** 0 **(C)** 1 **(D)** 2

Ans: (D) 2 **1**

8. The length of the perpendicular drawn from the point (4, –7, 3) on the y-axis is

(A) 3 units **(B)** 4 units **(C)** 5 units **(D)** 7 units

Ans: (C) 5 units **1**

9. If A and B are two independent events with $P(A)=\frac{1}{3}$ and $P(B)=\frac{1}{4}$, then $P(B'|A)$ is equal to

(A) $\frac{1}{4}$ **(B)** $\frac{1}{3}$ **(C)** $\frac{3}{4}$ **(D)** 1

Ans: (C) $\frac{3}{4}$ **1**

10. The corner points of the feasible region determined by the system of linear inequalities are (0, 0), (4,0), (2, 4) and (0, 5). If the maximum value of $z=ax+by$, where a, b > 0 occurs at both (2, 4) and (4,0), then

(A) a = 2b **(B)** 2a = b **(C)** a = b **(D)** 3a = b

Ans: (A) $a=2b$ **1**

Fill in the blanks in questions numbers 11 to 15

11. A relation R in a set A is called _____, if $(a_1, a_2) \in R$ implies $(a_2, a_1) \in R$, for all $a_1, a_2 \in A$.

Ans: Symmetric **1**

12. The greatest integer function defined by $f(x)=[x], 0<x<2$ is not differentiable at x = _____.

Ans: 1

13. If A is a matrix of order 3 × 2, then the order of the matrix A′ is _____.

Ans: 2 × 3 **1**

OR

A square matrix A is said to be skew-symmetric, if _____

Ans: $A=-A'\left(\text{or, } A'=-A\right)$ **1**

14. The equation of the normal to the curve $y^2=8x$ at the origin is _____

Ans: $y=0$ **1**

OR

The radius of a circle is increasing at the uniform rate of 3 cm/s. At the instant when the radius of the circle is 2 cm, its area increases at the rate of _____ cm^2/s.

Ans: 12π **1**

15. The position vectors of two points A and B are $\overrightarrow{OA} = 2\hat{i} - \hat{j} - \hat{k}$ and $\overrightarrow{OB} = 2\hat{i} - \hat{j} + 2\hat{k}$, respectively. The position vector of a point P which divides the line segment joining A and B in the ratio 2 : 1 is _____

Ans: $2\hat{i} - \hat{j} + \hat{k}$ **1**

Question numbers 16 to 20 are very short answer type questions

16. If $A = \begin{bmatrix} 2 & 0 & 0 \\ -1 & 2 & 3 \\ 3 & 3 & 5 \end{bmatrix}$, then find A (adj A).

Ans: $A \cdot adj(A) = |A|\, I$ **1/2**

$\therefore\ A \cdot adj\,(A) = 2I$ or $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$ **1/2**

17. Find $\int x^4 \log x\, dx$

Ans: $\int x^4 \cdot \log x\, dx = \log x \cdot \frac{x^5}{5} - \int \frac{1}{x} \cdot \frac{x^5}{5} dx$ **1/2**

$= \frac{x^5 \cdot \log x}{5} - \frac{x^5}{25} + c$ **1/2**

OR

Find $\int \frac{2x}{\sqrt[3]{x^2+1}} dx$

Ans: Let, $x^2 + 1 = t \quad \therefore \quad 2x\, dx = dt$ **1/2**

$$\int \frac{2x}{\sqrt[3]{x^2+1}} dx = \int \frac{1}{\sqrt[3]{t}} dt = \int t^{-1/3} dt = \frac{3}{2} t^{2/3} + c$$

$= \frac{3}{2}\left(x^2+1\right)^{2/3} + c$ **1/2**

18. Evaluate $\int_1^3 |2x-1|\,dx$.

Ans: $\int_1^3 12x-11dx = \int_1^3 (2x-1)\,dx = \left[\frac{1}{4}(2x-1)^2\right]_1^3$ **1/2**

$= 6$ **1/2**

19. Two cards are drawn at random and one-by-one without replacement from a well-shuffled pack of 52 playing cards. Find the probability that one card is red and the other is black.

Ans: $\frac{^{26}C_1 \times {}^{26}C_1}{^{52}C_2} = \frac{26}{51}$ **1/2+1/2**

20. Find $\int \frac{dx}{\sqrt{9-4x^2}}$.

Ans: $\int \frac{dx}{\sqrt{9-4x^2}} = \int \frac{dx}{\sqrt{3^2-(2x)^2}}$ **1/2**

$= \frac{1}{2}\sin^{-1}\left(\frac{2x}{3}\right)+c$ **1/2**

SECTION-B

Question numbers 21 to 26 carry 2 marks each.

21. Prove that $\sin^{-1}\left(2x\sqrt{1-x^2}\right) = 2\cos^{-1}x, \frac{1}{\sqrt{2}} \le x \le 1$

Ans: Put $x = \cos\theta \Leftrightarrow \theta = \cos^{-1}x$ **1/2**

L.H.S. $= \sin^{-1}\left(2x\sqrt{1-x^2}\right)$

$= \sin^{-1}(2\cos\theta\ \sin\theta) = \sin^{-1}(\sin 2\theta) = 2\theta = 2\cos^{-1}x = \text{R.H.S.}$ **$1\frac{1}{2}$**

OR

Consider a bijective function $f: R_+ \to (7,\infty)$ given by $f(x) = 16x^2 + 24x + 7$, where R_+ is the set of all positive real numbers. Find the inverse function of f.

Ans: Let $y = f(x) = 16x^2 + 24x + 7 = (4x+3)^2 - 2$ **1**

$\Rightarrow\ f^{-1}(y) = x = \frac{\sqrt{y+2}-3}{4}$ **1**

22. If $x = at^2, y = 2at$, then find $\frac{d^2y}{dx^2}$.

Ans: $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{2a}{2at} = \frac{1}{t}$ **1**

$\frac{d^2y}{dx^2} = -\frac{1}{t^2} \cdot \frac{dt}{dx} = -\frac{1}{t^2} \cdot \frac{1}{2at} = -\frac{1}{2a\,t^3}$ **1**

23. Find the points on the curve $y = x^3 - 3x^2 - 4x$ at which the tangent lines are parallel to the line $4x + y - 3 = 0$.

Ans: $\frac{dy}{dx} = -4 \Rightarrow 3x^2 - 6x - 4 = -4$ **1**

$\Rightarrow 3x(x-2) = 0 \;\therefore\; x = 0 ; x = 2$ **1/2**

Points on the curve are (0, 0), (2, –12) **1/2**

24. Find a unit vector perpendicular to each of the vectors $\vec{a}$ and $\vec{b}$ where $\vec{a} = 5\hat{i} + 6\hat{j} - 2\hat{k}$ and $\vec{b} = 7\hat{i} + 6\hat{j} + 2\hat{k}$.

Ans: $\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 5 & 6 & -2 \\ 7 & 6 & 2 \end{vmatrix} = 24\hat{i} - 24\hat{j} - 12\hat{k}$ **1**

Unit vector perpendicular to both $\vec{a}$ and $\vec{b}$ is $\frac{2}{3}\hat{i} - \frac{2}{3}\hat{j} - \frac{1}{3}\hat{k}$ **1**

OR

Find the volume of the parallelopiped whose adjacent edges are represented by $2\vec{a}, -\vec{b}$ and $3\vec{c}$, where $\vec{a} = \hat{i} - \hat{j} + 2\hat{k}$, $\vec{b} = 3\hat{i} + 4\hat{j} - 5\hat{k}$ and $\vec{c} = 2\hat{i} - \hat{j} + 3\hat{k}$

Ans: Volume of the parallelopiped $= \begin{vmatrix} 2 & -2 & 4 \\ -3 & -4 & 5 \\ 6 & -3 & 9 \end{vmatrix}$ **1**

$= |-24| = 24$ **1**

25. Find the value of k so that the lines $x = -y = kz$ and $x - 2 = 2y + 1 = -z + 1$ are perpendicular to each other.

Ans: The lines, $\frac{x}{1} = \frac{y}{-1} = \frac{z}{\frac{1}{k}}$ and $\frac{x-2}{1} = \frac{y+\frac{1}{2}}{\frac{1}{2}} = \frac{z-1}{-1}$ **1**

are perpendicular $\;\therefore\; 1 - \frac{1}{2} - \frac{1}{k} = 0 \Rightarrow k = 2$ **1**

26. The probability of finding a green signal on a busy crossing X is 30%. What is the probability of finding a green signal on X on two consecutive days out of three?

Ans: Probability of green signal on crossing $X = \frac{30}{100} = \frac{3}{10}$

Probability of not a green signal on crossing $X = 1 - \frac{3}{10} = \frac{7}{10}$ **1**

Probability of a green signal on X on two concecutative days out of three

$= \frac{3}{10} \times \frac{3}{10} \times \frac{7}{10} + \frac{7}{10} \times \frac{3}{10} \times \frac{3}{10} = \frac{63}{500}$ **1**

SECTION-C

Question numbers 27 to 32 carry 4 marks each.

27. Let N be the set of natural numbers and R be the relation on N × N defined by (a, b) R (c, d) iff ad = bc for all a, b, c, d ∈ N. Show that R is an equivalence relation.

Ans: Reflexive: For any (a, b) ∈ N × N

$a \cdot b = b \cdot a$

$\therefore$ (a, b) R (a, b) thus R is reflexive **1**

Symmetric: For (a, b), (c, d) ∈ N × N

(a, b) R (c, d) $\Rightarrow a \cdot d = b \cdot c$

$\Rightarrow c \cdot b = d \cdot a$

$\Rightarrow$ (c, d) R (a, b) $\therefore$ R is symmetric **$1\frac{1}{2}$**

Transitive : For any (a, b), (c, d), (e, f), ∈ N × N

(a, b) R (c, d) and (c, d) R (e, f)

$\Rightarrow a \cdot d = b \cdot c$ and $c \cdot f = d \cdot e$

$\Rightarrow a \cdot d \cdot c \cdot f = b \cdot c \cdot d \cdot e \Rightarrow a \cdot f = b \cdot e$

$\therefore$ (a, b) R (e, f), $\therefore$ R is transitive **$1\frac{1}{2}$**

$\therefore$ R is an equivalance Relation

28. If $y = e^{x^2 \cos x} + (\cos x)^x$, then find $\frac{dy}{dx}$.

Ans. Let $u = (\cos x)^x \Rightarrow y = e^{x^2 \cdot \cos x} + u$

$\therefore \frac{dy}{dx} = e^{x^2 \cdot \cos x}(2x \cdot \cos x - x^2 \cdot \sin x) + \frac{du}{dx}$ **$1\frac{1}{2}$**

$\log u = \log (\cos x)^x \Rightarrow \log u = x \,.\, \log(\cos x)$

Differentiate w.r.t. "x"

$$\frac{1}{u}\frac{du}{dx} = \log(\cos x) - x\tan x \Rightarrow \frac{du}{dx} = (\cos x)^x \{\log(\cos x) - x\tan x\}$$ **2**

Therefore,

$$\frac{dy}{dx} = e^{x^2\cdot\cos x}(2x\cdot\cos x - x^2\cdot\sin x) + (\cos x)^x\{\log(\cos x) - x\tan x\}$$ **1/2**

29. Find $\int \sec^3 x\,dx$.

Ans. $\int \sec^3 x\,dx = \int \sec x\cdot\sec^2 x\,dx = \int \sqrt{1+\tan^2 x}\cdot\sec^2 x\,dx$ **$1\frac{1}{2}$**

(Put $\tan x = t$; $\sec^2 x\,dx = dt$) **1/2**

$$= \int \sqrt{1+t^2}\,dt$$

$$= \frac{t}{2}\sqrt{1+t^2} + \frac{1}{2}\log\left|t+\sqrt{1+t^2}\right| + c$$ **$1\frac{1}{2}$**

$$= \frac{\sec x\cdot\tan x}{2} + \frac{1}{2}\log|\tan x + \sec x| + c$$ **1/2**

30. Find the general solution of the differential equation $ye^y dx = (y^3 + 2xe^y)dy$.

Ans. $y\cdot e^y dx = \left(y^3 + 2xe^y\right)dy \Rightarrow y\cdot e^y\frac{dy}{dx} = y^3 + 2xe^y$

$$\therefore \frac{dx}{dy} - \frac{2}{y}x = y^2\cdot e^{-y}$$ **1**

I.F. (Integrating factor) $= e^{-2\int\frac{1}{y}dy} = e^{-2\log y} = e^{\log\frac{1}{y^2}} = \frac{1}{y^2}$ **1**

$\therefore$ Solution is

$$x\cdot\frac{1}{y^2} = \int y^2\cdot e^{-y}\cdot\frac{1}{y^2}dy + c = \int e^{-y}dy + c$$ **1**

$$\Rightarrow \frac{x}{y^2} = -e^{-y} + c \quad \text{or} \quad x = -y^2e^{-y} + cy^2$$ **1**

OR

Find the particular solution of the differential equation

$x\frac{dy}{dx} = y - x\tan\left(\frac{y}{x}\right)$, given that $y = \frac{\pi}{4}$ at $x = 1$.

Ans. The differential equation can be written as:

$$\frac{dy}{dx} = \frac{y}{x} - \tan\frac{y}{x}, \text{ let } y = vx \therefore \frac{dy}{dx} = v + x\frac{dv}{dx}$$ **1**

$\Rightarrow \quad v + x\frac{dv}{dx} = v - \tan v \Rightarrow \cot v\, dv = -\frac{1}{x}dx$

Integrate both sides

$\log \sin v = -\log|x| + \log c \Rightarrow \log \sin\frac{y}{x} = \log\frac{c}{x}$ **2**

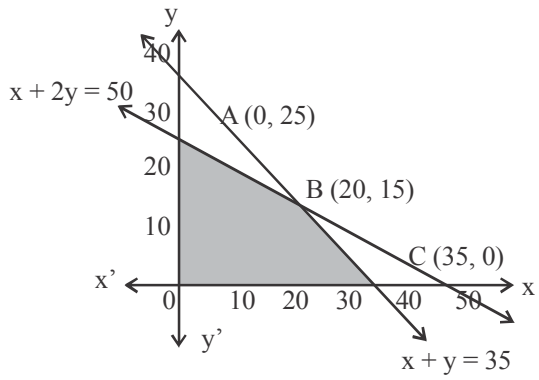
$\Rightarrow \quad x \cdot \sin\frac{y}{x} = c$, Put $y = \frac{\pi}{4}$ and $x = 1$

$\Rightarrow \quad \sin\frac{\pi}{4} = c$ or $c = \frac{1}{\sqrt{2}}$ **1/2**

$\therefore$ Particular solution is $x \cdot \sin\left(\frac{y}{x}\right) = \frac{1}{\sqrt{2}}$ **1/2**

31. A furniture trader deals in only two items – chairs and tables. He has ₹ 50,000 to invest and a space to store at most 35 items. A chair costs him ₹ 1000 and a table costs him ₹ 2000. The trader earns a profit of ₹ 150 and ₹ 250 on a chair and table, respectively. Formulate the above problem as an LPP to maximise the profit and solve it graphically.

Ans.



Let No. of chairs = x, No. of tables = y

Then L.P. P. is:

Maximize (Profit) : Z = 150x + 250y **1**

Subject to : $x + y \leq 35$

$1000x + 2000y \leq 50000 \Rightarrow x + 2y \leq 50$ **1**

$x, y \geq 0$

Correct graph **$1\frac{1}{2}$**

Corner:	Value of Z	
A(0, 25)	₹ 6250	
B(20, 15)	₹ 6750 (Max)	**1/2**
C(35, 0)	₹ 5250	

$\therefore$ Max (z) = ₹ 6750

Number of chairs = 20, Tables = 15

32. There are two bags, I and II. Bag I contains 3 red and 5 black balls and Bag II contains 4 red and 3 black balls. One ball is transferred randomly from Bag I to Bag II and then a ball is drawn randomly from Bag II. If the ball so drawn is found to be black in colour, then find the probability that the transferred ball is also black.

Ans. E_1 = Event that the ball transfered from Bag I is Black

E_2 = Event that the ball transfered from Bag I is Red **1/2**

A = Event that the ball drawn from Bag II is Black

$P(E_1) = \frac{5}{8}; P(E_2) = \frac{3}{8}; P\left(\frac{A}{E_1}\right) = \frac{4}{8} = \frac{1}{2}; P\left(\frac{A}{E_2}\right) = \frac{3}{8}$ **2**

Required Probability:

$$P\left(\frac{E_1}{A}\right)=\frac{P(E_1)\cdot P\left(\frac{A}{E_1}\right)}{P(E_1)\cdot P\left(\frac{A}{E_1}\right)+P(E_2)\cdot P\left(\frac{A}{E_2}\right)}=\frac{\frac{5}{8}\cdot\frac{1}{2}}{\frac{5}{8}\cdot\frac{1}{2}+\frac{3}{8}\cdot\frac{3}{8}}=\frac{20}{29}$$ $1\frac{1}{2}$

OR

An urn contains 5 red, 2 white and 3 black balls. Three balls are drawn, one-by-one, at random without replacement. Find the probability distribution of the number of white balls. Also, find the mean and the variance of the number of white balls drawn.

Ans. Let X = No. of white balls = 0, 1, 2

X :	0	1	2	1/2
P(X) :	$\frac{8}{10}\times\frac{7}{9}\times\frac{6}{8}=\frac{7}{15}$	$3\times\frac{8}{10}\times\frac{7}{9}\times\frac{2}{8}=\frac{7}{15}$	$3\times\frac{2}{10}\times\frac{1}{9}\times\frac{8}{8}=\frac{1}{15}$	$1\frac{1}{2}$
X · P(X) :	0	$\frac{7}{15}$	$\frac{2}{15}$	1/2
$X^2P(X)$:	0	$\frac{7}{15}$	$\frac{4}{15}$	

Mean = $\sum XP(X)=\frac{9}{15}=\frac{3}{5}$ 1/2

Variance = $\sum X^2P(x)-\left[\sum X\,P(X)^2\right]=\frac{11}{15}-\left[\frac{3}{5}\right]^2=\frac{28}{75}$ 1

SECTION-D

Question numbers 33 to 36 carry 6 marks each.

33. If $A=\begin{bmatrix}1 & 2 & -3\\ 3 & 2 & -2\\ 2 & -1 & 1\end{bmatrix}$, then find A^{-1} and use it to solve the following system of the equations:

$x+2y-3z=6$

$3x+2y-2z=3$

$2x-y+z=2$

Ans. $|A|=7$; $\text{adj}(A)=\begin{bmatrix}0 & 1 & 2\\ -7 & 7 & -7\\ -7 & 5 & -4\end{bmatrix}$; $A^{-1}=\frac{1}{|A|}\text{adj}\,A=\frac{1}{7}\begin{bmatrix}0 & 1 & 2\\ -7 & 7 & -7\\ -7 & 5 & -4\end{bmatrix}$ $1+1\frac{1}{2}+\frac{1}{2}$

The system of equations in Matrix form can be written as :

$A \cdot X = B$, where $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}; B = \begin{bmatrix} 6 \\ 3 \\ 2 \end{bmatrix}$ **1**

$X = A^{-1}B \Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{7}\begin{bmatrix} 0 & 1 & 2 \\ -7 & 7 & -7 \\ -7 & 5 & -4 \end{bmatrix}\begin{bmatrix} 6 \\ 3 \\ 2 \end{bmatrix} = \frac{1}{7}\begin{bmatrix} 7 \\ -35 \\ -35 \end{bmatrix} = \begin{bmatrix} 1 \\ -5 \\ -5 \end{bmatrix}$ **1**

$\therefore \; x = 1, y = -5, z = -5$ **1**

OR

Using properties of determinants, prove that

$$\begin{vmatrix} (b+c)^2 & a^2 & bc \\ (c+a)^2 & b^2 & ca \\ (a+b)^2 & c^2 & ab \end{vmatrix} = (a-b)(b-c)(c-a)(a+b+c)(a^2+b^2+c^2)$$

Ans. $\begin{vmatrix} (b+c)^2 & a^2 & bc \\ (c+a)^2 & b^2 & ca \\ (a+b)^2 & c^2 & ab \end{vmatrix}$

$= \begin{vmatrix} b^2+c^2 & a^2 & bc \\ c^2+a^2 & b^2 & ca \\ a^2+b^2 & c^2 & ab \end{vmatrix}$ $(C_1 \to C_1 - 2C_3)$ **1**

$= \begin{vmatrix} a^2+b^2+c^2 & a^2 & bc \\ a^2+b^2+c^2 & b^2 & ca \\ a^2+b^2+c^2 & c^2 & ab \end{vmatrix}$ $(C_1 \to C_1 + C_2)$ **1**

$= \begin{vmatrix} a^2+b^2+c^2 & a^2 & bc \\ 0 & b^2-a^2 & ca-bc \\ 0 & c^2-a^2 & ab-bc \end{vmatrix}$ $(R_2 \to R_2 - R_1, R_3 \to R_3 - R_1)$ **2**

$= (b-a)(c-a)\begin{vmatrix} a^2+b^2+c^2 & a^2 & bc \\ 0 & b+a & -c \\ 0 & c+a & -b \end{vmatrix}$ **1**

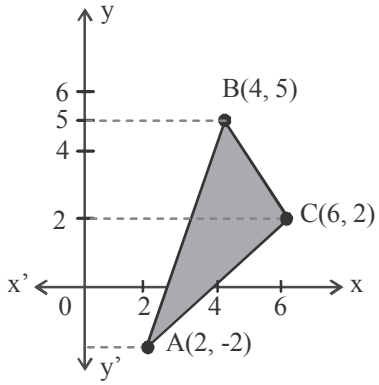
Expand along C_1

$= \left(a^2+b^2+c^2\right)(b-a)(c-a)\left(-b^2-ab+c^2+ac\right)$

$= (a-b)(b-c)(c-a)(a+b+c)\left(a^2+b^2+c^2\right)$ **1**

34. Using integration, find the area of the region bounded by the triangle whose vertices are (2, –2), (4,5) and (6,2).

Ans.



Let A(2, –2) ; B(4, 5) ; C(6, 2)

Equations of the lines

$$AB : x = \frac{2}{7}(y+9)$$

$$BC : x = -\frac{2}{3}(y-11)$$

$$AC : x = y + 4$$

1 1/2

Correct graph **1/2**

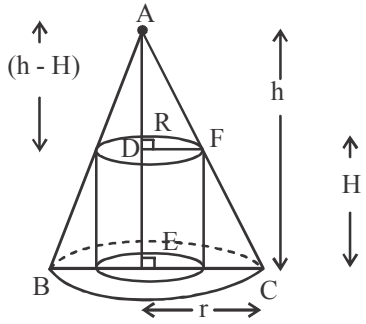
$$ar(\Delta ABC) = \int_{-2}^{2}(y+4)dy + \left(\frac{-2}{3}\right)\int_{2}^{5}(y-11)dy - \int_{-2}^{5}\frac{2}{7}(y+9)dy$$ **2**

$$= \frac{1}{2}\left[(y+4)^2\right]_{-2}^{2} - \frac{1}{3}\left[(y-11)^2\right]_{2}^{5} - \frac{1}{7}\left[(y+9)^2\right]_{-2}^{5}$$ **1 1/2**

$$= 16 + 15 - 21 = 10$$ **1/2**

35. Show that the height of the right circular cylinder of greatest volume which can be inscribed in a right circular cone of height h and radius r is one-third of the height of the cone, and the greatest volume of the cylinder is $\frac{4}{9}$ times the volume of the cone.

Ans.



Let H = Height of cylinder

R = Radius of cylinder

$$\text{Volume of cone} = \frac{\pi}{3}r^2h$$ **1/2**

$$V = \text{Volume of cylinder} = \pi R^2 H$$ **1/2**

$$\Delta ADF \sim \Delta AEC \Rightarrow \frac{h-H}{h} = \frac{R}{r} \Rightarrow R = \frac{r}{R}(h-H)$$ **1**

$$\therefore V = \pi \cdot H \cdot \frac{r^2}{h^2}(h-H)^2 = \frac{\pi r^2}{h^2}\left(H^3 - 2hH^2 + Hh^2\right)$$ **1**

$$V'(H) = \frac{\pi r^2}{h^2}\left(3H^2 - 4hH + h^2\right),\ V'(h) = 0 \Rightarrow H = \frac{h}{3}$$ **1+1**

$$V''(H) = \frac{\pi r^2}{h^2}(6H - 4h),\ V''\left(H = \frac{h}{3}\right) = \frac{\pi r^2}{h^2}(-2h) < 0$$ **1/2**

$$\therefore V \text{ is max iff } H = \frac{h}{3} \text{ and } R = \frac{2r}{3}$$

$$\frac{\text{Volume of cylinder}}{\text{Volume of cone}} = \frac{3\pi R^2 H}{\pi r^2 h} = 3 \cdot \frac{4r^2}{9} \cdot \frac{h}{3} \cdot \frac{1}{r^2 h} = \frac{4}{9}$$ **1/2**

36. Find the equation of the plane that contains the point A(2,1,–1) and is perpendicular to the line of intersection of the planes $2x + y - z = 3$ and $x + 2y + z = 2$. Also find the angle between the plane thus obtained and the y-axis.

Ans. Let equation of the required plane be:

$a(x - 2) + b(y - 1) + c(z + 1) = 0$ **$1\frac{1}{2}$**

Also : $2a + b - c = 0$

$a + 2b + c = 0$

Solving: $\frac{a}{3} = \frac{b}{-3} = \frac{c}{3} = k \Rightarrow a = 3k,\ b = -3k,\ c = 3k$ **$1\frac{1}{2}$**

$\therefore$ Equation of plane is : $3k(x - 2) - 3k(y - 1) + 3k(z + 1) = 0$

$\Rightarrow x - y + z = 0$ **$1\frac{1}{2}$**

Let angle between y-axis and plane = θ

then, $\sin\theta = \left|\frac{0-1+0}{\sqrt{1+1+1}}\right| = \left|\frac{-1}{\sqrt{3}}\right| \Rightarrow \theta = \sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$ **$1\frac{1}{2}$**

OR

Find the distance of the point P(–2, –4, 7) from the point of intersection Q of the line $\vec{r} = (3\hat{i} - 2\hat{j} + 6\hat{k}) + \lambda(2\hat{i} - \hat{j} + 2\hat{k})$ and the plane $\vec{r} \cdot (\hat{i} - \hat{j} + \hat{k}) = 6$. Also write the vector equation of the line PQ.

Ans. General point on line is: $\vec{r} = (3 + 2\lambda)\hat{i} + (-2 - \lambda)\hat{j} + (6 + 2\lambda)\hat{k}$ **1**

For the point of intersection:

$\left[(3 + 2\lambda)\hat{i} + (-2 - \lambda)\hat{j} + (6 + 2\lambda)\hat{k}\right] \cdot \left(\hat{i} - \hat{j} + \hat{k}\right) = 6$ **1**

$\Rightarrow 3 + 2\lambda + 2 + \lambda + 6 + 2\lambda = 6 \Rightarrow \lambda = -1$ **1**

$\therefore Q(\hat{i} - \hat{j} + 4\hat{k}) = Q(1, -1, 4)$ **1**

$PQ = 3\sqrt{3}$, equation of the line PQ : $\vec{r} = -2\hat{i} - 4\hat{j} + 7\hat{k} + \mu\left(3\hat{i} + 3\hat{j} - 3\hat{k}\right)$ **1+1**

Senior School Certificate Examination-2020
Marking Scheme - MATHEMATICS
Subject Code: 041 Paper Code: 65/2/1

***General instructions*:-**

1. You are aware that evaluation is the most important process in the actual and correct assessment of the candidates. A small mistake in evaluation may lead to serious problems which may affect the future of the candidates, education system and teaching profession. To avoid mistakes, it is requested that before starting evaluation, you must read and understand the spot evaluation guidelines carefully. **Evaluation is a 10-12 days mission for all of us. Hence, it is necessary that you put in your best efforts in this process.**
2. Evaluation is to be done as per instructions provided in the Marking Scheme. It should not be done according to one's own interpretation or any other consideration. Marking Scheme should be strictly adhered to and religiously followed. **However, while evaluating, answers which are based on latest information or knowledge and/or are innovative, they may be assessed for their correctness otherwise and marks be awarded to them.**
3. The Head-Examiner must go through the first five answer books evaluated by each evaluator on the first day, to ensure that evaluation has been carried out as per the instructions given in the Marking Scheme. The remaining answer books meant for evaluation shall be given only after ensuring that there is no significant variation in the marking of individual evaluators.
4. Evaluators will mark(√) wherever answer is correct. For wrong answer 'X"be marked. Evaluators will not put right kind of mark while evaluating which gives an impression that answer is correct and no marks are awarded. This is most common mistake which evaluators are committing.
5. If a question has parts, please award marks on the right-hand side for each part. Marks awarded for different parts of the question should then be totaled up and written in the left-hand margin and encircled. This may be followed strictly.
6. If a question does not have any parts, marks must be awarded in the left-hand margin and encircled. This may also be followed strictly.
7. If a student has attempted an extra question, answer of the question deserving more marks should be retained and the other answer scored out.
8. No marks to be deducted for the cumulative effect of an error. It should be penalized only once.
9. A full scale of marks 0 - 80 has to be used. Please do not hesitate to award full marks if the answer deserves it.
10. Every examiner has to necessarily do evaluation work for full working hours i.e. 8 hours every day and evaluate 20 answer books per day in main subjects and 25 answer books per day in other subjects (Details are given in Spot Guidelines).
11. Ensure that you do not make the following common types of errors committed by the Examiner in the past:-
    - Leaving answer or part thereof unassessed in an answer book.
    - Giving more marks for an answer than assigned to it.
    - Wrong totaling of marks awarded on a reply
    - Wrong transfer of marks from the inside pages of the answer book to the title page.
    - Wrong question wise totaling on the title page.
    - Wrong totaling of marks of the two columns on the title page.
    - Wrong grand total.
    - Marks in words and figures not tallying.
    - Wrong transfer of marks from the answer book to online award list.
    - Answers marked as correct, but marks not awarded. (Ensure that the right tick mark is correctly and clearly indicated. It should merely be a line. Same is with the X for incorrect answer.)
    - Half or a part of answer marked correct and the rest as wrong, but no marks awarded.
12. While evaluating the answer books if the answer is found to be totally incorrect, it should be marked as cross (X) and awarded zero (0)Marks.
13. Any unassessed portion, non-carrying over of marks to the title page, or totaling error detected by the candidate shall damage the prestige of all the personnel engaged in the evaluation work as also of the Board. Hence, in order to uphold the prestige of all concerned, it is again reiterated that the instructions be followed meticulously and judiciously.
14. The Examiners should acquaint themselves with the guidelines given in the Guidelines for spot Evaluation before starting the actual evaluation.
15. Every Examiner shall also ensure that all the answers are evaluated, marks carried over to the title page, correctly totaled and written in figures and words.
16. The Board permits candidates to obtain photocopy of the Answer Book on request in an RTI application and also separately as a part of the re-evaluation process on payment of the processing charges.

QUESTION PAPER CODE 65/2/1

EXPECTED ANSWER/VALUE POINTS

SECTION – A

Question Numbers 1 to 20 carry 1 mark each.

Question Numbers 1 to 10 are multiple choice type questions. Select the correct option.

Q.No. **Marks**

1. The relation R in the set {1, 2, 3} given by R = {(1, 2), (2, 1), (1, 1)} is

(A) symmetric and transitive, but not reflexive

(B) reflexive and symmetric, but not transitive

(C) symmetric, but neither reflexive nor transitive

(D) an equivalence relation

Ans: (C) symmetric, but neither reflexive nor transitive **1**

2. $\tan^{-1}3 + \tan^{-1}\lambda = \tan^{-1}\frac{3+\lambda}{1-3\lambda}$ is valid for what values of λ?

(A) $\lambda \in \left(-\frac{1}{3}, \frac{1}{3}\right)$ **(b)** $\lambda > \frac{1}{3}$

(C) $\lambda < \frac{1}{3}$ **(d)** All real values of λ

Ans: (C) $\lambda < \frac{1}{3}$ **1**

3. If A is a non-singular square matrix of order 3 such that $A^2 = 3A$, then value of |A| is

(A) –3 **(B)** 3 **(C)** 9 **(D)** 27

Ans: (D) 27 **1**

4. The function f : R → R given by $f(x) = -|x-1|$ is

(A) continuous as well as differentiable at x = 1

(B) not continuous but differentiable at x = 1

(C) continuous but not differentiable at x = 1

(D) neither continuous nor differentiable at x = 1

Ans: (C) continuous but not differentiable at x = 1 **1**

5. Let A = {1, 3, 5}. Then the number of equivalence relations in A containing (1, 3) is

(A) 1 **(B)** 2 **(C)** 3 **(D)** 4

Ans: (B) 2 **1**

6. The interval in which the function f given by $f(x) = x^2e^{-x}$ is strictly increasing, is

(A) $(-\infty, \infty)$ **(B)** $(-\infty, 0)$ **(C)** $(2, \infty)$ **(D)** $(0, 2)$

Ans: (D) $(0, 2)$ **1**

7. If $|\vec{a}| = 4$ and $-3 \le \lambda \le 2$, then $|\lambda\vec{a}|$ lies in

(A) $[0, 12]$ **(B)** $[2, 3]$ **(C)** $[8, 12]$ **(D)** $[-12, 8]$

Ans: (A) $[0, 12]$ **1**

8. The vector $3\hat{i} - \hat{j} + 2\hat{k}$, $2\hat{i} + \hat{j} + 3\hat{k}$ and $\hat{i} + \lambda\hat{j} - \hat{k}$ are coplanar if value of λ is

(A) -2 **(B)** 0

(C) 2 **(D)** Any real number

Ans: (A) -2 **1**

9. The area of a triangle formed by vertices O, A and B, where $\overrightarrow{OA} = \hat{i} + 2\hat{j} + 3\hat{k}$ $\hat{j}$ and $\overrightarrow{OB} = -3\hat{i} - 2\hat{j} + \hat{k}$ is

(A) $3\sqrt{5}$ sq. units **(B)** $5\sqrt{5}$ sq. units

(C) $6\sqrt{5}$ sq. units **(D)** 4 sq. units

Ans: (A) $3\sqrt{5}$ sq. units **1**

10. The coordinates of the foot of the perpendicular drawn from the point $(2, -3, 4)$ on the y-axis is

(A) $(2, 3, 4)$ **(B)** $(-2, -3, -4)$

(C) $(0, -3, 0)$ **(D)** $(2, 0, 4)$

Ans: (C) $(0, -3, 0)$ **1**

Fill in the blanks in questions numbers 11 to 15.

11. The range of the principal value branch of the function $y = \sec^{-1} x$ is _____.

Ans: $[0, \pi] - \left\{\frac{\pi}{2}\right\}$ **1**

OR

The principal value of $\cos^{-1}\left(-\frac{1}{2}\right)$ is _____.

Ans: $\frac{2\pi}{3}$ **1**

12. Given a skew-symmetric matrix $A = \begin{bmatrix} 0 & a & 1 \\ -1 & b & 1 \\ -1 & c & 0 \end{bmatrix}$, the value of $(a + b + c)^2$ is _____

Ans: 0 **1**

13. The distance between parallel planes $2x + y - 2z - 6 = 0$ and $4x + 2y - 4z = 0$ _____ units.

Ans: 2 **1**

OR

If P(1,0,–3) is the foot of the perpendicular from the origin to the plane, then the cartesian equation of the plane is _____

Ans: $x - 3z = 10$ **1**

14. If the radius of the circle is increasing at the rate of 0.5 cm/s, then the rate of increase of its circumference is _____

Ans: π cm/s **1**

15. The corner points of the feasible region of an LPP are (0,0), (0,8), (2,7), (5,4) and (6,0). The maximum profit $P = 3x + 2y$ occurs at the point _____

Ans: (5, 4) **1**

Question numbers 16 to 20 are very short answer type questions.

16. Differentiate $\sec^2(x^2)$ with respect to x^2.

Ans: Let $x^2 = u$, differentiating $\sec^2 u$ w.r.t u, putting $u = x^2$ **1/2**

$2\sec^2 x^2 \tan x^2$ **1/2**

OR

If $y = f(x^2)$ and $f'(x) = e^{\sqrt{x}}$, then find $\frac{dy}{dx}$.

Ans: $\frac{dy}{dx} = f'(x^2)2x$ **1/2**

$= 2xe^x$ **1/2**

17. Find the value of k, so that the function $f(x) = \begin{cases} kx^2 + 5, & \text{if } x \le 1 \\ 2 \quad , & \text{if } x > 1 \end{cases}$

is continuous at x = 1.

Ans: L.H.L. is k + 5 **1/2**

getting $k = -3$ **1/2**

18. Evaluate: $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x\cos^2 x\,dx$.

Ans: Let $f(x) = x\cos^2 x$ $\quad f(-x) = -f(x)$ or f is odd **1/2**

$\therefore \int_{-\pi/2}^{\pi/2} x\cos^2 x\,dx = 0$ **1/2**

19. Find the general solution of the differential equation $e^{y-x}\frac{dy}{dx} = 1$.

Ans: Given differential equation is $e^y dy = e^x\,dx$ **1/2**

Integrating to get $e^y = e^x + C$ **1/2**

20. Find the coordinates of the point where the line $\frac{x-1}{3} = \frac{y+4}{7} = \frac{z+4}{2}$ cuts the line xy-plane.

Ans: Putting z = 0 in given equation gives $\frac{x-1}{3} = \frac{y+4}{7} = 2$ **1/2**

Coordinates of required point are (7, 10, 0) **1/2**

SECTION-B

Question numbers 21 to 26 carry 2 marks each.

21. If $A = \begin{bmatrix} -3 & 2 \\ 1 & -1 \end{bmatrix}$ and $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, find scalar k so that $A^2 + I = kA$.

Ans: $A^2 = \begin{bmatrix} 11 & -8 \\ -4 & 3 \end{bmatrix}$ **1**

$A^2 + I = kA \Rightarrow \begin{bmatrix} 12 & -8 \\ -4 & 4 \end{bmatrix} = \begin{bmatrix} -3k & 2k \\ k & -k \end{bmatrix}$ **1/2**

$k = -4$ **1/2**

22. If $f(x)=\sqrt{\frac{\sec x-1}{\sec x+1}}$, find $f'\left(\frac{\pi}{3}\right)$.

Ans: $f(x) = \sqrt{\frac{\frac{1}{\cos x}-1}{\frac{1}{\cos x}+1}} = \tan\frac{x}{2}$ **1**

$f'(x) = \frac{1}{2}\sec^2\frac{x}{2}$ **1/2**

$f'\left(\frac{\pi}{3}\right) = \frac{2}{3}$ **1/2**

OR

Find $f'(x)$ if $f(x)=(\tan x)^{\tan x}$.

Ans: Taking log on both sides. $\log f(x) = \tan x \log \tan x$ **1/2**

differentiating to get $\frac{f'(x)}{f(x)} = \sec^2 x + \sec^2 x \log \tan x$ **1**

Thus, $f'(x) = (\tan x)^{\tan x} \cdot \sec^2 x(1 + \log \tan x)$ **1/2**

23. Find $\int \frac{\tan^3 x}{\cos^3 x}dx$.

Ans: Given Integral is $I = \int \frac{\sin^3 x}{\cos^6 x}dx$ **1/2**

Put $\cos x = t$

$\sin x\, dx = -dt$ **1/2**

$= \int\left[\frac{-1}{t^6}+\frac{1}{t^4}\right]dt$

$= \frac{t^{-5}}{5}-\frac{t^{-3}}{3}+c$ **1/2**

$= \frac{1}{5(\cos x)^5}-\frac{1}{3(\cos x)^3}+c$ or $\frac{\sec^5 x}{5}-\frac{\sec^3 x}{3}+c$ **1/2**

24. Find a vector $\vec{r}$ equally inclined to the three axes and whose magnitude is $3\sqrt{3}$ units.

Ans: Let the vector $\vec{r} = a\hat{i} + a\hat{j} + a\hat{k}$ **1/2**

$\therefore \quad \sqrt{3a^2} = 3\sqrt{3}$ **1**

required vector is $3\hat{i} + 3\hat{j} + 3\hat{k}$ or $-3\hat{i} - 3\hat{j} - 3\hat{k}$ **1/2**

OR

Find the angle between unit vectors $\vec{a}$ and $\vec{b}$ so that $\sqrt{3}\,\vec{a}-\vec{b}$ is also a unit vector.

Ans: Using $\left|\sqrt{3}\,\vec{a}-\vec{b}\right|=1$ i.e. $\left|\sqrt{3}\,\vec{a}-\vec{b}\right|^2=1$ **1/2**

& getting $\vec{a}\cdot\vec{b}=\frac{\sqrt{3}}{2}$ **1**

getting angle $\frac{\pi}{6}$ or $30°$ **1/2**

25. Find the points of intersection of the line $\vec{r}=2\hat{i}-\hat{j}+2\hat{k}+\lambda(3\hat{i}+4\hat{j}+2\hat{k})$ and the plane $\vec{r}\cdot(\hat{i}-\hat{j}+\hat{k})=5$.

Ans: Any point on line is $(2+3\lambda,-1+4\lambda,2+2\lambda)$ **1**

Putting in equation of plane to get $\lambda = 0$ **1/2**

Thus point of intersection is (2, –1, 2) **1/2**

26. A purse contains 3 silver and 6 copper coins and a second purse contains 4 silver and 3 copper coins. If a coin is drawn at random from one of the two purses, find the probability that it is a silver coin.

Ans: E_1 : coin is drawn from purse 1.
E_2 : coin is drawn from purse 2. **1/2**
A : Silver coin is drawn

$$P(E_1)=\frac{1}{2},\ P(E_2)=\frac{1}{2},\ P\left(\frac{A}{E_1}\right)=\frac{3}{9},\ P\left(\frac{A}{E_2}\right)=\frac{4}{7}$$ **1**

$$P(A)=\frac{1}{2}\times\frac{3}{9}+\frac{1}{2}\times\frac{4}{7}$$

$$=\frac{19}{42}$$ **1/2**

SECTION-C

Question numbers 27 to 32 carry 4 marks each.

27. Check whether the relation R in the set N of natural numbers given by

$R=\{(a,b):a \text{ is divisor of } b\}$

is reflexive, symmetric or transitive. Also determine whether R is an equivalence relation.

Ans: For reflexive

Let $a \in N$ clearly a divides a $\therefore$ $(a, a) \in R$

$\therefore$ R is reflexive **1**

For symmetric

$(1, 2) \in R$ but $(2, 1) \notin R$ **1**

$\therefore$ R is not symmetric

For transitive

Let $(a, b), (b, c) \in R$

$\therefore$ a divides b and b divides c

$\Rightarrow$ a divides c $\quad \therefore \ (a, c) \in R$ **1**

R is transitive

As R is not symmetric $\therefore$ It is not an equivalence relation **1**

OR

Prove that $\tan^{-1}\frac{1}{4}+\tan^{-1}\frac{2}{9}=\frac{1}{2}\sin^{-1}\left(\frac{4}{5}\right)$.

Ans: LHS $= \tan^{-1}\left(\dfrac{\frac{1}{4}+\frac{2}{9}}{1-\frac{1}{4}\cdot\frac{2}{9}}\right) = \tan^{-1}\frac{1}{2}$ **2**

$= \frac{1}{2}\cdot 2\tan^{-1}\frac{1}{2}$ **1**

$= \frac{1}{2}\sin^{-1}\left(\dfrac{2\times\frac{1}{2}}{1+\frac{1}{4}}\right)=\frac{1}{2}\sin^{-1}\left(\frac{4}{5}\right)$ **1**

= RHS

28. If $\tan^{-1}\left(\frac{y}{x}\right)=\log\sqrt{x^2+y^2}$, prove that $\frac{dy}{dx}=\frac{x+y}{x-y}$.

Ans: Differentiating both sides w.r.t. x to get

$$\frac{1}{1+\frac{y^2}{x^2}}\cdot\frac{x\frac{dy}{dx}-y}{x^2} = \frac{1}{\sqrt{x^2+y^2}}\cdot\frac{2x+2y\frac{dy}{dx}}{2\sqrt{x^2+y^2}}$$ **$1\frac{1}{2}+1\frac{1}{2}$**

Simplyfying we get $x\frac{dy}{dx}-y=x+y\frac{dy}{dx}$ **$\frac{1}{2}$**

getting $\frac{dy}{dx}=\frac{x+y}{x-y}$ **$\frac{1}{2}$**

OR

If $y=e^{a\cos^{-1}x}, -1<x<1$, then show that $\left(1-x^2\right)\frac{d^2y}{dx^2}-x\frac{dy}{dx}-a^2y=0$

Ans: $\frac{dy}{dx} = \frac{-ae^{a\cos^{-1}x}}{\sqrt{1-x^2}}$ **1**

$\sqrt{1-x^2}\frac{dy}{dx} = -ae^{a\cos^{-1}x}$ **$\frac{1}{2}$**

Differentiating again & getting

$\sqrt{1-x^2}\frac{d^2y}{dx^2} - \frac{x}{\sqrt{1-x^2}}\cdot\frac{dy}{dx} = \frac{a^2e^{a\cos^{-1}x}}{\sqrt{1-x^2}}$ **2**

$(1-x^2)\frac{d^2y}{dx^2} - x\frac{dy}{dx} - a^2y = 0$ **$\frac{1}{2}$**

29. Find $\int \frac{x^3+1}{x^3-x}dx$

Ans: Writing Integral as $I = \int \frac{x^3 - x + x + 1}{x^3 - x}dx$ **1**

$= \int 1dx + \int \frac{1}{x(x-1)}\,dx$ **1**

$= x + \int\left(\frac{1}{x-1} - \frac{1}{x}\right)dx$ **1**

$= x + \log|x-1| - \log|x| + c$ **1**

30. Solve the following differential equation:

$(1+e^{y/x})dy + e^{y/x}\left(1-\frac{y}{x}\right)dx = 0 \ (x \neq 0)$

Ans: Writing given differential equation as $\frac{dy}{dx} = \frac{e^{y/x}(y/x-1)}{1+e^{y/x}}$ **1**

Putting $y = vx$ & $\frac{dy}{dx} = v + x\frac{dv}{dx}$ **$\frac{1}{2}$**

to get $x\frac{dv}{dx} = \frac{-(e^v+v)}{1+e^v}$ **1**

$\Rightarrow \int \frac{e^v+1}{e^v+v}dv = -\int\frac{dx}{x}$

$\log|e^v+v| = -\log|x| + \log c$ **1**

$e^{y/x} + \frac{y}{x} = \frac{c}{x}$ or $xe^{y/x} + y = c$ **$\frac{1}{2}$**

31. Find the shortest distance between the lines

$\vec{r} = 2\hat{i} - \hat{j} + \hat{k} + \lambda(3\hat{i} - 2\hat{j} + 5\hat{k})$

$\vec{r} = 3\hat{i} + 2\hat{j} - 4\hat{k} + \mu(4\hat{i} - \hat{j} + 3\hat{k})$

Ans: $\vec{a}_2 - \vec{a}_1 = \hat{i} + 3\hat{j} - 5\hat{k}$ **1**

$$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & -2 & 5 \\ 4 & -1 & 3 \end{vmatrix} = -\hat{i} + 11\hat{j} + 5\hat{k}$$ **2**

Using, shortest distance $= \left| \dfrac{(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)}{|\vec{b}_1 \times \vec{b}_2|} \right|$

$= \dfrac{7}{\sqrt{147}}$ units or $\dfrac{1}{\sqrt{3}}$ units **1**

32. A cotton industry manufactures pedestal lamps and wooden shades. Both the products require machine time as well as craftsman time in the making. The number of hour(s) required for producing 1 unit of each and the corresponding profit is given in the following table

Items	Machine time	Craftsman time	Profit (in ₹)
Pedestal lamp	1.5 hours	3 hours	30
Wooden shades	3 hours	1 hour	20

In a day, the factory has availability of not more than 42 hours of machine time and 24 hours of craftsman time.

Assuming that all items manufactured are sold, how should the manufacture schedule his daily production in order to maximize the profit? Formulate it as an LPP and solve it graphically.

Ans: Let number of pedestal lamps = x
number of wooden shades = y

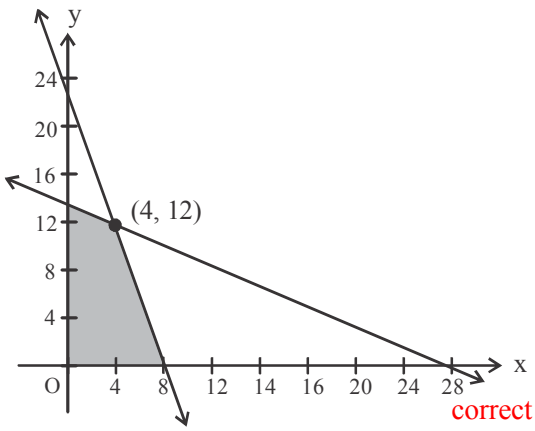
Maximize Profit Z = 30x + 20y **$\frac{1}{2}$**

Subject to constraints

$1.5x + 3y \leq 42$ **$1\frac{1}{2}$**

$3x + y \leq 24$

$x \geq 0, y \geq 0$



correct graph **1**

getting corners points & values of Z

(0, 0)	0
(8, 0)	240
(4, 12)	360
(0, 14)	280

1/2

Maximum profit = ₹ 360 where x = 4, y = 12 **1/2**

SECTION-D

Question numbers 33 to 36 carry 6 marks each.

33. $A = \begin{bmatrix} 5 & -1 & 4 \\ 2 & 3 & 5 \\ 5 & -2 & 6 \end{bmatrix}$, find A^{-1} and use it to solve the following system of equations:

$5x - y + 4z = 5$

$2x + 3y + 5z = 2$

$5x - 2y + 6z = -1$

Ans: $|A| = 51$ **1**

Cofactors: $\begin{matrix} A_{11} = 28 & A_{12} = 13 & A_{13} = -19 \\ A_{21} = -2 & A_{22} = 10 & A_{23} = 5 \\ A_{31} = -17 & A_{32} = -17 & A_{33} = 17 \end{matrix}$ **2**

$$A^{-1} = \frac{1}{51}\begin{bmatrix} 28 & -2 & -17 \\ 13 & 10 & -17 \\ -19 & 5 & 17 \end{bmatrix}$$ **1**

Given system is AX = B where $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} B = \begin{bmatrix} 5 \\ 2 \\ -1 \end{bmatrix}$ **1/2**

$$\Rightarrow X = A^{-1}B = \frac{1}{51}\begin{bmatrix} 28 & -2 & -17 \\ 13 & 10 & -17 \\ -19 & 5 & 17 \end{bmatrix}\begin{bmatrix} 5 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \\ -2 \end{bmatrix}$$ **1**

$x = 3, y = 2, z = -2$ **1/2**

OR

If x, y, z are different and $\begin{vmatrix} x & x^2 & 1+x^3 \\ y & y^2 & 1+y^3 \\ z & z^2 & 1+z^3 \end{vmatrix} = 0$, then using properties of determinants

show that $1 + xyz = 0$.

Ans: Writing L.H.S as $\begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix} + \begin{vmatrix} x & x^2 & x^3 \\ y & y^2 & y^3 \\ z & z^2 & z^3 \end{vmatrix} = 0$ **1**

$$\Rightarrow \begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix} + xyz \begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = 0$$ **1**

getting

$$(1+xyz)\begin{vmatrix} x & x^2 & 1 \\ y & y^2 & 1 \\ z & z^2 & 1 \end{vmatrix} = 0$$ **2**

Applying $R_2 \to R_2 - R_1$, $R_3 \to R_3 - R_1$

$$(1+xyz)\begin{vmatrix} x & x^2 & 1 \\ y-x & (y+x)(y-x) & 0 \\ z-x & (z+x)(z-x) & 0 \end{vmatrix} = 0$$ **1**

Expanding and simplifying to get

$\Rightarrow (1 + xyz)(x - y)(y - z)(z - x) = 0$ **1/2**

As x, y, z are different **1/2**

$\therefore \; 1 + xyz = 0$

34. Amongst all open (from the top) right circular cylindrical boxes of volume 125π cm^3, find the dimensions of the box which has the least surface area.

Ans: Let radius = r & height = h

$$\pi r^2 h = 125\pi \quad \text{or} \quad h = \frac{125}{r^2}$$ **1**

Surface Area, $S = 2\pi r h + \pi r^2$

$$S = \frac{250\pi}{r} + \pi r^2$$ **1**

$$\frac{dS}{dr} = \frac{-250\pi}{r^2} + 2\pi r$$ **1**

$$\frac{dS}{dr} = 0 \Rightarrow r = 5$$ **1/2+1**

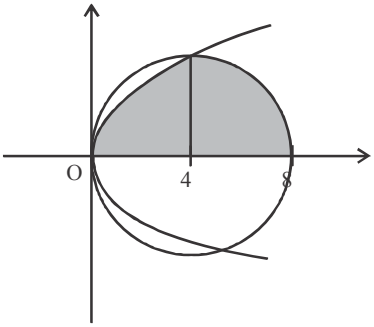
$$\frac{d^2S}{dr^2} = \frac{500\pi}{r^3} + 2\pi > 0$$ so s is least **1**

when r = 5 cm , gives h = 5 cm **1/2**

35. Using integration, find the area lying above x-axis and included between the circle $x^2 + y^2 = 8x$ and inside the parabola $y^2 = 4x$.

Ans. Correct figure **1**

x-coordinate of point of intersection is 4, 0 **1**



$$\text{Required Area} = \int_0^4 2\sqrt{x}\,dx + \int_4^8 \sqrt{16-(x-4)^2}\,dx$$ **2**

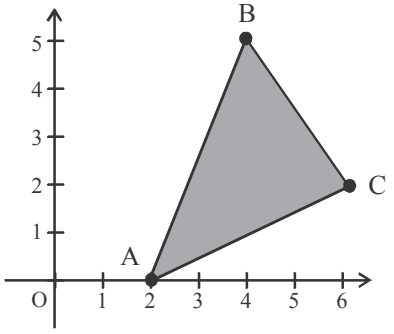
$$= \frac{4}{3}x^{\frac{3}{2}}\Big|_0^4 + \frac{x-4}{2}\sqrt{16-(x-4)^2} + 8\sin^{-1}\frac{x-4}{4}\Big|_4^8$$ **1**

$$= \frac{32}{3} + 4\pi$$ **1**

OR

Using the method of integration, find the area of the ΔABC, coordinates of whose vertices are A(2, 0), B(4, 5) and C(6, 3).

Ans. Correct figure **1**



Equation of AB : $y = \frac{5}{2}(x-2)$ **1/2**

Equation of BC : $y = 9 - x$ **1/2**

Equation of AC : $y = \frac{3}{4}(x-2)$ **1/2**

$$\text{Required Area} = \frac{5}{2}\int_2^4 (x-2)\,dx + \int_4^6 (9-x)\,dx - \frac{3}{4}\int_2^6 (x-2)\,dx$$ **2**

$$= \frac{5}{2}\frac{(x-2)^2}{2}\Big|_2^4 + \frac{(9-x)^2}{-2}\Big|_4^6 - \frac{3}{4}\frac{(x-2)^2}{2}\Big|_2^6$$ **1**

$$= 7$$ **1/2**

36. Find the probability distribution of the random variable X, which denotes the number of doublets in four throws of a pair of dice. Hence, find the mean of the number of doublets (X).

Ans. Let X denote the number of doublets P(doublet) = $\frac{1}{6}$, P(not a doublet) = $\frac{5}{6}$ **1**

X	0	1	2	3	4	**1**
P(X)	$\frac{625}{1296}$	$\frac{500}{1296}$	$\frac{150}{1296}$	$\frac{20}{1296}$	$\frac{1}{1296}$	**$2\frac{1}{2}$**
X·P(X)	0	$\frac{500}{1296}$	$\frac{300}{1296}$	$\frac{60}{1296}$	$\frac{4}{1296}$	**1**

mean = $\sum X \cdot P(X) = \frac{864}{1296}$ or $\frac{2}{3}$ **1/2**

Senior School Certificate Examination-2020
Marking Scheme - MATHEMATICS
Subject Code: 041 Paper Code: 65/3/1

***General instructions*:-**

1. You are aware that evaluation is the most important process in the actual and correct assessment of the candidates. A small mistake in evaluation may lead to serious problems which may affect the future of the candidates, education system and teaching profession. To avoid mistakes, it is requested that before starting evaluation, you must read and understand the spot evaluation guidelines carefully. **Evaluation is a 10-12 days mission for all of us. Hence, it is necessary that you put in your best efforts in this process.**
2. Evaluation is to be done as per instructions provided in the Marking Scheme. It should not be done according to one's own interpretation or any other consideration. Marking Scheme should be strictly adhered to and religiously followed. **However, while evaluating, answers which are based on latest information or knowledge and/or are innovative, they may be assessed for their correctness otherwise and marks be awarded to them.**
3. The Head-Examiner must go through the first five answer books evaluated by each evaluator on the first day, to ensure that evaluation has been carried out as per the instructions given in the Marking Scheme. The remaining answer books meant for evaluation shall be given only after ensuring that there is no significant variation in the marking of individual evaluators.
4. Evaluators will mark(√) wherever answer is correct. For wrong answer 'X"be marked. Evaluators will not put right kind of mark while evaluating which gives an impression that answer is correct and no marks are awarded. This is most common mistake which evaluators are committing.
5. If a question has parts, please award marks on the right-hand side for each part. Marks awarded for different parts of the question should then be totaled up and written in the left-hand margin and encircled. This may be followed strictly.
6. If a question does not have any parts, marks must be awarded in the left-hand margin and encircled. This may also be followed strictly.
7. If a student has attempted an extra question, answer of the question deserving more marks should be retained and the other answer scored out.
8. No marks to be deducted for the cumulative effect of an error. It should be penalized only once.
9. A full scale of marks 0 - 80 has to be used. Please do not hesitate to award full marks if the answer deserves it.
10. Every examiner has to necessarily do evaluation work for full working hours i.e. 8 hours every day and evaluate 20 answer books per day in main subjects and 25 answer books per day in other subjects (Details are given in Spot Guidelines).
11. Ensure that you do not make the following common types of errors committed by the Examiner in the past:-
    - Leaving answer or part thereof unassessed in an answer book.
    - Giving more marks for an answer than assigned to it.
    - Wrong totaling of marks awarded on a reply
    - Wrong transfer of marks from the inside pages of the answer book to the title page.
    - Wrong question wise totaling on the title page.
    - Wrong totaling of marks of the two columns on the title page.
    - Wrong grand total.
    - Marks in words and figures not tallying.
    - Wrong transfer of marks from the answer book to online award list.
    - Answers marked as correct, but marks not awarded. (Ensure that the right tick mark is correctly and clearly indicated. It should merely be a line. Same is with the X for incorrect answer.)
    - Half or a part of answer marked correct and the rest as wrong, but no marks awarded.
12. While evaluating the answer books if the answer is found to be totally incorrect, it should be marked as cross (X) and awarded zero (0)Marks.
13. Any unassessed portion, non-carrying over of marks to the title page, or totaling error detected by the candidate shall damage the prestige of all the personnel engaged in the evaluation work as also of the Board. Hence, in order to uphold the prestige of all concerned, it is again reiterated that the instructions be followed meticulously and judiciously.
14. The Examiners should acquaint themselves with the guidelines given in the Guidelines for spot Evaluation before starting the actual evaluation.
15. Every Examiner shall also ensure that all the answers are evaluated, marks carried over to the title page, correctly totaled and written in figures and words.
16. The Board permits candidates to obtain photocopy of the Answer Book on request in an RTI application and also separately as a part of the re-evaluation process on payment of the processing charges.

QUESTION PAPER CODE 65/3/1

EXPECTED ANSWER/VALUE POINTS

SECTION – A

Question Numbers 1 to 20 carry 1 mark each.

Question Numbers 1 to 10 are multiple choice type questions. Select the correct option.

Q.No. **Marks**

1. If f and g are two functions from R to R defined as $f(x) = |x| + x$ and $g(x) = |x| - x$, then $f \circ g(x)$ for $x < 0$ is

(A) $4x$ **(B)** $2x$ **(C)** 0 **(D)** $-4x$

Ans: (D) $-4x$ **1**

2. The principal value of $\cot^{-1}\left(-\sqrt{3}\right)$ is

(A) $-\frac{\pi}{6}$ **(B)** $\frac{\pi}{6}$ **(C)** $\frac{2\pi}{3}$ **(D)** $\frac{5\pi}{6}$

Ans: (D) $\frac{5\pi}{6}$ **1**

3. If $A = \begin{bmatrix} -2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$, then the value of $|\text{adj } A|$ is

(A) 64 **(B)** 16 **(C)** 0 **(D)** -8

Ans: (A) 64 **1**

4. The maximum value of slope of the curve $y = -x^3 + 3x^2 + 12x - 5$ is

(A) 15 **(B)** 12 **(C)** 9 **(D)** 0

Ans: (A) 15 **1**

5. $\int \frac{e^x(1+x)}{\cos^2(xe^x)}dx$ is equal to

(A) $\tan(xe^x)+c$ **(B)** $\cot(xe^x)+c$

(C) $\cot(e^x)+c$ **(D)** $\tan\left[e^x(1+x)\right]+c$

Ans: (A) $\tan(xe^x)+c$ **1**

6. The degree of the differential equation $x^2\frac{d^2y}{dx^2} = \left(x\frac{dy}{dx} - y\right)^3$

(A) 1 **(B)** 2 **(C)** 3 **(D)** 6

Ans: (A) 1 **1**

7. The value of p for which $p(\hat{i}+\hat{j}+\hat{k})$ is a unit vector is

(A) 0 **(B)** $\frac{1}{\sqrt{3}}$ **(C)** 1 **(D)** $\sqrt{3}$

Ans: (B) $\frac{1}{\sqrt{3}}$ **1**

8. The coordinates of the foot of the perpendicular drawn from the point (–2, 8, 7) on the XZ-plane is

(A) (–2, –8, 7) **(B)** (2, 8, –7) **(C)** (–2, 0, 7) **(D)** (0, 8, 0)

Ans: (C) (–2, 0, 7) **1**

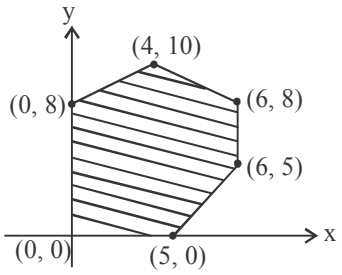
9. The vector equation of XY-plane is

(A) $\vec{r} \cdot \hat{k} = 0$ **(B)** $\vec{r} \cdot \hat{j} = 0$ **(C)** $\vec{r} \cdot \hat{i} = 0$ **(D)** $\vec{r} \cdot \vec{n} = 1$

Ans: (A) $\vec{r} \cdot \hat{k} = 0$ **1**

10. The feasible region for an LPP is shown below:

Let z = 3x– 4y be the objective function. Minimum of z occurs at



(A) (0, 0) **(B)** (0, 8) **(C)** (5, 0) **(D)** (4, 10)

Ans: (B) (0, 8) **1**

Fill in the blanks in questions numbers 11 to 15

11. If $y = \tan^{-1} x + \cot^{-1} x$, $x \in R$, then $\frac{dy}{dx}$ is equal to _____.

Ans: 0 **1**

OR

If $\cos(xy) = k$, where k is a constant and $xy \neq n\pi$, $n \in Z$,

then $\frac{dy}{dx}$ is equal to _____.

Ans: $-\frac{y}{x}$ **1**

12. The value of λ so that the function f defined by $f(x) = \begin{cases} \lambda x, & \text{if } x \le \pi \\ \cos x, & \text{if } x > \pi \end{cases}$

is continuous at $x = \pi$ is _____

Ans: $-\frac{1}{\pi}$ **1**

13. The equation of the tangent to the curve $y = \sec x$ at the point $(0, 1)$ is _____.

Ans: $y = 1$ **1**

14. The area of the parallelogram whose diagonals are $2\hat{i}$ and $-3\hat{k}$ is _____ square units.

Ans: 3 **1**

OR

The value of λ for which the vectors $2\hat{i} - \lambda\hat{j} + \hat{k}$ and $\hat{i} + 2\hat{j} - \hat{k}$ are orthogonal is _____.

Ans: $\frac{1}{2}$ **1**

15. A bag contains 3 black, 4 red and 2 green balls. If three balls are drawn simultaneously at random, then the probability that the balls are of different colours is _____

Ans: $\frac{2}{7}$ **1**

Question numbers 16 to 20 are very short answer type questions

16. Construct a 2×2 matrix $A = [a_{ij}]$ whose elements are given by $a_{ij} = |(i)^2 - j|$.

Ans: $\begin{bmatrix} 0 & 1 \\ 3 & 2 \end{bmatrix}$ $\frac{1}{2}$ mark for any two correct = **1**

17. Differentiate $\sin^2\left(\sqrt{x}\right)$ with respect to x.

Ans: $\frac{\sin\left(2\sqrt{x}\right)}{2\sqrt{x}}$ or $\frac{\sin\sqrt{x}\cos\sqrt{x}}{\sqrt{x}}$ **1**

18. Find the interval in which the function f given by $f(x) = 7 - 4x - x^2$ is strictly increasing.

Ans: $f'(x) = -4 - 2x$ **1/2**

$\Rightarrow$ f(x) is increasing on $(-\infty, -2)$ **1/2**

19. Evaluate: $\int_{-2}^{2} |x|\,dx$.

Ans: $\int_{-2}^{2} |x|dx = -\int_{-2}^{0} xdx + \int_{0}^{2} xdx = 4$ **1/2+1/2**

OR

Find $\int \frac{dx}{9+4x^2}$

Ans: $\int \frac{dx}{9+4x^2} = \frac{1}{6}\tan^{-1}\frac{2x}{3} + c$ **1/2+1/2**

20. An unbiased coin is tossed 4 times. Find the probability of getting at least one head.

Ans: $1-\left(\frac{1}{2}\right)^4 = \frac{15}{16}$ **1/2+1/2**

SECTION-B

Question numbers 21 to 26 carry 2 marks each.

21. Solve for x: $\sin^{-1}4x + \sin^{-1}3x = -\frac{\pi}{2}$

Ans: $\sin^{-1}(4x) + \sin^{-1}(3x) = -\frac{\pi}{2}$

$\Rightarrow \quad \sin^{-1}(4x) = -\frac{\pi}{2} - \sin^{-1}(3x)$

$\Rightarrow \quad 4x = -\sin\left(\frac{\pi}{2} + \sin^{-1}3x\right)$

$= -\cos(\sin^{-1}3x)$ **1**

$\Rightarrow \quad -4x = \sqrt{1-9x^2}$ **1/2**

$\Rightarrow \quad 16x^2 = 1 - 9x^2$

$\Rightarrow \quad 25x^2 = 1$

$\Rightarrow \quad x^2 = \frac{1}{25} \Rightarrow x = \pm\frac{1}{5}$

As $\sin^{-1}4x + \sin^{-1}3x < 0, x \neq \frac{1}{5}$ **1/2**

So, $x = -\frac{1}{5}$

OR

Express $\tan^{-1}\left(\frac{\cos x}{1-\sin x}\right), -\frac{3\pi}{2} < x < \frac{\pi}{2}$ in the simplest form.

Ans: $\tan^{-1}\left(\frac{\cos x}{1-\sin x}\right) = \tan^{-1}\left(\frac{\sin\left(\frac{\pi}{2}-x\right)}{1-\cos\left(\frac{\pi}{2}-x\right)}\right)$ **1**

$= \tan^{-1}\left[\cot\left(\frac{\pi}{4}-\frac{x}{2}\right)\right]$

$= \tan^{-1}\left[\tan\left(\frac{\pi}{2}-\frac{\pi}{4}+\frac{x}{2}\right)\right] = \frac{\pi}{4}+\frac{x}{2}$ **1**

22. Express $A = \begin{bmatrix} 4 & -3 \\ 2 & -1 \end{bmatrix}$ as a sum of a symmetric and a skew symmetric matrix.

Ans: $A = \begin{bmatrix} 4 & -3 \\ 2 & -1 \end{bmatrix} \Rightarrow A^T = \begin{bmatrix} 4 & 2 \\ -3 & -1 \end{bmatrix}$ **1/2**

$P = \frac{A + A^T}{2} = \frac{1}{2}\begin{bmatrix} 8 & -1 \\ -1 & -2 \end{bmatrix}$ **1/2**

$Q = \frac{A - A^T}{2} = \frac{1}{2}\begin{bmatrix} 0 & -5 \\ 5 & 0 \end{bmatrix}$ **1/2**

Now, $A = P + Q$ **1/2**

$P + Q = \frac{1}{2}\begin{bmatrix} 8 & -6 \\ 4 & -2 \end{bmatrix} = \begin{bmatrix} 4 & -3 \\ 2 & -1 \end{bmatrix} = A$

23. If $y^2 \cos\left(\frac{1}{x}\right) = a^2$, then find $\frac{dy}{dx}$.

Ans: $y^2 \cos\left(\frac{1}{x}\right) = a^2$

Then $2y\frac{dy}{dx}\cdot\cos\left(\frac{1}{x}\right) - y^2 \sin\left(\frac{1}{x}\right)\left(-\frac{1}{x^2}\right) = 0$ **1**

$\Rightarrow \quad 2y\cdot\cos\left(\frac{1}{x}\right)\frac{dy}{dx} = -\frac{y^2}{x^2}\sin\left(\frac{1}{x}\right)$

$\therefore \quad \frac{dy}{dx} = -\frac{y}{2x^2}\tan\left(\frac{1}{x}\right)$ **1**

24. Show that for any two non-zero vectors $\vec{a}$ and $\vec{b}$, $|\vec{a}+\vec{b}| = |\vec{a}-\vec{b}|$ iff $\vec{a}$ and $\vec{b}$ are perpendicular vectors.

Ans: $|\vec{a}+\vec{b}| = |\vec{a}-\vec{b}|$

$\Rightarrow \quad |\vec{a}+\vec{b}|^2 = |\vec{a}-\vec{b}|^2$

$\Rightarrow \quad 4\,\vec{a}\cdot\vec{b} = 0$ or $\vec{a}\cdot\vec{b} = 0$ or $\vec{a} \perp \vec{b}$ **1**

Let $\vec{a} \perp \vec{b}$

Then $\vec{a}\cdot\vec{b} = 0$

Thus, $|\vec{a}+\vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2$ and $|\vec{a}-\vec{b}|^2 = |\vec{a}|^2 + |\vec{b}|^2$

$\Rightarrow \quad |\vec{a}+\vec{b}| = |\vec{a}-\vec{b}|$ **1**

OR

Show that the vectors $2\hat{i}-\hat{j}+\hat{k}, 3\hat{i}+7\hat{j}+\hat{k}$ and $5\hat{i}+6\hat{j}+2\hat{k}$ form the sides of a right-angled triangle.

Ans: Let $\vec{a} = 2\hat{i}-\hat{j}+\hat{k}$, $\vec{b} = 3\hat{i}+7\hat{j}+\hat{k}$ and $\vec{c} = 5\hat{i}+6\hat{j}+2\hat{k}$

Since $\vec{c} = \vec{a}+\vec{b}$, three vectors form a triangle. **1**

Also, $\vec{a}\cdot\vec{b} = 0$.

So, triangle is a right angled triangle. **1**

25. Find the coordinates of the point where the line through (-1, 1, –8) and (5, –2, 10) crosses the ZX-plane.

Ans: Let the line segment AB is cut by ZX-plane in the ratio 1 : λ.
So, y-coordinate is zero. **1**

i.e., $\dfrac{-2+\lambda}{1+\lambda} = 0$ i.e. $\lambda = 2$

$\therefore$ The point of intersection is (1, 0, –2) **1**

26. If A and B are two events such that P(A) = 0.4, P(B) = 0.3 and $P(A \cup B) = 0.6$, then find $P(B' \cap A)$.

Ans: $P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.1$ **1**

$P(B' \cap A) = P(A) - P(A \cap B) = 0.3$ **1**

SECTION-C

Question numbers 27 to 32 carry 4 marks each.

27. Show that the function f: $(-\infty, 0) \to (-1, 0)$ defined by

$f(x) = \dfrac{x}{1+|x|}$, $x \in (-\infty, 0)$ is one-one and onto.

Ans: Let $x_1, x_2 \in (-\infty, 0)$ such that $f(x_1) = f(x_2)$

i.e., $\dfrac{x_1}{1+|x_1|} = \dfrac{x_2}{1+|x_2|}$

$\Rightarrow \quad \dfrac{x_1}{1-x_1} = \dfrac{x_2}{1-x_2}$ **1**

$\Rightarrow \quad x_1 - x_1x_2 = x_2 - x_1x_2$

$\Rightarrow \quad x_1 = x_2$

$\therefore$ f is one-one. **1**

Let $y \in (-1, 0)$ such that $y = \dfrac{x}{1+|x|}$

$\Rightarrow \quad y = \dfrac{x}{1-x}$

$\Rightarrow \quad x = \dfrac{y}{1+y}$ **1**

For each $y \in (-1, 0)$, there exists $x \in (-\infty, 0)$,

such that $f(x) = f\left(\dfrac{y}{1+y}\right) = \dfrac{\frac{y}{1+y}}{1+\left|\frac{y}{1+y}\right|}$

$= \dfrac{\frac{y}{1+y}}{1-\frac{y}{1+y}} = y$

Hence f is onto. **1**

OR

Show that the relation R in the set A = {1, 2, 3, 4, 5, 6} given by R = {(a, b) : |a – b| is divisible by 2} is an equivalence relation.

Ans: **Reflexive:** $|a-a| = 0$, which is divisible by 2 for all $a \in A$.

$\therefore$ $(a, a) \in R \Rightarrow$ R is reflexive. **1**

Symmetric: Let $(a, b) \in R$ i.e., $|a-b| = 2\lambda$, $\lambda \in \omega$

then $|b-a| = |-(a-b)| = |a-b| = 2\lambda$

$\Rightarrow$ (b, a) $\in$ R $\Rightarrow$ R is symmetric. **1**

Transitive : Let (a, b), (b, c) $\in$ R i.e., $|a-b| = 2\lambda$, $|b-c| = 2\mu$

$a - c = (a-b) + (b-c) = \pm 2\lambda \pm 2\mu = \pm 2(\lambda + \mu)$

$|a-c| = 2|\lambda + \mu|$, which is divisible by 2

$\Rightarrow$ (a, c) $\in$ R $\Rightarrow$ R is transitive. **1**

Hence R is an equivalence relation. **1**

28. If $y = x^3(\cos x)^x + \sin^{-1}\sqrt{x}$, find $\frac{dy}{dx}$.

Ans: Let $u = x^3(\cos x)^x$ and $v = \sin^{-1}\sqrt{x}$ so that $y = u + v$

$\log u = 3\log x + x\log(\cos x)$ **1/2**

$\Rightarrow \quad \frac{1}{u}\frac{du}{dx} = \frac{3}{x} - x\tan x + \log\cos x$ **1**

$\Rightarrow \quad \frac{du}{dx} = x^3(\cos x)^x\left[\frac{3}{x} - x\tan x + \log\cos x\right]$ .... (i) **1/2**

and $v = \sin^{-1}\sqrt{x} \Rightarrow \frac{dv}{dx} = \frac{1}{2\sqrt{x}\sqrt{1-x}}$ ... (ii) **1**

$\frac{dy}{dx} = \frac{du}{dx} + \frac{dv}{dx}$

$\Rightarrow \quad \frac{dy}{dx} = x^3(\cos x)^x\left[\frac{3}{x} - x\tan x + \log\cos x\right] + \frac{1}{2\sqrt{x - x^2}}$ **1**

29. Evaluate: $\int_{-1}^{5} (|x| + |x+1| + |x-5|)\,dx$

Ans: If $x \in [-1, 0] \Rightarrow f(x) = -x + x + 1 - x + 5 = 6 - x$ **1**

If $x \in [0, 5] \Rightarrow f(x) = x + x + 1 - x + 5 = x + 6$ **1**

$\therefore \int_{-1}^{5} (|x| + |x+1| + |x-5|)\,dx = \int_{-1}^{0} (6-x)dx + \int_{0}^{5} (x+6)dx$ **1**

$= \left[\frac{(6-x)^2}{-2}\right]_{-1}^{0} + \left[\frac{(x+6)^2}{2}\right]_{0}^{5}$

$= \frac{13}{2} + \frac{85}{2} = 49$ **1**

30. Find the general solution of the differential equation $x^2y\,dx - \left(x^3 + y^3\right)dy = 0$

Ans: $\dfrac{dx}{dy} = \dfrac{x^3 + y^3}{x^2y}$

Put $x = vy \Rightarrow \dfrac{dx}{dy} = v + y\cdot\dfrac{dv}{dy}$ **1**

$\therefore \quad v + y\dfrac{dv}{dy} = \dfrac{y^3(v^3+1)}{y^3v^2}$ **1**

$\Rightarrow \quad y\dfrac{dv}{dy} = \dfrac{1}{v^2}$

$\Rightarrow \quad v^2dv = \dfrac{dy}{y}$ **1**

Integrating both sides, we get

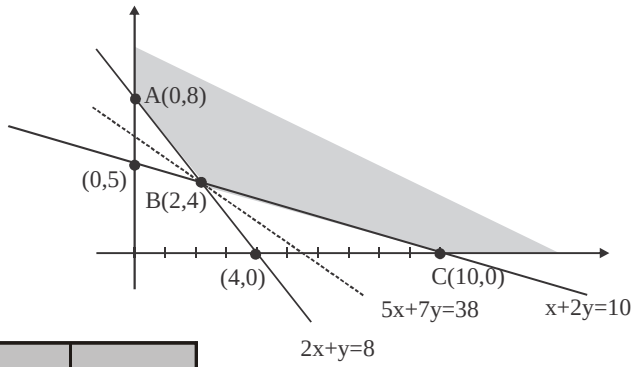
$\dfrac{v^3}{3} = \log y + c \Rightarrow \dfrac{x^3}{3y^3} = \log y + c$ **1**

$\Rightarrow \quad x^3 = 3y^3\log y + 3cy^3$

31. Solve the following LPP graphically:
Minimise z = 5x + 7y
subject to the constraints

$2x + y \geq 8$
$x + 2y \geq 10$
$x, y \geq 0$

Ans: **1+1**



Corner Points	Z	
A (0, 8)	56	
B (2, 4)	38	← Smallest value
C (10, 0)	50	

$1\frac{1}{2}$

To verify whether the smallest value of z = 38 is the minimum value we draw open half plane.

5x + 7y < 38. Since there is no common point with the possible feasible region except (2, 4).

Hence minimum value of z = 38 at x = 2 and y = 4. **1/2**

32. A bag contains two coins, one biased and the other unbiased. When tossed, the biased coin has a 60% chance of showing heads. One of the coins is selected at random and on tossing it shows tails. What is the probability it was an unbiased coin?

Ans: Let E_1 be the event that unbiased coin is tossed.
E_2 be the event that biased coin is tossed.
A be the event that coin tossed shows tail **1**

$$P(E_1)=\frac{1}{2}, P(E_2)=\frac{1}{2}, P(A \mid E_1)=\frac{1}{2}, P(A \mid E_2)=\frac{2}{5}$$ **1**

$$P(E_1 \mid A)=\frac{P(E_1)\cdot P(A \mid E_1)}{P(E_1)\cdot P(A \mid E_1)+P(E_2)\cdot P(A \mid E_2)}$$ **1**

$$=\frac{\frac{1}{2}\times\frac{1}{2}}{\frac{1}{2}\times\frac{1}{2}+\frac{1}{2}\times\frac{2}{5}}=\frac{5}{9}$$ **1**

OR

The probability distribution of a random variable X, where k is a constant is given below:

$$P(X=x)=\begin{cases} 0{\cdot}1, & \text{if} \quad x=0 \\ k\,x^2, & \text{if} \quad x=1 \\ k\,x, & \text{if} \quad x=2 \text{ or } 3 \\ 0, & \text{otherwise} \end{cases}$$

Determine

(a) the value of k

(b) $P(x \le 2)$

(c) Mean of the variable X.

Ans:

x_i	P_i
0	0.1
1	k
2	2k
3	3k

(i) $\sum P_i = 1$

$\Rightarrow 0{\cdot}1 + 6k = 1$

$\Rightarrow \quad k = \frac{3}{20}$ **1**

(ii) $P(x \le 2) = 0.1 + 3k$

$= \frac{1}{10} + \frac{9}{20} = \frac{11}{20}$ **1 1/2**

(iii) Mean $= \sum P_i x_i = 14k = \frac{21}{10}$ **1 1/2**

SECTION-D

Question numbers 33 to 36 carry 6 marks each.

33. Solve the following system of equations by matrix method:

$$x - y + 2z = 7$$
$$2x - y + 3z = 12$$
$$3x + 2y - z = 5$$

Ans: Writing given equations in matrix form

$$\begin{bmatrix} 1 & -1 & 2 \\ 2 & -1 & 3 \\ 3 & 2 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 7 \\ 12 \\ 5 \end{bmatrix}$$

Which is of the form AX = B **1**

Here $|A| = -2 \neq 0$ **1**

$$A^{-1} = \frac{1}{-2}\begin{bmatrix} -5 & 3 & -1 \\ 11 & -7 & 1 \\ 7 & -5 & 1 \end{bmatrix}$$ **2**

$$\therefore \quad X = A^{-1}B = \frac{1}{-2}\begin{bmatrix} -5 & 3 & -1 \\ 11 & -7 & 1 \\ 7 & -5 & 1 \end{bmatrix}\begin{bmatrix} 7 \\ 12 \\ 5 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}$$ **1**

$$\Rightarrow \quad x = 2,\ y = 1,\ z = 3$$ **1**

OR

Obtain the inverse of the following matrix using elementary operations:

$$A = \begin{bmatrix} 2 & 1 & -3 \\ -1 & -1 & 4 \\ 3 & 0 & 2 \end{bmatrix}$$

Ans: Using elementary row transformation,

$$A = IA \Rightarrow \begin{bmatrix} 2 & 1 & -3 \\ -1 & -1 & 4 \\ 3 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot A$$ **1**

Operating $R_1 \to R_1 + R_2$

$$\begin{bmatrix} 1 & 0 & 1 \\ -1 & -1 & 4 \\ 3 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

[4 marks for correct operations]

$R_2 \to R_2 + R_1$, $R_3 \to R_3 - 3R_1$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & -1 & 5 \\ 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 0 \\ -3 & -3 & 1 \end{bmatrix} \cdot A$$

$R_2 \to -R_2$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -5 \\ 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ -1 & -2 & 0 \\ -3 & -3 & 1 \end{bmatrix} \cdot A$$

$R_1 \to R_1 + R_3$, $R_2 \to R_2 - 5R_3$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} = \begin{bmatrix} -2 & -2 & 1 \\ 14 & 13 & -5 \\ -3 & -3 & 1 \end{bmatrix} \cdot A$$

$R_3 \to -R_3$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -2 & -2 & 1 \\ 14 & 13 & -5 \\ 3 & 3 & -1 \end{bmatrix} \cdot A$$

$$\Rightarrow A^{-1} = \begin{bmatrix} -2 & -2 & 1 \\ 14 & 13 & -5 \\ 3 & 3 & -1 \end{bmatrix}$$

1

34. Find the points on the curve $9y^2 = x^2$, where the normal to the curve makes equal intercepts with both the axes. Also find the equation of the normals.

Ans: Equation of given curve, $9y^2 = x^3$ ... (i)

$$\Rightarrow \quad 18y\frac{dy}{dx} = 3x^2 \Rightarrow \frac{dy}{dx} = \frac{x^2}{6y}$$

1/2

Slope of normal = $\dfrac{-6y}{x^2}$

1/2

$$-\frac{6y}{x^2} = \pm 1 \quad \text{(given)}$$

$\Rightarrow \quad y = \pm\frac{x^2}{6}$ ... (ii) **1**

From (i) & (ii), we get

$9\cdot\frac{x^4}{36} = x^3 \Rightarrow x^3(x-4) = 0 \Rightarrow x = 0, 4$ (x = 0 is rejected)

$x = 4, y^2 = \frac{64}{9} \Rightarrow y = \pm\frac{8}{3}$ **1**

Point of contacts are $\left(4, \frac{8}{3}\right), \left(4, \frac{-8}{3}\right)$ **$1\frac{1}{2}$**

Equation of normal at $\left(4, \frac{8}{3}\right)$ is $y - \frac{8}{3} = -(x-4)$

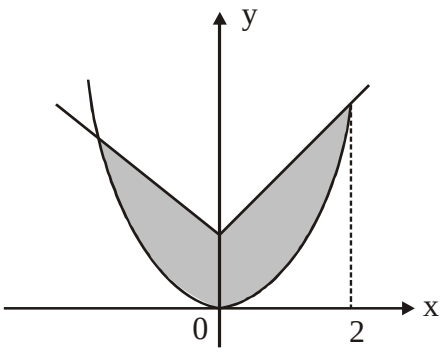
$\Rightarrow 3x + 3y - 20 = 0$ **1**

and equation of normal at $\left(4, -\frac{8}{3}\right)$ is $y + \frac{8}{3} = -(x-4)$

$\Rightarrow 3x + 3y = 20$ **1/2**

35. Find the area of the following region using integration: $\left\{(x, y) : y \le |x| + 2, y \ge x^2\right\}$.

Ans: **[Correct figure and shade (2)]**



$y = x^2$

$y = |x| + 2 = x + 2, \text{ if } x \ge 0$

$= -x + 2, \text{ if } x < 0$

Solving, $y = x^2$ and $y = x + 2$

$x^2 = x + 2 \Rightarrow x^2 - x - 2 = 0$

$\Rightarrow \quad (x-2)(x+1) = 0$

$\Rightarrow$ x = 2, –1 (x = –1 rejected) **1**

Required area $= 2\left[\int_0^2 (x+2)dx - \int_0^2 x^2 dx\right]$ **1**

$= 2\left[\frac{(x+2)^2}{2} - \frac{x^3}{3}\right]_0^2$ **1**

$= 2\left[6 - \frac{8}{3}\right] = \frac{20}{3}$ sq. units **1**

OR

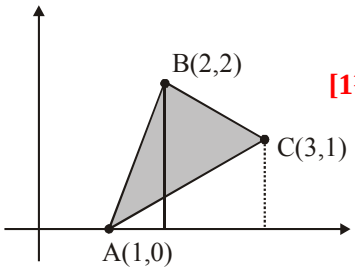
Using integration, find the area of a triangle whose vertices are (1,0), (2, 2) and (3,1).

Ans:

Equations of AB; $y = 2x - 2$

BC; $y = 4 - x$

AC; $y = \frac{1}{2}x - \frac{1}{2}$



[1½ marks for correct equations]

[1 mark for figure and shade]

Required area $= 2\int_1^2 (x-1)dx + \int_2^3 (4-x)dx - \frac{1}{2}\int_1^3 (x-1)dx$ **$1\frac{1}{2}$**

$= 2\left[\frac{(x-1)^2}{2}\right]_1^2 - \left[\frac{(4-x)^2}{2}\right]_2^3 - \frac{1}{2}\left[\frac{(x-1)^2}{2}\right]_1^3$ **1**

$= 1 + \frac{3}{2} - 1 = \frac{3}{2}$ sq. units **1**

36. Show that the lines

$\frac{x-2}{1} = \frac{y-2}{3} = \frac{z-3}{1}$ and $\frac{x-2}{1} = \frac{y-3}{4} = \frac{z-4}{2}$ intersect.

Also, find the coordinates of the point of intersection. Find the equation of the plane containing the two lines.

Ans: $\frac{x-2}{1} = \frac{y-2}{3} = \frac{z-3}{1} = \lambda$ (say)

and $\frac{x-2}{1} = \frac{y-3}{3} = \frac{z-4}{2} = \mu$ (say)

Arbitrary points on the lines are

$(\lambda + 2, 3\lambda + 2, \lambda + 3)$ and $(\mu + 2, 4\mu + 3, 2\mu + 4)$

$\Rightarrow$ $\lambda + 2 = \mu + 2$, and $\lambda + 3 = 2\mu + 4$

$\Rightarrow$ $\lambda = \mu$, solving we get $\lambda = -1$, $\mu = -1$ **1**

$\lambda = -1$, $\mu = -1$ satisfying y-coordinates $3\lambda + 2 = 4\mu + 3$ **1**

$\therefore$ Point of intersection is $(1, -1, 2)$ **1**

Equation of plane passing through two given lines are **1**

$$\begin{vmatrix} x-2 & y-2 & z-3 \\ 1 & 3 & 1 \\ 1 & 4 & 2 \end{vmatrix} = 0$$ **1**

$\Rightarrow \quad 2x - y + z - 5 = 0$ **1**

Senior School Certificate Examination-2020
Marking Scheme - MATHEMATICS
Subject Code: 041 Paper Code: 65/4/1

***General instructions*:-**

1. You are aware that evaluation is the most important process in the actual and correct assessment of the candidates. A small mistake in evaluation may lead to serious problems which may affect the future of the candidates, education system and teaching profession. To avoid mistakes, it is requested that before starting evaluation, you must read and understand the spot evaluation guidelines carefully. **Evaluation is a 10-12 days mission for all of us. Hence, it is necessary that you put in your best efforts in this process.**
2. Evaluation is to be done as per instructions provided in the Marking Scheme. It should not be done according to one's own interpretation or any other consideration. Marking Scheme should be strictly adhered to and religiously followed. **However, while evaluating, answers which are based on latest information or knowledge and/or are innovative, they may be assessed for their correctness otherwise and marks be awarded to them.**
3. The Head-Examiner must go through the first five answer books evaluated by each evaluator on the first day, to ensure that evaluation has been carried out as per the instructions given in the Marking Scheme. The remaining answer books meant for evaluation shall be given only after ensuring that there is no significant variation in the marking of individual evaluators.
4. Evaluators will mark(√) wherever answer is correct. For wrong answer 'X"be marked. Evaluators will not put right kind of mark while evaluating which gives an impression that answer is correct and no marks are awarded. This is most common mistake which evaluators are committing.
5. If a question has parts, please award marks on the right-hand side for each part. Marks awarded for different parts of the question should then be totaled up and written in the left-hand margin and encircled. This may be followed strictly.
6. If a question does not have any parts, marks must be awarded in the left-hand margin and encircled. This may also be followed strictly.
7. If a student has attempted an extra question, answer of the question deserving more marks should be retained and the other answer scored out.
8. No marks to be deducted for the cumulative effect of an error. It should be penalized only once.
9. A full scale of marks 0 - 80 has to be used. Please do not hesitate to award full marks if the answer deserves it.
10. Every examiner has to necessarily do evaluation work for full working hours i.e. 8 hours every day and evaluate 20 answer books per day in main subjects and 25 answer books per day in other subjects (Details are given in Spot Guidelines).
11. Ensure that you do not make the following common types of errors committed by the Examiner in the past:-
    - Leaving answer or part thereof unassessed in an answer book.
    - Giving more marks for an answer than assigned to it.
    - Wrong totaling of marks awarded on a reply
    - Wrong transfer of marks from the inside pages of the answer book to the title page.
    - Wrong question wise totaling on the title page.
    - Wrong totaling of marks of the two columns on the title page.
    - Wrong grand total.
    - Marks in words and figures not tallying.
    - Wrong transfer of marks from the answer book to online award list.
    - Answers marked as correct, but marks not awarded. (Ensure that the right tick mark is correctly and clearly indicated. It should merely be a line. Same is with the X for incorrect answer.)
    - Half or a part of answer marked correct and the rest as wrong, but no marks awarded.
12. While evaluating the answer books if the answer is found to be totally incorrect, it should be marked as cross (X) and awarded zero (0)Marks.
13. Any unassessed portion, non-carrying over of marks to the title page, or totaling error detected by the candidate shall damage the prestige of all the personnel engaged in the evaluation work as also of the Board. Hence, in order to uphold the prestige of all concerned, it is again reiterated that the instructions be followed meticulously and judiciously.
14. The Examiners should acquaint themselves with the guidelines given in the Guidelines for spot Evaluation before starting the actual evaluation.
15. Every Examiner shall also ensure that all the answers are evaluated, marks carried over to the title page, correctly totaled and written in figures and words.
16. The Board permits candidates to obtain photocopy of the Answer Book on request in an RTI application and also separately as a part of the re-evaluation process on payment of the processing charges.

XII MATHEMATICS

QUESTION PAPER CODE 65/4/1

EXPECTED ANSWER/VALUE POINTS

Q. No.	VALUE POINTS	Marks
	SECTION - A	
Question Numbers 1 to 20 carry 1 mark each.		
Q. Nos. 1 to 10 are multiple choice questions of 1 mark each. Select the correct option:		
1	The value of $\sin^{-1}\left(\cos\frac{3\pi}{5}\right)$ is (a) $\frac{\pi}{10}$ (b) $\frac{3\pi}{5}$ (c) $\frac{-\pi}{10}$ (d) $\frac{-3\pi}{5}$ **Answer:** $(c)-\frac{\pi}{10}$	1
2	If $A=\begin{bmatrix}2 & -3 & 4\end{bmatrix}, B=\begin{bmatrix}3\\2\\2\end{bmatrix}, X=\begin{bmatrix}1 & 2 & 3\end{bmatrix}$ and $Y=\begin{bmatrix}2\\3\\4\end{bmatrix}$, then $AB+XY$ equals (a) $[28]$ (b) $[24]$ (c) 28 (d) 24 **Answer:** $(a)[28]$	1
3	If $\begin{vmatrix}2 & 3 & 2\\ x & x & x\\ 4 & 9 & 1\end{vmatrix}+3=0$, then the value of x is (a) 3 (b) 0 (c) -1 (d) 1 **Answer:** $(c)-1$	1
4	$\int_0^{\frac{\pi}{8}}\tan^2(2x)\,dx$ is equal to (a) $\frac{4-\pi}{8}$ (b) $\frac{4+\pi}{8}$ (c) $\frac{4-\pi}{4}$ (d) $\frac{4-\pi}{2}$ **Answer:** $(a)\frac{4-\pi}{8}$	1
5	If $\vec{a}.\vec{b}=\frac{1}{2}\lvert\vec{a}\rvert\lvert\vec{b}\rvert$, then the angle between $\vec{a}$ and $\vec{b}$ is is (a) 0^0 (b) 30^0 (c) 60^0 (d) 90^0 **Answer:** $(c)60^0$	1

6	The two lines $x = ay + b, z = cy + d$; and $x = a'y + b', z = c'y + d'$ are perpendicular to each other, if (a) $\frac{a}{a'} + \frac{c}{c'} = 1$ (b) $\frac{a}{a'} + \frac{c}{c'} = -1$ (c) $aa' + cc' = 1$ (d) $aa' + cc' = -1$ **Answer:** $(d)\ aa' + cc' = -1$	1
7	The two planes $x - 2y + 4z = 10$ and $18x + 17y + kz = 50$ are perpendicular, if k is equal to (a) -4 (b) 4 (c) 2 (d) -2 **Answer:** $(b)\ 4$	1
8	In a LPP, if the objective function $z = ax + by$ has the same maximum value on two corner points of a feasible region, then the number of points at which z_{max} occurs is (a) 0 (b) 2 (c) finite (d) infinite **Answer:** (d) infinite	1
9	From the set $\{1, 2, 3, 4, 5\}$, two numbers a and b $(a \neq b)$ are chosen at random. The probability that $\frac{a}{b}$ is an integer is (a) $\frac{1}{3}$ (b) $\frac{1}{4}$ (c) $\frac{1}{2}$ (d) $\frac{3}{5}$ **Answer:** $(b)\ \frac{1}{4}$	1
10	A bag contains 3 white, 4 black and 2 red balls. If 2 balls are drawn at random (without replacement), then the probability that both the balls are white is (a) $\frac{1}{18}$ (b) $\frac{1}{36}$ (c) $\frac{1}{12}$ (d) $\frac{1}{24}$ **Answer:** $(c)\ \frac{1}{12}$	1
In Q. Nos. 11 to 15, fill in the blanks with correct word/sentence:		
11	If $f : R \to R$ be given by $f(x) = (3 - x^3)^{\frac{1}{3}}$, then $fof(x) =$ _____. **Answer:** x	1
12	If $\begin{bmatrix} x+y & 7 \\ 9 & x-y \end{bmatrix} = \begin{bmatrix} 2 & 7 \\ 9 & 4 \end{bmatrix}$, then $x.y =$ _____. **Answer: - 3**	1
13	The number of points of discontinuity of f defined by $f(x) = \lvert x \rvert - \lvert x + 1 \rvert$ is	

	_____. **Answer:** Zero	1
14	The slope of the tangent to the curve $y = x^3 - x$ at the point $(2,6)$ is _____. **Answer:** 11 **OR** The rate of change of the area of a circle with respect to its radius r , when r=3 cm, is _____. **Answer:** 6π cm^2/cm	1 1
15	If $\vec{a}$ is a non-zero vector, then $(\vec{a}.\hat{i})i+(\vec{a}.\hat{j})j+(\vec{a}.\hat{k})k$ equals _____. **Answer:** $\vec{a}$ **OR** The projection of the vector $\hat{i}-\hat{j}$ on the vector $\hat{i}+\hat{j}$ is _____. **Answer:** 0	1 1
Q. 16 to 20 are very short answer questions.		
16	Find adj$A$, if $A=\begin{bmatrix}2 & -1\\ 4 & 3\end{bmatrix}$. **Answer:** adj$A=\begin{bmatrix}3 & 1\\ -4 & 2\end{bmatrix}$ ($\frac{1}{2}$ mark for any two correct co-factors)	1
17	Find $\int \frac{2^{x+1}-5^{x-1}}{10^x}dx$ **Answer:** $I=\int\left(2\left(5^{-x}\right)-\frac{1}{5}\left(2^{-x}\right)\right)dx=-\frac{2}{5^x\log 5}+\frac{1}{5\left(2^x\right)\log 2}+C$	1
18	Evaluate $\int_0^{2\pi}\lvert\sin x\rvert dx$ **Answer:** $I=4\int_0^{\frac{\pi}{2}}\sin x\,dx=4$	$\frac{1}{2}+\frac{1}{2}$
19	If $\int_0^a \frac{dx}{1+4x^2}=\frac{\pi}{8}$, then find the value of a. **Answer:**	

	$\int_0^a \frac{dx}{(2x)^2+1}=\frac{\pi}{8}$ $\Rightarrow \frac{1}{2}\left[\tan^{-1}(2x)\right]_0^a=\frac{\pi}{8}$ $\Rightarrow a=\frac{1}{2}$ **OR** Find $\int \frac{dx}{\sqrt{x}+x}$ **Answer:** Put $\sqrt{x}=t$ $\therefore \int \frac{dx}{\sqrt{x}+x}=2\log\left(1+\sqrt{x}\right)+C$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$
20	Show that the function $y=ax+2a^2$ is a solution of the differential equation $2\left(\frac{dy}{dx}\right)^2+x\left(\frac{dy}{dx}\right)-y=0$. **Answer:** $y=ax+2a^2 \Rightarrow \frac{dy}{dx}=a$ $\text{LHS}=2\left(\frac{dy}{dx}\right)^2+x\left(\frac{dy}{dx}\right)-y$ $=2(a)^2+x(a)-\left(ax+2a^2\right)=0=\text{RHS}$	$\frac{1}{2}$ $\frac{1}{2}$
	SECTION – B	
Q. Nos. 21 to 26 carry 2 marks each.		
21	Check if the relation R on the set $A=\{1,2,3,4,5,6\}$ defined as $R=\{(x,y): y \text{ is divisible by } x\}$ is (i) symmetric (ii) transitive. **Answer:** (i) As $(2,4)\in R$ but $(4,2)\notin R \Rightarrow R$ is not symmetric. (ii) Let $(a,b)\in R$ and $(b,c)\in R$ $\Rightarrow b=\lambda a$ and $c=\mu b$ Now, $c=\mu b=\mu(\lambda a)\Rightarrow (a,c)\in R$ $\Rightarrow R$ is transitive. **OR** Prove that: $\frac{9\pi}{8}-\frac{9}{4}\sin^{-1}\left(\frac{1}{3}\right)=\frac{9}{4}\sin^{-1}\left(\frac{2\sqrt{2}}{3}\right)$ **Answer:**	1 1

	$\text{LHS} = \frac{9\pi}{8} - \frac{9}{4}\sin^{-1}\frac{1}{3}$ $= \frac{9}{4}\left[\frac{\pi}{2} - \sin^{-1}\frac{1}{3}\right] = \frac{9}{4}\cos^{-1}\frac{1}{3}$ $= \frac{9}{4}\sin^{-1}\left(\sqrt{1-\left(\frac{1}{3}\right)^2}\right) = \frac{9}{4}\sin^{-1}\left(\frac{2\sqrt{2}}{3}\right) = \text{RHS}$	1 1
22	Find the value of $\frac{dy}{dx}$ at $\theta = \frac{\pi}{3}$, if $x = \cos\theta - \cos 2\theta$, $y = \sin\theta - \sin 2\theta$. **Answer:** $\frac{dx}{d\theta} = -\sin\theta + 2\sin 2\theta$ $\frac{dy}{d\theta} = \cos\theta - 2\cos 2\theta$ $\therefore \frac{dy}{dx} = \frac{\cos\theta - 2\cos 2\theta}{-\sin\theta + 2\sin 2\theta}$ $\therefore \left.\frac{dy}{dx}\right\|_{\theta=\frac{\pi}{3}} = \sqrt{3}$	 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$
23	Show that the function f defined by $f(x) = (x-1)e^x + 1$ an increasing function for all $x > 0$. **Answer:** $f'(x) = xe^x$ Now $x > 0$ and $e^x > 0$ for all x $\therefore f'(x) > 0 \Rightarrow f$ is an increasing function.	 1 $\frac{1}{2}$ $\frac{1}{2}$
24	Find $\|\vec{a}\|$ and $\|\vec{b}\|$, if $\|\vec{a}\| = 2\|\vec{b}\|$ and $(\vec{a}+\vec{b}).(\vec{a}-\vec{b}) = 12$. **Answer:** $(\vec{a}+\vec{b})(\vec{a}-\vec{b}) = 12 \Rightarrow \|\vec{a}\|^2 - \|\vec{b}\|^2 = 12$ $\Rightarrow 3\|\vec{b}\|^2 = 12 \Rightarrow \|\vec{b}\| = 2$ Now, $\|\vec{a}\|^2 = 12 + \|\vec{b}\|^2 = 16 \Rightarrow \|\vec{a}\| = 4$ **OR** Find the unit vector perpendicular to each of the vectors $\vec{a} = 4\hat{i} + 3\hat{j} + \hat{k}$ and $\vec{b} = 2\hat{i} - \hat{j} + 2\hat{k}$ **Answer:**	 1 1

	$\vec{a}\times\vec{b}=7\hat{i}-6\hat{j}-10\hat{k}$ and $\left\|\vec{a}\times\vec{b}\right\|=\sqrt{185}$ Required unit vector $=\frac{1}{\sqrt{185}}\left(7\hat{i}-6\hat{j}-10\hat{k}\right)$	$1+\frac{1}{2}$ $\frac{1}{2}$
25	Find the equation of the plane with intercept 3 on the y-axis and parallel to xz – plane. **Answer:** Let required plane parallel to xz -plane is $y=k$ Given y -intercept is $3 \Rightarrow k=3$ $\Rightarrow$ Equation of required plane is $y=3$	 1 1
26	Find $\left[P(B\mid A)+P(A\mid B)\right]$, if $P(A)=\frac{3}{10}, P(B)=\frac{2}{5}$ and $P(A\cup B)=\frac{3}{5}$. **Answer:** $P(A\cap B)=\frac{3}{10}+\frac{2}{5}-\frac{3}{5}=\frac{1}{10}$ Now, $P(B\mid A)+P(A\mid B)$ $=\frac{P(A\cap B)}{P(A)}+\frac{P(A\cap B)}{P(B)}$ $=\frac{1}{3}+\frac{1}{4}=\frac{7}{12}$	 $\frac{1}{2}$ $\frac{1}{2}$ 1
	SECTION – C	
Q. Nos. 27 to 32 carry 4 marks each.		
27	Prove that the relation R on Z, defined by $R=\{(x,y):(x-y)\text{ is divisible by }5\}$ is an equivalence relation. **Answer:** For reflexive $x-x=0$, for every $x\in Z$ is divisible by $5 \Rightarrow (x,x)\in R$ For symmetric $(x,y)\in R\Rightarrow x-y$ is divisible by $5\Rightarrow y-x$ is divisible by 5 $\Rightarrow (y,x)\in R\Rightarrow R$ is symmetric. For transitive Let $(x,y)\in R$ and $(y,z)\in R$ $(x,y)\in R\Rightarrow x-y=5\lambda \quad ...(i)$ $(y,z)\in R\Rightarrow y-z=5\mu \quad ...(ii)$ adding (i) and (ii), $x-z=5(\lambda+\mu)=5k$ $\Rightarrow (x,z)\in R\Rightarrow R$ is transitive. Hence R is an equivalence relation.	 1 1 2
28	If $y=\sin^{-1}\left(\frac{\sqrt{1+x}+\sqrt{1-x}}{2}\right)$, then show that $\frac{dy}{dx}=\frac{-1}{2\sqrt{1-x^2}}$	

Answer:

Put $x=\cos 2\theta \Rightarrow \theta=\frac{1}{2}\cos^{-1}x$ — 1

$\therefore y=\sin^{-1}\left(\frac{\sqrt{2}\cos\theta+\sqrt{2}\sin\theta}{2}\right)=\sin^{-1}\left(\sin\left(\frac{\pi}{4}+\theta\right)\right)$ — 2

$\Rightarrow y=\frac{\pi}{4}+\theta=\frac{\pi}{4}+\frac{1}{2}\cos^{-1}x$ — $\frac{1}{2}$

$\Rightarrow \frac{dy}{dx}=\frac{-1}{2\sqrt{1-x^2}}$ — $\frac{1}{2}$

OR

Verify the Rolle's Theorem for the function $f(x)=e^x\cos x$ in $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$.

Answer:

f is continuous in $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$.

f is differentiable in $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ with $f'(x)=e^x(\cos x-\sin x)$ — 2

Also, $f\left(\frac{-\pi}{2}\right)=f\left(\frac{\pi}{2}\right)=0$

All conditions of Rolle's Theorem are satisfied. *So*, there exist $c\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$

such that $f'(c)=0\Rightarrow e^c(\cos c-\sin c)=0$ — 1

$\Rightarrow c=\frac{\pi}{4}\in\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ — 1

29 Evaluate: $\int_0^{\pi}\frac{x\sin x}{1+\cos^2 x}dx$

Answer:

$I=\int_0^{\pi}\frac{x\sin x}{1+\cos^2 x}dx \quad ...(i)$

$\Rightarrow I=\int_0^{\pi}\frac{(\pi-x)\sin(\pi-x)}{1+\cos^2(\pi-x)}dx=\int_0^{\pi}\frac{(\pi-x)\sin x}{1+\cos^2 x}dx \quad ...(ii)$ — 1

Adding (i) and (ii)

$2I=\int_0^{\pi}\frac{\pi\sin x}{1+\cos^2 x}dx$ — 1

Putting $\cos x=t$ gives $I=\frac{\pi}{2}\int_{-1}^{1}\frac{dt}{1+t^2}$ — 1

$\Rightarrow I=\frac{\pi}{2}\left[\tan^{-1}t\right]_{-1}^{1}=\frac{\pi^2}{4}$ — 1

30 For the differential equation given below, find a particular solution satisfying the given condition.

$(x+1)\frac{dy}{dx}=2e^{-y}+1$; $y=0$ when $x=0$

Answer:

Given differential equation can be written as

$\frac{dy}{2e^{-y}+1}=\frac{dx}{x+1}$ — 1

$\Rightarrow \int \frac{e^y}{2+e^y}dy=\int \frac{dx}{x+1}$

$\Rightarrow \log\left|2+e^y\right|=\log\left|x+1\right|+\log C$ — 2

$\Rightarrow 2+e^y=C(x+1)$

when $x=0, y=0 \Rightarrow C=3$ — $\frac{1}{2}$

$\therefore$ Required solution is $2+e^y=3(x+1)$ or $e^y=3x+1$ — $\frac{1}{2}$

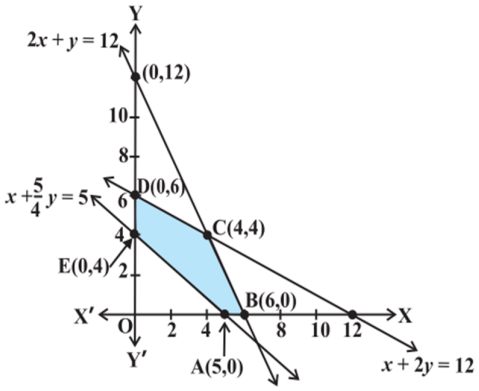
31 A manufacturer has three machines I, II and III installed in his factory. Machines I and II are capable of being operated for at most 12 hours whereas machine III must be operated for at least 5 hours a day . He produces only two items M and N each requiring the use of all the three machines. The number of hours required for producing 1 unit of each of M and N on the three machines are given in the following table:

Items	Number of hours required on Machines		
	I	II	III
M	1	2	1
N	2	1	1.25

He makes a profit of ₹ 600 and ₹ 400 on items M and N respectively. How many of each item should he produce so as to maximize his profit assuming that she can sell all the items that he produced? What will be the maximum profit?

Answer:

Let x units of item M and y units of item N are produced.



For correct graph : — $1\frac{1}{2}$ marks

Maximize $Z=600x+400y$

subject to

$x+2y\le 12$

$2x+y\le 12$ — $1\frac{1}{2}$

$x+1.25y\ge 5$

$x\ge 0, y\ge 0$

Corner points values:

$Z_{A(5,0)}=3000, Z_{B(6,0)}=3600$

$Z_{C(4,4)}=4000, Z_{D(0,6)}=2400$ — $\frac{1}{2}$

$Z_{E(0,4)}=1600$

$\therefore$ 4 units each of M and N must be produced to get maximum profit of Rs. 4,000 — $\frac{1}{2}$

32 A coin is biased so that the head is three times as likely to occur as tail. If the coin is tossed twice, find the probability distribution of number of tails. Hence find the mean of the number of tails.

Answer:

$P(\text{Head})=\frac{3}{4}, P(\text{Tail})=\frac{1}{4}$ — 1

Let $X=$ number of tails. Clearly X can be $0,1,2$ — $\frac{1}{2}$

Probability distribution is given by

X	0	1	2
$P(X)$	$\frac{9}{16}$	$\frac{6}{16}$	$\frac{1}{16}$

$1\frac{1}{2}$

$\text{Mean}=\sum X.P(X)=\frac{1}{2}$ — 1

OR

Suppose that 5 men out of 100 and 25 women out of 1000 are good orators. Assuming that there are equal number of men and women, find the probability of choosing a good orator.

Answer:

Let M be an event of choosing a man and N be an event of choosing a women. — $\frac{1}{2}$

A be an event of choosing a good orator.

$P(M)=P(W)=\frac{1}{2};$ — 2

$P(A|M)=\frac{5}{100}=\frac{1}{20}, P(A|W)=\frac{25}{1000}=\frac{1}{40}$

$P(A)=P(A|M).P(M)+P(A|W).P(W)$

$=\frac{1}{20}\times\frac{1}{2}+\frac{1}{40}\times\frac{1}{2}=\frac{3}{80}$ — $1+\frac{1}{2}$

SECTION – D

Q. Nos. 33 to 36 carry 6 marks each.

33 Using properties of determinants prove that: $\begin{vmatrix} a-b & b+c & a \\ b-c & c+a & b \\ c-a & a+b & c \end{vmatrix} = a^3+b^3+c^3-3abc.$

Answer:

$\text{LHS}=\Delta=\begin{vmatrix} a-b & b+c & a \\ b-c & c+a & b \\ c-a & a+b & c \end{vmatrix}$

$C_3 \to C_3+C_2$

$\Delta=\begin{vmatrix} a-b & b+c & a+b+c \\ b-c & c+a & a+b+c \\ c-a & a+b & a+b+c \end{vmatrix}$ — 1

taking $(a+b+c)$ common from C_3 and applying $R_1 \to R_1 - R_2$, $R_2 \to R_2 - R_3$

$$\Delta = (a+b+c)\begin{vmatrix} a-2b+c & b-a & 0 \\ b-2c+a & c-b & 0 \\ c-a & a+b & 1 \end{vmatrix}$$ 1+1+1

expanding along C_3,

$\Delta = (a+b+c)(a^2+b^2+c^2-ab-bc-ca)$ 1

$= a^3+b^3+c^3-3abc = \text{RHS}$ 1

OR

If $A = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix}$, then show that $A^3 - 4A^2 - 3A + 11I = O$. Hence find A^{-1}.

Answer:

$$A^2 = \begin{bmatrix} 9 & 7 & 5 \\ 1 & 4 & 1 \\ 8 & 9 & 9 \end{bmatrix}$$ 1

$$A^3 = \begin{bmatrix} 28 & 37 & 26 \\ 10 & 5 & 1 \\ 35 & 42 & 34 \end{bmatrix}$$ 1

$LHS = A^3 - 4A^2 - 3A + 11I$

$$= \begin{bmatrix} 28 & 37 & 26 \\ 10 & 5 & 1 \\ 35 & 42 & 34 \end{bmatrix} - 4\begin{bmatrix} 9 & 7 & 5 \\ 1 & 4 & 1 \\ 8 & 9 & 9 \end{bmatrix} - 3\begin{bmatrix} 1 & 3 & 2 \\ 2 & 0 & -1 \\ 1 & 2 & 3 \end{bmatrix} + 11\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O$$ 2

Now, $A^{-1} = -\frac{1}{11}(A^2 - 4A - 3I)$ 1

$$= -\frac{1}{11}\begin{bmatrix} 2 & -5 & -3 \\ -7 & 1 & 5 \\ 4 & 1 & -6 \end{bmatrix}$$ 1

34 Find the intervals on which the function $f(x) = (x-1)^3(x-2)^2$ is (a) strictly increasing (b) strictly decreasing.

Answer:

$f(x)=(x-1)^3(x-2)^2$

$\Rightarrow f'(x)=(x-1)^2(x-2)(5x-8)$ — 2

(a) for strictly increasing, $f'(x)>0$

$\Rightarrow (x-1)^2(x-2)(5x-8)>0$ — 1

$\Rightarrow (x-2)(5x-8)>0 \quad (\text{as } x\neq 1)$

$\Rightarrow x\in\left(-\infty,\frac{8}{5}\right)\cup(2,\infty) \quad (\text{as } x\neq 1)$ — $1\frac{1}{2}$

$\therefore x\in(-\infty,1)\cup\left(1,\frac{8}{5}\right)\cup(2,\infty)$

(b) for strictly decreasing, $f'(x)<0$

$\Rightarrow x\in\left(\frac{8}{5},2\right)$ — $1\frac{1}{2}$

OR

Find the dimensions of the rectangle of perimeter 36 cm which will sweep out a volume as large as possible, when revolved about one of its side. Also, find the maximum volume.

Answer:

Let sides of a rectangle are x cm and y cm.

Given $2x+2y=36\Rightarrow y=18-x$ — 1

Volume, $V=\pi x^2y=\pi x^2(18-x)=\pi\left(18x^2-x^3\right)$ — 1

$\Rightarrow \frac{dV}{dx}=\pi\left(36x-3x^2\right)$ — 1

For maxima/minima, put $\frac{dV}{dx}=0$

$\Rightarrow x=12$ cm $(\because x\neq 0)$ — 1

Again, $\frac{d^2V}{dx^2}=\pi(36-6x)$

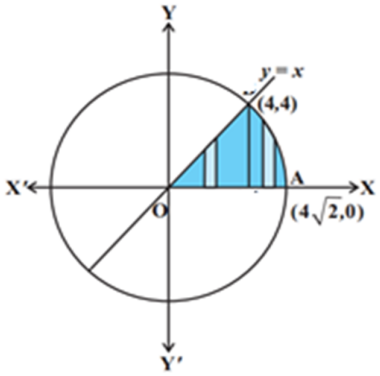
$\Rightarrow \left.\frac{d^2V}{dx^2}\right|_{x=12\,\text{cm}}=-36\pi<0$ — 1

$\therefore$ Volume is maximum when $x=12$ cm.

Also, $y=(18-x)$ cm $=6$ cm

Dimension of rectangle are $12\text{ cm}\times 6\text{cm}$

Maximum volume $=\pi x^2y=864\pi$ cm^3 — $\frac{1}{2}+\frac{1}{2}$

35	Find the area of the region lying in the first quadrant and enclosed by the x-axis, the line $y = x$ and the circle $x^2 + y^2 = 32$. **Answer:** Point of intersection of $y = x$ and $x^2 + y^2 = 32$ in first quadrant is $(4,4)$.	1
	 Correct figure	1
	$\text{Area} = \int_0^4 x\,dx + \int_4^{4\sqrt{2}} \sqrt{32 - x^2}\,dx$	1
	$= \left[\frac{x^2}{2}\right]_0^4 + \left[\frac{x}{2}\sqrt{32 - x^2} + 16\sin^{-1}\left(\frac{x}{4\sqrt{2}}\right)\right]_4^{4\sqrt{2}}$	2
	$= 8 + (4\pi - 8) = 4\pi$	1
36	Show that the lines $\vec{r} = \vec{a} + \lambda\vec{b}$ and $\vec{r} = \vec{b} + \mu\vec{a}$ are coplanar and the plane containing them is given by $\vec{r}.(\vec{a} \times \vec{b}) = 0$. **Answer:** Two lines $\vec{r} = \vec{a}_1 + \lambda\vec{b}_1$ and $\vec{r} = \vec{a}_2 + \mu\vec{b}_2$ are coplanar if $(\vec{a}_2 - \vec{a}_1).(\vec{b}_1 \times \vec{b}_2) = 0$ Now, $(\vec{a}_2 - \vec{a}_1).(\vec{b}_1 \times \vec{b}_2)$ $= (\vec{b} - \vec{a}).(\vec{b} \times \vec{a}) \quad (\because \vec{a}_1 = \vec{a}, \vec{b}_1 = \vec{b}, \vec{a}_2 = \vec{b}, \vec{b}_2 = \vec{a})$ $= \vec{b}.(\vec{b} \times \vec{a}) - \vec{a}.(\vec{b} \times \vec{a})$ $= \left[\vec{b} \quad \vec{b} \quad \vec{a}\right] - \left[\vec{a} \quad \vec{b} \quad \vec{a}\right] = 0$ Hence the lines are coplanar.	4
	Equation of plane containing them is given by, $(\vec{r} - \vec{a}).(\vec{b} \times \vec{a}) = 0 \quad [\because (\vec{r} - \vec{a}).\vec{n} = 0]$ $\Rightarrow \vec{r}.(\vec{b} \times \vec{a}) - \vec{a}.(\vec{b} \times \vec{a}) = 0$ $\Rightarrow \vec{r}.(\vec{b} \times \vec{a}) = 0 \quad (\because \left[\vec{a} \quad \vec{b} \quad \vec{a}\right] = 0)$	2

Senior School Certificate Examination-2020
Marking Scheme - MATHEMATICS
Subject Code: 041 Paper Code: 65/5/1

***General instructions*:-**

1. You are aware that evaluation is the most important process in the actual and correct assessment of the candidates. A small mistake in evaluation may lead to serious problems which may affect the future of the candidates, education system and teaching profession. To avoid mistakes, it is requested that before starting evaluation, you must read and understand the spot evaluation guidelines carefully. **Evaluation is a 10-12 days mission for all of us. Hence, it is necessary that you put in your best efforts in this process.**
2. Evaluation is to be done as per instructions provided in the Marking Scheme. It should not be done according to one's own interpretation or any other consideration. Marking Scheme should be strictly adhered to and religiously followed. **However, while evaluating, answers which are based on latest information or knowledge and/or are innovative, they may be assessed for their correctness otherwise and marks be awarded to them.**
3. The Head-Examiner must go through the first five answer books evaluated by each evaluator on the first day, to ensure that evaluation has been carried out as per the instructions given in the Marking Scheme. The remaining answer books meant for evaluation shall be given only after ensuring that there is no significant variation in the marking of individual evaluators.
4. Evaluators will mark(√) wherever answer is correct. For wrong answer 'X"be marked. Evaluators will not put right kind of mark while evaluating which gives an impression that answer is correct and no marks are awarded. This is most common mistake which evaluators are committing.
5. If a question has parts, please award marks on the right-hand side for each part. Marks awarded for different parts of the question should then be totaled up and written in the left-hand margin and encircled. This may be followed strictly.
6. If a question does not have any parts, marks must be awarded in the left-hand margin and encircled. This may also be followed strictly.
7. If a student has attempted an extra question, answer of the question deserving more marks should be retained and the other answer scored out.
8. No marks to be deducted for the cumulative effect of an error. It should be penalized only once.
9. A full scale of marks 0 - 80 has to be used. Please do not hesitate to award full marks if the answer deserves it.
10. Every examiner has to necessarily do evaluation work for full working hours i.e. 8 hours every day and evaluate 20 answer books per day in main subjects and 25 answer books per day in other subjects (Details are given in Spot Guidelines).
11. Ensure that you do not make the following common types of errors committed by the Examiner in the past:-
    - Leaving answer or part thereof unassessed in an answer book.
    - Giving more marks for an answer than assigned to it.
    - Wrong totaling of marks awarded on a reply
    - Wrong transfer of marks from the inside pages of the answer book to the title page.
    - Wrong question wise totaling on the title page.
    - Wrong totaling of marks of the two columns on the title page.
    - Wrong grand total.
    - Marks in words and figures not tallying.
    - Wrong transfer of marks from the answer book to online award list.
    - Answers marked as correct, but marks not awarded. (Ensure that the right tick mark is correctly and clearly indicated. It should merely be a line. Same is with the X for incorrect answer.)
    - Half or a part of answer marked correct and the rest as wrong, but no marks awarded.
12. While evaluating the answer books if the answer is found to be totally incorrect, it should be marked as cross (X) and awarded zero (0)Marks.
13. Any unassessed portion, non-carrying over of marks to the title page, or totaling error detected by the candidate shall damage the prestige of all the personnel engaged in the evaluation work as also of the Board. Hence, in order to uphold the prestige of all concerned, it is again reiterated that the instructions be followed meticulously and judiciously.
14. The Examiners should acquaint themselves with the guidelines given in the Guidelines for spot Evaluation before starting the actual evaluation.
15. Every Examiner shall also ensure that all the answers are evaluated, marks carried over to the title page, correctly totaled and written in figures and words.
16. The Board permits candidates to obtain photocopy of the Answer Book on request in an RTI application and also separately as a part of the re-evaluation process on payment of the processing charges.

XII MATHEMATICS

QUESTION PAPER CODE 65/5/1

EXPECTED ANSWER/VALUE POINTS

Q. No.	VALUE POINTS	Marks
	SECTION - A	
Question Numbers 1 to 20 carry 1 mark each.		
Q. Nos. 1 to 10 are multiple choice questions of 1 mark each. Select the correct option:		
1	If A is a square matrix of order 3, such that $A(\text{adj}A) = 10I$, then $\lvert\text{adj}A\rvert$ is equal to (a) 1 (b) 10 (c) 100 (d) 101 **Answer: (c)** 100	1
2	If A is a 3 x 3 matrix such that $\lvert A\rvert = 8$, then $\lvert 3A\rvert$ equals. (a) 8 (b) 24 (c) 72 (d) 216 Answer: **(d)** 216	1
3	If $y = Ae^{5x} + Be^{-5x,}$ then $\frac{d^2y}{dx^2}$ is equal to (a) $25y$ (b) $5y$ (c) $-25y$ (d) $15y$ **Answer: (a)** $25y$	1
4	$\int x^2 e^{x^3} dx$ equals (a) $\frac{1}{3}e^{x^3} + C$ (b) $\frac{1}{3}e^{x^4} + C$ (c) $\frac{1}{2}e^{x^3} + C$ (d) $\frac{1}{2}e^{x^2} + C$ **Answer: (a)** $\frac{1}{3}e^{x^3} + C$	1
5	If $\hat{i}, \hat{j}, \hat{k}$ are unit vectors along three mutually perpendicular directions, then (a) $\hat{i}.\hat{j} = 1$ (b) $\hat{i} \times \hat{j} = 1$ (c) $\hat{i}.\hat{k} = 0$ (d) $\hat{i} \times \hat{k} = 0$ **Answer: (c)** $\hat{i}.\hat{k} = 0$	1
6	ABCD is a rhombus whose diagonals intersect at E. Then $\overrightarrow{EA} + \overrightarrow{EB} + \overrightarrow{EC} + \overrightarrow{ED}$ equals (a) $\vec{0}$ (b) $\overrightarrow{AD}$ (c) $2\overrightarrow{BC}$ (d) $2\overrightarrow{AD}$ **Answer: (a)** $\vec{0}$	1
7	The lines $\frac{x-2}{1} = \frac{y-3}{1} = \frac{4-z}{k}$ and $\frac{x-1}{k} = \frac{y-4}{2} = \frac{z-5}{-2}$ are mutually perpendicular if the value of k is (a) $-\frac{2}{3}$ (b) $\frac{2}{3}$ (c) -2 (d) 2 **Answer: (a)** $-\frac{2}{3}$	1

8	The graph of the inequality $2x+3y>6$ is (a) half plane that contains the origin (b) half plane that neither contains the origin nor the points of the line $2x+3y=6$ (c) whole XOY-plane excluding the points on the line $2x+3y=6$ (d) entire XOY-plane **Answer: (b)** half plane that neither contains the origin nor the points of the line $2x+3y=6$	1
9	A card is picked at random from a pack of 52 playing cards. Given that the picked card is a queen, the probability of this card to be a card of spade is (a) $\frac{1}{3}$ (b) $\frac{4}{13}$ (c) $\frac{1}{4}$ (d) $\frac{1}{2}$ **Answer: (c)** $\frac{1}{4}$	1
10	A die is thrown once. Let A be the event that the number obtained is greater than 3. Let B be the event that the number obtained is less than 5. Then, $P(A \cup B)$ is (a) $\frac{2}{5}$ (b) $\frac{3}{5}$ (c) 0 (d) 1 **Answer: (d)** 1	1
In Q. Nos. 11 to 15, fill in the blanks with correct word/sentence:		
11	A relation in a set A is called _____ relation, if each element of A is related to itself. **Answer:** reflexive	1
12	If A + B = $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ and A – 2B = $\begin{bmatrix} -1 & 1 \\ 0 & -1 \end{bmatrix}$, then A = _____ **Answer:** $\begin{bmatrix} \frac{1}{3} & \frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} \end{bmatrix}$	1
13	The least value of the function $f(x)=ax+\frac{b}{x}$ $(a>0, b>0, x>0)$ is_____. **Answer:** $2\sqrt{ab}$	1
14	The integrating factor of the differential equation $x\frac{dy}{dx}+2y=x^2$ is _____. **Answer:** x^2 **OR** The degree of the differential equation $1+\left(\frac{dy}{dx}\right)^2=x$ is _____. **Answer:** 2	1 1

15	The vector equation of a line which passes through the points $(3,4,-7)$ and $(1,-1,6)$ is_____. **Answer:** $\vec{r}=\left(3\hat{i}+4\hat{j}-7\hat{k}\right)+\lambda\left(-2\hat{i}-5\hat{j}+13\hat{k}\right)$ Or, $\vec{r}=\left(3\hat{i}+4\hat{j}-7\hat{k}\right)+\lambda\left(2\hat{i}+5\hat{j}-13\hat{k}\right)$	1
	OR	
	The line of shortest distance between two skew lines is _____ to both the lines. **Answer:** perpendicular	1
Q. 16 to 20 are very short answer questions.		
16	Find the value of $\sin^{-1}\left[\sin\left(-\frac{17\pi}{8}\right)\right]$. **Answer:** $\sin^{-1}\left[\sin\left(-\frac{17\pi}{8}\right)\right]=-\sin^{-1}\left[\sin\left(2\pi+\frac{\pi}{8}\right)\right]$ $=-\frac{\pi}{8}$	$\frac{1}{2}$ $\frac{1}{2}$
17	For $A=\begin{bmatrix}3 & -4\\ 1 & -1\end{bmatrix}$, write A^{-1}. **Answer:** $\lvert A\rvert=1$ $A^{-1}=\begin{bmatrix}-1 & 4\\ -1 & 3\end{bmatrix}$	1/2 1/2
18	If the function f defined as $f(x)=\begin{cases}\frac{x^2-9}{x-3} & ,x\neq 3\\ k & ,x=3\end{cases}$ is continuous at $x=3$, find the value of k. **Answer:** $\lim_{x\to 3}\frac{x^2-9}{x-3}=6,\ \therefore\quad k=6$	$\frac{1}{2}+\frac{1}{2}$
19	If $f(x)=x^4-10$, then find the approximate value of $f(2.1)$. **Answer:** $f(2.1)\simeq f(2)+(0.1)\,f'(2)$ $=9.2$	1/2 1/2
	OR	
	Find the slope of the tangent to the curve $y=2\sin^2(3x)$ at $x=\frac{\pi}{6}$. **Answer:** $\frac{dy}{dx}=6\sin 6x$ $\therefore$ slope of tangent $=0$	1/2 1/2
20	Find the value of $\int_1^4\lvert x-5\rvert dx$. **Answer:** $\int_1^4\lvert x-5\rvert dx=\int_1^4(5-x)\,dx=\frac{15}{2}$	1/2+1/2

SECTION – B

Q. Nos. 21 to 26 carry 2 marks each.

21 If $f(x)=\frac{4x+3}{6x-4}, x\neq\frac{2}{3}$, then show that $(fof)(x)=x$, for all $x\neq\frac{2}{3}$. Also, write inverse of f.

Answer:

$$(fof)(x)=f\left(\frac{4x+3}{6x-4}\right)=\frac{4\left(\frac{4x+3}{6x-4}\right)+3}{6\left(\frac{4x+3}{6x-4}\right)-4}=\frac{34x}{34}=x$$ $1\frac{1}{2}$

Now, $(fof)(x)=x\Rightarrow f^{-1}=f$ or $f^{-1}(x)=\frac{4x+3}{6x-4}$ $\frac{1}{2}$

OR

Check if the relation R in the set **R** of real numbers defined as $R=\{(a,b):a<b\}$ is (i) symmetric, (ii) transitive.

Answer:

$(i)\ 1,2\in\mathbb{R}$ such that $1<2\Rightarrow(1,2)\in R$,
but since 2 is not less than $1\Rightarrow(2,1)\notin R$.
Hence R is not symmetric. — 1

(ii) Let $(a,b)\in R$ and $(b,c)\in R, \therefore a<b$ and $b<c$
$\Rightarrow a<c\Rightarrow(a,c)\in R\therefore R$ is transitive. — 1

22 Find $\int\frac{x}{x^2+3x+2}dx$

Answer:

$$\int\frac{x}{x^2+3x+2}dx=\int\frac{x}{(x+1)(x+2)}dx=\int\left(\frac{-1}{x+1}+\frac{2}{x+2}\right)dx$$ 1

$$=-\log|x+1|+2\log|x+2|+C$$ 1

23 If $x=a\cos\theta; y=b\sin\theta$, then find $\frac{d^2y}{dx^2}$.

Answer:

$$\frac{dx}{d\theta}=-a\sin\theta, \frac{dy}{d\theta}=b\cos\theta\Rightarrow\frac{dy}{dx}=-\frac{b}{a}\cot\theta$$ $\frac{1}{2}+\frac{1}{2}$

$$\frac{d^2y}{dx^2}=\frac{b}{a}\operatorname{cosec}^2\theta\left(\frac{-1}{a\sin\theta}\right)=-\frac{b}{a^2}\operatorname{cosec}^3\theta$$ $\frac{1}{2}+\frac{1}{2}$

OR

Find the differential of $\sin^2 x$ w.r.t. $e^{\cos x}$.

Answer:

Let $y=\sin^2 x$ and $z=e^{\cos x}\therefore\frac{dy}{dx}=2\sin x\cos x$ and $\frac{dz}{dx}=-\sin x.e^{\cos x}$ $\frac{1}{2}+\frac{1}{2}$

$$\therefore\frac{dy}{dz}=\frac{2\sin x\cos x}{-\sin x\ e^{\cos x}}=\frac{-2\cos x}{e^{\cos x}}\ or\ -2\cos x\,e^{-\cos x}$$ $\frac{1}{2}+\frac{1}{2}$

24	Evaluate $\int_1^2 \left[\frac{1}{x}-\frac{1}{2x^2}\right]e^{2x}dx$ **Answer:** Put $2x=t, \therefore dx=\frac{1}{2}dt$ $\therefore I=\int_1^2 \left[\frac{1}{x}-\frac{1}{2x^2}\right]e^{2x}dx=\int_2^4 \left[\frac{1}{t}-\frac{1}{t^2}\right]e^{t}dt$ $=\left[\frac{1}{t}e^t\right]_2^4=\frac{e^4}{4}-\frac{e^2}{2}$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}+\frac{1}{2}$
25	Find the value of $\int_0^1 x(1-x)^n\,dx$. **Answer:** $\int_0^1 x(1-x)^n\,dx=\int_0^1 (1-x)(1-1+x)^n\,dx=\int_0^1 \left(x^n-x^{n+1}\right)dx$ $=\left[\frac{x^{n+1}}{n+1}-\frac{x^{n+2}}{n+2}\right]_0^1=\frac{1}{n+1}-\frac{1}{n+2}$ or $\frac{1}{(n+1)(n+2)}$	1 $\frac{1}{2}+\frac{1}{2}$
26	Given two independent events A and B such that P(A) = 0.3 and P(B) = 0.6, find $P\left(A' \cap B'\right)$. **Answer:** $P\left(A' \cap B'\right)=P\left(A'\right)P\left(B'\right)$ $=(0.7)(0.4)=0.28$	1 1
	SECTION – C	
Q. Nos. 27 to 32 carry 4 marks each.		
27	Solve for x : $\sin^{-1}(1-x)-2\sin^{-1}x=\frac{\pi}{2}$. **Answer:** $\sin^{-1}(1-x)-2\sin^{-1}x=\frac{\pi}{2}\Rightarrow(1-x)=\sin\left(\frac{\pi}{2}+2\sin^{-1}x\right)$ $\Rightarrow(1-x)=\cos\left(2\sin^{-1}x\right)\Rightarrow 1-x=1-2x^2$ $\therefore 2x^2-x=0\Rightarrow x=0, x=\frac{1}{2}$ since $x=\frac{1}{2}$ does not satisty the given equation $\therefore x=0$ is the required solution.	$1\frac{1}{2}$ 1 1 $\frac{1}{2}$
28	If $y=(\log x)^x+x^{\log x}$, then find $\frac{dy}{dx}$	

	Answer: $y=(\log x)^x+x^{\log x}=u+v\Rightarrow\frac{dy}{dx}=\frac{du}{dx}+\frac{dv}{dx}$ $\therefore \log u=x\log(\log x)$ and $\log v=(\log x)^2$ $\frac{du}{dx}=(\log x)^x\left[\frac{1}{\log x}+\log(\log x)\right]$ and $\frac{dv}{dx}=x^{\log x}.\frac{2\log x}{x}$ $\Rightarrow\frac{dy}{dx}=(\log x)^x\left[\frac{1}{\log x}+\log(\log x)\right]+x^{\log x}.\frac{2\log x}{x}$	1 $\frac{1}{2}$ 1+1 $\frac{1}{2}$
29	Solve the differential equation: $x\sin\left(\frac{y}{x}\right)\frac{dy}{dx}+x-y\sin\left(\frac{y}{x}\right)=0$, given that $x=1$ when $y=\frac{\pi}{2}$. **Answer:** Given differential equation gives $\frac{dy}{dx}=\frac{y\sin\left(\frac{y}{x}\right)-x}{x\sin\left(\frac{y}{x}\right)}$ $\frac{y}{x}=v\Rightarrow y=vx$ and $\frac{dy}{dx}=v+x\frac{dv}{dx}$ $\therefore v+x\frac{dv}{dx}=\frac{v\sin v-1}{\sin v}\Rightarrow x\frac{dv}{dx}=-\frac{1}{\sin v}$ $\Rightarrow\int\sin v\,dv=-\int\frac{1}{x}dx$ $\therefore\cos v=\log\lvert x\rvert+C$ or $\cos\left(\frac{y}{x}\right)=\log\lvert x\rvert+C$ Given $x=1$ when $y=\frac{\pi}{2}\Rightarrow C=0$ $\therefore\cos\left(\frac{y}{x}\right)=\log\lvert x\rvert$ is the required solution.	$\frac{1}{2}$ $\frac{1}{2}$ 1 $\frac{1}{2}$ 1 $\frac{1}{2}$
30	If $\vec{a}=\hat{i}+2\hat{j}+3\hat{k}$ and $\vec{b}=2\hat{i}+4\hat{j}-5\hat{k}$ represent two adjacent sides of a parallelogram, find unit vectors parallel to the diagonals of the parallelogram. **Answer:** Diagonal vectors are: $\vec{a}+\vec{b}=3\hat{i}+6\hat{j}-2\hat{k}$ and $\vec{a}-\vec{b}=-\hat{i}-2\hat{j}+8\hat{k}$ (or, $\vec{b}-\vec{a}=\hat{i}+2\hat{j}-8\hat{k}$) $\therefore$ unit vectors are $\frac{(\vec{a}+\vec{b})}{\lvert\vec{a}+\vec{b}\rvert}=\frac{3}{7}\hat{i}+\frac{6}{7}\hat{j}-\frac{2}{7}\hat{k}$ and $\frac{(\vec{a}-\vec{b})}{\lvert\vec{a}-\vec{b}\rvert}=-\frac{1}{\sqrt{69}}\hat{i}-\frac{2}{\sqrt{69}}\hat{j}+\frac{8}{\sqrt{69}}\hat{k}$ **OR** Using vectors, find the area of the triangle ABC with vertices A (1, 2, 3), B(2, –1,4) and C(4, 5, –1).	1+1 1+1

Answer:

$$\text{area of } \Delta ABC = \frac{1}{2}\left|\text{cross product of any two side vectors}\right| \quad 1$$

$$\overrightarrow{AB} = \hat{i} - 3\hat{j} + \hat{k} \text{ and } \overrightarrow{BC} = 2\hat{i} + 6\hat{j} - 5\hat{k} \quad \frac{1}{2} + \frac{1}{2}$$

$$\overrightarrow{AB} \times \overrightarrow{BC} = 9\hat{i} + 7\hat{j} + 12\hat{k} \quad 1$$

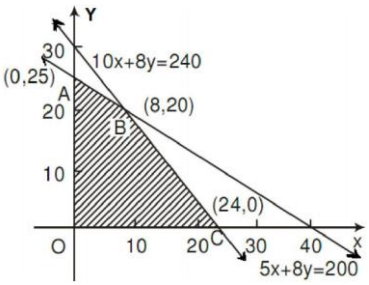
$$\therefore \text{area of } \Delta ABC = \frac{1}{2}\sqrt{81 + 49 + 144} = \frac{1}{2}\sqrt{274} \quad 1$$

31 A company manufactures two types of novelty souvenirs made of plywood. Souvenirs of type A requires 5 minutes each for cutting and 10 minutes each for assembling. Souvenirs of type B require 8 minutes each for cutting and 8 minutes each for assembling. Given that total time for cutting is 3 hours 20 minutes and for assembling 4 hours. The profit for type A souvenir is ₹100 each and for type B souvenir, profit is ₹120 each. How many souvenirs of each type should the company manufacture in order to maximize the profit? Formulate the problem as an LPP and solve it graphically.

Answer:

Let the company manufacture ' x ' number of souvenirs of Type A

And, ' y ' number of souvenirs of Type B



$\therefore$ LPP is: Maximise $P = 100x + 120y$ $\quad \frac{1}{2}$

subject to $5x + 8y \le 200$

$10x + 8y \le 240$ $\quad 1$

$x \ge 0, y \ge 0$

Correct Graph $\quad 1\frac{1}{2}$

P(A) = ₹ 3,000

P(B) = ₹ 3,200 (Max.)

P(C) = ₹2,400

$\therefore$ For Maximum profit, No. of souvenirs of Type A = 8

No. of souvenirs of Type B = 20 $\quad 1$

32 Three rotten apples are mixed with seven fresh apples. Find the probability distribution of the number of rotten apples, if three apples are drawn one by one with replacement. Find the mean of the number of rotten apples.

Answer:

Let X represents the number of rotten apples drawn. $\quad \frac{1}{2}$

X :	0	1	2	3
P (X) :	$\frac{7}{10}.\frac{7}{10}.\frac{7}{10}$	$3.\frac{7}{10}.\frac{7}{10}.\frac{3}{10}$	$3.\frac{7}{10}.\frac{3}{10}.\frac{3}{10}$	$\frac{3}{10}.\frac{3}{10}.\frac{3}{10}$
	$= \frac{343}{1000}$	$= \frac{441}{1000}$	$= \frac{189}{1000}$	$= \frac{27}{1000}$
X.P(X):	0	$\frac{441}{1000}$	$\frac{378}{1000}$	$\frac{81}{1000}$

2 $\quad 1$

Mean = $\sum X\,P(X) = \frac{900}{1000} = \frac{9}{10}$ $\quad \frac{1}{2}$

OR

In a shop X, 30 tins of ghee of type A and 40 tins of ghee of type B which look alike, are kept for sale. While in a shop Y, similar 50 tins of ghee of type A and 60 tins of ghee of type B are there. One tin of ghee is purchased from one of the randomly selected shop and is found to be of type B. Find the probability that it is purchased from shop Y.

Answer:

E_1 : selecting shop X

E_2 : selecting shop Y

A : purchased tin is of type B — $\frac{1}{2}$

$P(E_1) = P(E_2) = \frac{1}{2}$ — 1

$P(A|E_1) = \frac{4}{7}, P(A|E_2) = \frac{6}{11}$

$$P(E_2|A) = \frac{P(E_2)P(A|E_2)}{P(E_1)P(A|E_1) + P(E_2)P(A|E_2)}$$

$$= \frac{\frac{1}{2}\cdot\frac{6}{11}}{\frac{1}{2}\cdot\frac{4}{7} + \frac{1}{2}\cdot\frac{6}{11}}$$ — 2

$$= \frac{21}{43}$$ — $\frac{1}{2}$

SECTION – D

Q. Nos. 33 to 36 carry 6 marks each.

33 Find the vector and Cartesian equations of the line which is perpendicular to the lines with equations $\frac{x+2}{1} = \frac{y-3}{2} = \frac{z+1}{4}$ and $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and passes through the point (1, 1, 1). Also find the angle between the given lines.

Answer:

Let equation of required line is $\frac{x-1}{a} = \frac{y-1}{b} = \frac{z-1}{c}$ ……..(i) — $\frac{1}{2}$

Since this line is perpendicular to $\frac{x+2}{1} = \frac{y-3}{2} = \frac{z+1}{4}$ and $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$,

$a + 2b + 4c = 0$ ……(ii)

$2a + 3b + 4c = 0$ ……(iii) — 1

Solving (ii) and (iii), $\frac{a}{-4} = \frac{b}{4} = \frac{c}{-1}$

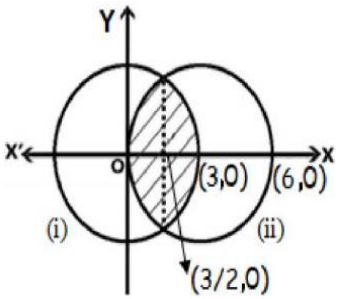
$\therefore$ DR′s of line in cartesian form is : $-4, 4, -1$ — 1

Equation of line in Cartesian form is: $\frac{x-1}{-4} = \frac{y-1}{4} = \frac{z-1}{-1}$ — 1

Vector form of line is $\vec{r} = (\hat{i} + j + k) + \lambda(-4\hat{i} + 4j - k)$ — 1

Let θ be the angle between given lines.

$$\cos\theta = \frac{1(2) + 2(3) + 4(4)}{\sqrt{1+4+16}\,\sqrt{4+9+16}} = \frac{24}{\sqrt{21}\,\sqrt{29}} \qquad \therefore \theta = \cos^{-1}\left(\frac{24}{\sqrt{21}\,\sqrt{29}}\right)$$ — $1 + \frac{1}{2}$

34	Using integration find the area of the region bounded between the two circles $x^2+y^2=9$ and $(x-3)^2+y^2=9$. **Answer:**	
	Correct Figure	1
	Point of intersection of, $x^2+y^2=9;(x-3)^2+y^2=9\Rightarrow(x-3)^2-x^2=0\Rightarrow x=3/2$	1
	 Required area $=2\left[\int_0^{\frac{3}{2}}\sqrt{9-(x-3)^2}\,dx+\int_{\frac{3}{2}}^{3}\sqrt{9-x^2}\,dx\right]$ $=4\left[\int_{\frac{3}{2}}^{3}\sqrt{9-x^2}\,dx\right]$	$1\frac{1}{2}$
	$=4\left[\frac{x}{2}\sqrt{9-x^2}+\frac{9}{2}\sin^{-1}\frac{x}{3}\right]_{\frac{3}{2}}^{3}=\left(6\pi-\frac{9\sqrt{3}}{2}\right)$	$1\frac{1}{2}+1$
	OR Evaluate the following integral as the limit of sums $\int_1^4\left(x^2-x\right)dx$. **Answer:**	
	$\int_1^4(x^2-x)\,dx=\lim_{h\to0}h\left[f(1)+f(1+h)+f(1+2h)+.....f\left(1+\overline{n-1}\,h\right)\right]$ where $f(x)=\left(x^2-x\right)$ and $nh=3$	1 1
	$\therefore\int_1^4(x^2-x)\,dx$ $=\lim_{h\to0}h\left[(1-1)+\left(1+h^2+2h-h-1\right)+\left(1+4h^2+4h-2h-1\right)+...+\left(1+(n-1)^2h^2+2(n-1)h-(n-1)h-1\right)\right]$	2
	$=\lim_{h\to0}h\left[h^2\left(1^2+2^2+3^2+.....+(n-1)^2\right)+h(1+2+3+.....+(n-1))\right]$	1
	$=\lim_{h\to0}\left[\frac{nh(nh-h)(2nh-h)}{6}+\frac{(nh(nh-h))}{2}\right]=9+\frac{9}{2}=\frac{27}{2}$	1
35	Find the minimum value of $(ax+by)$, where $xy=c^2$. **Answer:**	
	Let $S=ax+by$, where $y=\frac{c^2}{x}\ \therefore\ S=ax+\frac{bc^2}{x}$	1
	$\frac{dS}{dx}=a-\frac{bc^2}{x^2}$	1
	$\frac{dS}{dx}=0\Rightarrow x^2=\frac{bc^2}{a}$ or $x=\sqrt{\frac{b}{a}}$.c	$1\frac{1}{2}$
	$\left.\frac{d^2S}{dx^2}\right\|_{x=\sqrt{\frac{b}{a}}.c}=\frac{2bc^2}{x^3}=2bc^2\left[\sqrt{\frac{a}{b}}\,\frac{1}{c}\right]^3>0$ for $a,b,c>0$ and $x=\sqrt{\frac{b}{a}}.c$	$1\frac{1}{2}$

	$\therefore \text{minimum value } = a\sqrt{\frac{b}{a}}.c + b.\frac{c^2}{c}\sqrt{\frac{a}{b}} = 2\sqrt{ab}.c$	1
36	If a, b, c are p^{th}, q^{th}, and r^{th} terms respectively of a G.P, then prove that $\begin{vmatrix} \log a & p & 1 \\ \log b & q & 1 \\ \log c & r & 1 \end{vmatrix} = 0$ **Answer:** $a = AR^{p-1},\ \ b = AR^{q-1},\ \ c = AR^{r-1}$	$1\frac{1}{2}$
	$\therefore \Delta = \begin{vmatrix} \log A + (p-1)\log R & p & 1 \\ \log A + (q-1)\log R & q & 1 \\ \log A + (r-1)\log R & r & 1 \end{vmatrix} = \log A \begin{vmatrix} 1 & p & 1 \\ 1 & q & 1 \\ 1 & r & 1 \end{vmatrix} + \log R \begin{vmatrix} p & p & 1 \\ q & q & 1 \\ r & r & 1 \end{vmatrix} - \log R \begin{vmatrix} 1 & p & 1 \\ 1 & q & 1 \\ 1 & r & 1 \end{vmatrix}$	1+1+1+1
	$= 0 + 0 + 0 = 0$	$\frac{1}{2}$
	OR	
	If $A = \begin{bmatrix} 2 & -3 & 5 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{bmatrix}$, then find A^{-1}. Using A^{-1}, solve the following system of equations: $2x - 3y + 5z = 11$ $3x + 2y - 4z = -5$ $x + y - 2z = -3$ **Answer:**	
	$\lvert A \rvert = 2(0) + 3(-2) + 5(1) = -1$	1
	$\Rightarrow A^{-1} = -\begin{bmatrix} 0 & -1 & 2 \\ 2 & -9 & 23 \\ 1 & -5 & 13 \end{bmatrix}$ or $\begin{bmatrix} 0 & 1 & -2 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{bmatrix}$ **(1 mark for any 4 correct co-factors)**	2
	Given equations can be written as $AX = B$, where $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $B = \begin{bmatrix} 11 \\ -5 \\ -3 \end{bmatrix}$	1
	$\therefore X = A^{-1}B = \begin{bmatrix} 0 & 1 & -2 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{bmatrix} \begin{bmatrix} 11 \\ -5 \\ -3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$	1
	$\Rightarrow x = 1,\ \ y = 2,\ \ z = 3$	1

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Senior School Certificate Examination-2020

Marking Scheme - Mathematics 65/C/1

***General instructions*:-**

1. You are aware that evaluation is the most important process in the actual and correct assessment of the candidates. A small mistake in evaluation may lead to serious problems which may affect the future of the candidates, education system and teaching profession. To avoid mistakes, it is requested that before starting evaluation, you must read and understand the spot evaluation guidelines carefully. **Evaluation is a 10-12 days mission for all of us. Hence, it is necessary that you put in your best efforts in this process.**
2. Evaluation is to be done as per instructions provided in the Marking Scheme. It should not be done according to one's own interpretation or any other consideration. Marking Scheme should be strictly adhered to and religiously followed. **However, while evaluating, answers which are based on latest information or knowledge and/or are innovative, they may be assessed for their correctness otherwise and marks be awarded to them.**
3. The Head-Examiner must go through the first five answer books evaluated by each evaluator on the first day, to ensure that evaluation has been carried out as per the instructions given in the Marking Scheme. The remaining answer books meant for evaluation shall be given only after ensuring that there is no significant variation in the marking of individual evaluators.
4. Evaluators will mark(√) wherever answer is correct. For wrong answer 'X"be marked. Evaluators will not put right kind of mark while evaluating which gives an impression that answer is correct and no marks are awarded. This is most common mistake which evaluators are committing.
5. If a question has parts, please award marks on the right-hand side for each part. Marks awarded for different parts of the question should then be totaled up and written in the left-hand margin and encircled. This may be followed strictly.
6. If a question does not have any parts, marks must be awarded in the left-hand margin and encircled. This may also be followed strictly.
7. If a student has attempted an extra question, answer of the question deserving more marks should be retained and the other answer scored out.
8. No marks to be deducted for the cumulative effect of an error. It should be penalized only once.
9. A full scale of marks 80 has to be used. Please do nothesitate to award full marks if the answer deserves it.
10. Every examiner has to necessarily do evaluation work for full working hours i.e. 8 hours every day and evaluate 20 answer books per day in main subjects and 25 answer books per day in other subjects (Details are given in Spot Guidelines).
11. Ensure that you do not make the following common types of errors committed by the Examiner in the past:-
    - Leaving answer or part thereof unassessed in an answer book.
    - Giving more marks for an answer than assigned to it.
    - Wrong totaling of marks awarded on a reply
    - Wrong transfer of marks from the inside pages of the answer book to the title page.
    - Wrong question wise totaling on the title page.
    - Wrong totaling of marks of the two columns on the title page.
    - Wrong grand total.
    - Marks in words and figures not tallying.
    - Wrong transfer of marks from the answer book to online award list.
    - Answers marked as correct, but marks not awarded. (Ensure that the right tick mark is correctly and clearly indicated. It should merely be a line. Same is with the X for incorrect answer.)
    - Half or a part of answer marked correct and the rest as wrong, but no marks awarded.
12. While evaluating the answer books if the answer is found to be totally incorrect, it should be marked as cross (X) and awarded zero (0) Marks.
13. Any unassessed portion, non-carrying over of marks to the title page, or totaling error detected by the candidate shall damage the prestige of all the personnel engaged in the evaluation work as also of the Board. Hence, in order to uphold the prestige of all concerned, it is again reiterated that the instructions be followed meticulously and judiciously.
14. The Examiners should acquaint themselves with the guidelines given in the Guidelines for spot Evaluation before starting the actual evaluation.
15. Every Examiner shall also ensure that all the answers are evaluated, marks carried over to the title page, correctly totaled and written in figures and words.
16. The Board permits candidates to obtain photocopy of the Answer Book on request in an RTI application and also separately as a part of the re-evaluation process on payment of the processing charges.

QUESTION PAPER CODE 65/C/1

EXPECTED ANSWER/VALUE POINTS

SECTION – A

Question Numbers 1 to 20 carry 1 mark each.

Question Numbers 1 to 10 are multiple choice type questions.
Select the correct option.

Q.No. **Marks**

1. The probability of solving a specific question independently by A and B are $\frac{1}{3}$ and $\frac{1}{5}$ respectively. If both try to solve the question independently, the probability that the question is solved is

(A) $\frac{7}{15}$ **(B)** $\frac{8}{15}$ **(C)** $\frac{2}{15}$ **(D)** $\frac{14}{15}$

Ans: (A) $\frac{7}{15}$ **1**

2. The objective function of an LPP is

(A) a constant **(B)** a linear function to be optimised

(C) an inequality **(D)** a quadratic expression

Ans: (B) a linear function to be optimised **1**

3. If the two lines

$L_1 : x = 5, \frac{y}{3-\alpha} = \frac{z}{-2}$

$L_2 : x = 2, \frac{y}{-1} = \frac{z}{2-\alpha}$

are perpendicular, then the value of α is

(A) $\frac{2}{3}$ **(B)** 3 **(C)** 4 **(D)** $\frac{7}{3}$

Ans: (D) $\frac{7}{3}$ **1**

4. If $\vec{a}, \vec{b}$ and $\vec{c}$ are the position vectors of the points A(2, 3, –4), B(3, –4, –5) and C(3, 2, –3) respectively, then $\left|\vec{a}+\vec{b}+\vec{c}\right|$ is equal

(A) $\sqrt{113}$ **(B)** $\sqrt{185}$ **(C)** $\sqrt{203}$ **(D)** $\sqrt{209}$

Ans: (D) $\sqrt{209}$ **1**

5. The order and degree of the differential equation of the family of parabolas having vertex at origin and axis along positive x-axis is

(A) 1, 1 **(B)** 1, 2 **(C)** 2, 1 **(D)** 2, 2

Ans: (A) 1, 1 **1**

6. $\int_0^1 \tan\left(\sin^{-1} x\right) dx$ equals

(A) 2 **(B)** 0 **(C)** –1 **(D)** 1

Ans: (D) 1 **1**

7. $\int \frac{e^x}{x+1}\left[1+(x+1)\log(x+1)\right] dx$ equals

(A) $\frac{e^x}{x+1}+c$ **(B)** $e^x \frac{x}{x+1}+c$

(C) $e^x \log(x+1) + e^x + c$ **(D)** $e^x \log(x+1) + c$

Ans: (D) $e^x \log(x+1) + c$ **1**

8. If $\sec^{-1}\left(\frac{1+x}{1-y}\right) = a$, then $\frac{dy}{dx}$ is equal to

(A) $\frac{x-1}{y-1}$ **(B)** $\frac{x-1}{y+1}$ **(C)** $\frac{y-1}{x+1}$ **(D)** $\frac{y+1}{x-1}$

Ans: (C) $\frac{y-1}{x+1}$ **1**

9. If $A = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$, then A^2 equals

(A) $\begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix}$ **(B)** $\begin{bmatrix} 2 & -2 \\ -2 & -2 \end{bmatrix}$ **(C)** $\begin{bmatrix} -2 & -2 \\ -2 & 2 \end{bmatrix}$ **(D)** $\begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix}$

Ans: (A) $\begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix}$ **1**

10. $\begin{vmatrix} 43 & 44 & 45 \\ 44 & 45 & 46 \\ 45 & 46 & 47 \end{vmatrix}$ equals

(A) 0 **(B)** –1 **(C)** 1 **(D)** 2

Ans: (A) 0 **1**

Fill in the blanks in questions numbers 11 to 15

11. Two angles of a triangle are $\cot^{-1}2$ and $\cot^{-1}3$. The third angle of the triangle is _____.

Ans: $\frac{3\pi}{4}$ or $135°$ **1**

12. A square matrix A is said to be singular if _____.

Ans: $|A| = 0$

OR

If $A = \begin{bmatrix} 3 & -5 \\ 2 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 17 \\ 0 & -10 \end{bmatrix}$, then $|AB|$ = _____.

Ans: $|AB| = -100$

13. If $y = \log x$, then $\frac{d^2y}{dx^2}$ = _____.

Ans: $\frac{d^2y}{dx^2} = -\frac{1}{x^2}$ **1**

14. The integrating factor of the differential equation $x\frac{dy}{dx} - y = \log x$ is _____

Ans: $\frac{1}{x}$ **1**

15. From a pack of 52 cards, 3 cards are drawn at random (without replacement). The probability that they are two red cards and one black card, is _____.

Ans: $\frac{13}{34}$ **1**

Question numbers 16 to 20 are very short answer type questions

16. Find the distance of the point (a, b, c) from the x-axis.

Ans: $\sqrt{b^2 + c^2}$ **1**

17. If $\vec{a} = 2\hat{i} - \hat{j} + 2\hat{k}$ and $\vec{b} = 5\hat{i} - 3\hat{j} - 4\hat{k}$, then find the ratio $\frac{\text{projection of vector } \vec{a} \text{ on vector } \vec{b}}{\text{projection of vector } \vec{b} \text{ on vector } \vec{a}}$.

Ans: $\frac{\text{projection of vector } \vec{a} \text{ on vector } \vec{b}}{\text{projection of vector } \vec{b} \text{ on vector } \vec{a}} = \frac{|\vec{a}|}{|\vec{b}|}$ **1/2**

$= \frac{3}{5\sqrt{2}}$ **1/2**

OR

Let $\hat{a}$ and $\hat{b}$ be two unit vectors. If the vectors $\vec{c} = \hat{a} + 2\hat{b}$ and $\vec{d} = 5\hat{a} - 4\hat{b}$ are perpendicular to each other, then find the angle between the vectors $\hat{a}$ and $\hat{b}$.

Ans: $\vec{c} \perp \vec{d} \Rightarrow \vec{c} \cdot \vec{d} = 0$ **1/2**

$\Rightarrow \hat{a} \cdot \hat{b} = \frac{1}{2}$

$\Rightarrow$ Angle between vectors $\hat{a}$ & $\hat{b} = \frac{\pi}{3}$ or $60°$ **1/2**

18. Solve the differential equation $\left(e^x + 1\right) y\, dy = e^x (y+1) dx$.

Ans: $\int \frac{e^x}{e^x + 1} dx = \int \frac{y}{y+1} dy$ **1/2**

$\Rightarrow \log (e^x + 1) = y - \log (y + 1) + c$ **1/2**

19. If $\begin{bmatrix} 4 & x+2 \\ 2x-3 & x+1 \end{bmatrix}$ is a symmetric matrix, then find the value of x.

Ans: $2x - 3 = x + 2$ **1/2**

$\Rightarrow \quad x = 5$ **1/2**

OR

If A is a square matrix such that $A^2 = A$, then find $(2 + A)^3 - 19\, A$.

Ans: $(2 + A)^3 - 19\, A = A^3 + 8 + 12A + 6A^2 - 19\, A$ **1/2**

$= 8$ **1/2**

20. If $f(x) = \frac{1-x}{1+x}$, then find $(f \circ f)(x)$.

Ans: $f \circ f(x) = f(f(x)) = \frac{1 - f(x)}{1 + f(x)}$ **1/2**

$= x$ **1/2**

SECTION-B

Question numbers 21 to 26 carry 2 marks each.

21. Let W denote the set of words in the English dictionary. Define the relation R by

R = {(x , y) ∈ W × W such that x and y have at least one letter in common}

Show that this relation R is reflexive and symmetric, but not transitive.

Ans: For any word $x \in W$

x and x have atleast one (all) letter in common

$\therefore (x, x) \in R, \forall x \in W \ \therefore$ R is reflexive — **1**

Symmetric : Let $(x, y) \in R$, $x, y \in W$

$\Rightarrow$ x and y have atleast one letter in common

$\Rightarrow$ y and x have atleast one letter in common

$\Rightarrow (y, x) \in R \ \therefore$ R is symmetric — **1/2**

Transitive : Taking example of three English dictionary words x, y, z, $\in W$ such that $(x, y), (y, z) \in R$ but $(x, z) \notin R$ — **1/2**

$\therefore$ R is not transitive

OR

Find the inverse of the function $f(x) = \left(\frac{4x}{3x+4}\right)$.

Ans: Let $y = f(x) = \frac{4x}{3x+4} \Rightarrow x = \frac{4y}{4-3y}$ — **$1\frac{1}{2}$**

$\therefore \ f^{-1}(y) = \frac{4y}{4-3y}\left(\text{or } f^{-1}(x) = \frac{4x}{4-3x}\right)$ — **1/2**

22. For the matrix $A = \begin{bmatrix} 2 & 3 \\ -4 & -6 \end{bmatrix}$, verify the following:

$A(\text{adj } A) = (\text{adj } A)\, A = |A|\, I$

Ans: $|A| = -12 + 12 = 0$, adj $A = \begin{bmatrix} -6 & -3 \\ 4 & 2 \end{bmatrix}$ — **$\frac{1}{2} + 1$**

$A \cdot (\text{adj } A) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = (\text{adj } A) \cdot A = |A|\, I$ — **$\frac{1}{2}$**

23. If $y = e^x + e^{-x}$, then show that $\frac{dy}{dx} = \sqrt{y^2 - 4}$

Ans: $\frac{dy}{dx} = e^x - e^{-x}$ — **1**

$= \sqrt{\left(e^x + e^{-x}\right)^2 - 4}$ — **1/2**

$= \sqrt{y^2 - 4}$ — **1/2**

24. Solve the following homogeneous differential equation: $x\frac{dy}{dx} = x + y$

Ans: Let $y = vx \therefore \frac{dy}{dx} = v + x\frac{dv}{dx}$ $\frac{1}{2} + \frac{1}{2}$

$\therefore x\left(v + x\frac{dv}{dx}\right) = x + vx \Rightarrow x\frac{dv}{dx} = 1$ $\frac{1}{2}$

$\therefore \int dv = \int \frac{1}{x} dx \Rightarrow v = \log|x| + c$

$\Rightarrow y = x(\log|x| + c)$ $\frac{1}{2}$

25. Show that $|\vec{a}|\vec{b} + |\vec{b}|\vec{a}$ is perpendicular to $|\vec{a}|\vec{b} - |\vec{b}|\vec{a}$, for any two non-zero vectors $\vec{a}$ and $\vec{b}$.

Ans: $\left(|\vec{a}|\vec{b} + |\vec{b}|\vec{a}\right) \cdot \left(|\vec{a}|\vec{b} - |\vec{b}|\vec{a}\right)$

$= \left(|\vec{a}|\vec{b}\right)^2 - \left(|\vec{b}|\vec{a}\right)^2$ 1

$= |\vec{a}|^2|\vec{b}|^2 - |\vec{b}|^2|\vec{a}|^2 = 0$

$\therefore \left(|\vec{a}|\vec{b} + |\vec{b}|\vec{a}\right) \perp \left(|\vec{a}|\vec{b} - |\vec{b}|\vec{a}\right)$ 1

26. A bag contains 19 tickets, numbered 1 to 19. A ticket is drawn at random and then another ticket is drawn without replacing the first one in the bag. Find the probability distribution of the number of even numbers on the ticket.

Ans: Let X = No. of even tickets drawn

X	0	1	2
P(X)	$\frac{^{10}C_2}{^{19}C_2} = \frac{5}{19}$	$\frac{^{10}C_1 \cdot ^{9}C_1}{^{19}C_2} = \frac{10}{19}$	$\frac{^{9}C_2}{^{19}C_2} = \frac{4}{19}$

$\frac{1}{2}$

$\frac{1}{2} + \frac{1}{2} + \frac{1}{2}$

OR

Find the probability distribution of the number of successes in two tosses of a die, when a success is defined as "number greater than 5".

Ans: X = No. of success = No. of times getting a number greater than 5

X	0	1	2
P(X)	$\frac{5}{5} \cdot \frac{5}{6} = \frac{25}{36}$	$2 \cdot \frac{1}{6} \cdot \frac{5}{6} = \frac{10}{36}$	$\frac{1}{6} \cdot \frac{1}{6} = \frac{1}{36}$

$\frac{1}{2}$

$\frac{1}{2} + \frac{1}{2} + \frac{1}{2}$

SECTION-C

Question numbers 27 to 32 carry 4 marks each.

27. Prove that $2\tan^{-1}\frac{1}{2}+\tan^{-1}\frac{1}{7}=\tan^{-1}\frac{31}{17}$.

Ans: $\text{LHS} = 2\tan^{-1}\frac{1}{2}+\tan^{-1}\frac{1}{7} = \tan^{-1}\left(\frac{4}{3}\right)+\tan^{-1}\left(\frac{1}{7}\right)$ **2**

$$= \tan^{-1}\left(\frac{\frac{4}{3}+\frac{1}{7}}{1-\frac{4}{3}\cdot\frac{1}{7}}\right)=\tan^{-1}\frac{31}{17}$$ **2**

$$= \text{RHS}$$

28. Using properties of determinants, prove that

$$\begin{vmatrix} 1+a^2-b^2 & 2ab & -2b \\ 2ab & 1-a^2+b^2 & 2a \\ 2b & -2a & 1-a^2-b^2 \end{vmatrix} = \left(1+a^2+b^2\right)^3$$

Ans. $\text{LHS} = \begin{vmatrix} 1+a^2-b^2 & 2ab & -2b \\ 2ab & 1-a^2+b^2 & 2a \\ 2b & -2a & 1-a^2-b^2 \end{vmatrix}$

$\left.\begin{array}{l}(\text{Applying } C_1 \to C_1 - bC_3 \text{ , } C_2 \to C_2 + aC_3) \\ = \begin{vmatrix} 1+a^2+b^2 & 0 & -2b \\ 0 & 1+a^2+b^2 & 2a \\ b\left(1+a^2+b^2\right) & -a\left(1+a^2+b^2\right) & 1-a^2-b^2 \end{vmatrix}\end{array}\right\}$ **2**

(Taking $\left(1+a^2+b^2\right)$ common from C_1 & C_2)

$$= \left(1+a^2+b^2\right)^2 \begin{vmatrix} 1 & 0 & -2b \\ 0 & 1 & 2a \\ b & -a & 1-a^2-b^2 \end{vmatrix}$$ **1/2**

$(\text{Applying } C_3 \to C_3 + 2bC_1 - 2aC_2)$

$$= \left(1+a^2+b^2\right)^2 \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ b & -a & 1+a^2+b^2 \end{vmatrix}$$ **1**

(Expand along C_3)

$= \left(1+a^2+b^2\right)^3 = \text{RHS}$ **1/2**

OR

Find the equation of the line joining A(1, 3) and B(0, 0), using determinants. Also, find k if D(k, 0) is a point such that the area of the ΔABD is 3 square units.

Ans: Equation of the line through A(1, 3) and B(0, 0) is

$$\begin{vmatrix} x & y & 1 \\ 1 & 3 & 1 \\ 0 & 0 & 1 \end{vmatrix} = 0 \Rightarrow 3x - y = 0$$ **2**

$$\frac{1}{2}\begin{vmatrix} 1 & 3 & 1 \\ 0 & 0 & 1 \\ k & 0 & 1 \end{vmatrix} = \pm 3 \Rightarrow k = \pm 2$$ **2**

29. Prove that three points A, B and C with position vectors $\vec{a}, \vec{b}$ and $\vec{c}$ respectively are collinear if and only if $\left(\vec{b} \times \vec{c}\right) + \left(\vec{c} \times \vec{a}\right) + \left(\vec{a} \times \vec{b}\right) = \vec{0}$

Ans. Points $A(\vec{a}), B(\vec{b})$ and $C(\vec{c})$ are collinear

$\Rightarrow \overrightarrow{AB} \times \overrightarrow{AC} = \vec{0}$ **1**

$\Rightarrow \left(\vec{b} - \vec{a}\right) \times \left(\vec{c} - \vec{a}\right) = \vec{0}$ **1**

$\Rightarrow \vec{b} \times \vec{c} - \vec{b} \times \vec{a} - \vec{a} \times \vec{c} + \vec{a} \times \vec{a} = \vec{0}$ **1**

$\Rightarrow \vec{a} \times \vec{b} + \vec{b} \times \vec{c} + \vec{c} \times \vec{a} = \vec{0}$

Similarly, converse can be proved **1**

30. Find the shortest distance between the following lines and hence write whether the lines are intersecting or not.

$\frac{x-1}{2} = \frac{y-1}{3} = z, \quad \frac{x+1}{5} = \frac{y-2}{1}, \quad z = 2$

Ans. Let $\vec{a}_1 = \hat{i} - \hat{j}$; $\vec{a}_2 = -\hat{i} + 2\hat{j} + 2\hat{k}$

$\vec{b}_1 = 2\hat{i} + 3\hat{j} + \hat{k}$; $\vec{b}_2 = 5\hat{i} + \hat{j}$ **1**

then, $\vec{a}_2 - \vec{a}_1 = -2\hat{i} + 3\hat{j} + 2\hat{k}$, $\vec{b}_1 \times \vec{b}_2 = -\hat{i} + 5\hat{j} - 13\hat{k}$ **1/2+1$\frac{1}{2}$**

$\therefore$ Shortest distance $= \left|\frac{\left(\vec{a}_2 - \vec{a}_1\right) \cdot \left(\vec{b}_1 \times \vec{b}_2\right)}{\left|\vec{b}_1 \times \vec{b}_2\right|}\right| = \frac{9}{\sqrt{195}} \neq 0$ **1/2**

$\therefore$ lines are not intersecting **1/2**

OR

Find the equation of the plane through the line of intersection of the planes $\vec{r}\cdot\left(\hat{i}+3\hat{j}\right)+6=0$ and $\vec{r}\cdot\left(3\hat{i}-\hat{j}-4\hat{k}\right)=0$, which is at a unit distance from the origin.

Ans: Equation of plane through the line of intersection of the two given planes is

$$\vec{r}\cdot\left[(1+3\lambda)\hat{i}+(3-\lambda)\hat{j}-4\lambda\hat{k}\right]=-6 \qquad 1\frac{1}{2}$$

As per the given condition

$$\left|\frac{-6}{\sqrt{(1+3\lambda)^2+(3-\lambda)^2+(-4\lambda)^2}}\right|=1 \Rightarrow \lambda=\pm 1 \qquad 1\frac{1}{2}$$

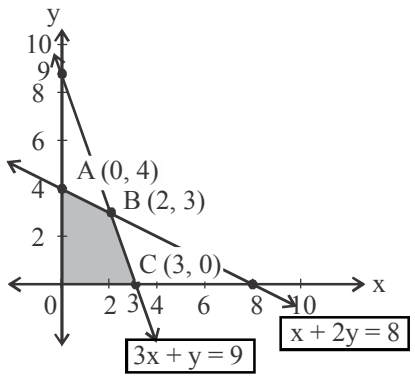
$\therefore$ Equation of plane is : $\vec{r}\cdot\left(4\hat{i}+2\hat{j}-4\hat{k}\right)+6=0$ **1/2**

or $\vec{r}\cdot\left(-2\hat{i}+4\hat{j}+4\hat{k}\right)+6=0$ **1/2**

31. A company produces two types of goods, A and B, that require gold and silver. Each unit of type A requires 3g of silver and 1g of gold, while that of type B requires 1g of silver and 2g of gold. The company can use at the most 9g of silver and 8 g of gold. If each unit of type A brings a profit of ₹ 120 and that of type B ₹ 150, then find the number of units of each type that the company should produce to maximise profit.

Formulate the above LPP and solve it graphically. Also, find the maximum profit.

Ans.



Let No. of goods type A = x,

Number of goods type B = y.

Then the L.P. P. is:

Maximize (Profit) : Z = 120x + 150y **1**

Subject to constraints :

$3x + y \leq 9$

$x + 2y \leq 8$ **1**

$x, y \geq 0$

Correct figure $1\frac{1}{2}$

Corner:	Value of Z (in ₹)
0(0, 0)	0
A(0, 4)	600
B(2, 3)	690 (Max)
C(3, 0)	360

1/2

$\therefore$ Max. profit = ₹ 690

when Good Type A = 2 units, Type B = 3 units

32. A bag contains 5 red and 4 black balls, a second bag contains 3 red and 6 black balls. One of the two bags is selected at random and two balls are drawn at random (without replacement), both of which are found to be red. Find the probability that these two balls are drawn from the second bag.

Ans. Let E_1 : Bag I is selected

E_2 : Bag II is selected

A : Two balls drawn at random both are red. **1/2**

$$P(E_1) = P(E_2) = \frac{1}{2}, P\left(\frac{A}{E_1}\right) = \frac{{}^5C_2}{{}^9C_2} = \frac{5}{18}, P\left(\frac{A}{E_2}\right) = \frac{{}^3C_2}{{}^9C_2} = \frac{1}{12}$$ **1/2+1+1**

$$P\left(\frac{E_1}{A}\right) = \frac{\frac{1}{2}\cdot\frac{1}{12}}{\frac{1}{2}\cdot\frac{5}{18}+\frac{1}{2}\cdot\frac{1}{12}} = \frac{3}{13}$$ **1**

SECTION-D

Question numbers 33 to 36 carry 6 marks each.

33. If $y = x^{\sin x} + \sin^{-1}\sqrt{x}$, then find $\frac{dy}{dx}$.

Ans. Let $u = x^{\sin x}$ $\therefore$ $y = u + \sin^{-1}\sqrt{x}$

$$\Rightarrow \frac{dy}{dx} = \frac{du}{dx} + \frac{1}{2\sqrt{x}\sqrt{1-x}} \quad \ldots (i)$$ **$1\frac{1}{2}$**

Also, $\log u = \sin x \cdot \log x$ **1**

$$\Rightarrow \frac{1}{u}\frac{du}{dx} = \cos x \cdot \log x + \frac{\sin x}{x}$$ **2**

$$\Rightarrow \frac{du}{dx} = x^{\sin x}\cdot\left(\cos x \cdot \log x + \frac{\sin x}{x}\right) \quad \ldots (ii)$$ **$\frac{1}{2}$**

Putting (ii) in (i) we get

$$\frac{dy}{dx} = x^{\sin x}\left(\cos x \cdot \log x + \frac{\sin x}{x}\right) + \frac{1}{2\sqrt{x}\sqrt{1-x}}$$ **1**

34. Evaluate : $\int_{\pi/6}^{\pi/3} \frac{\sin x + \cos x}{\sqrt{\sin 2x}}\, dx$

Ans. Let $\sin x - \cos x = t \quad \therefore \quad (\sin x + \cos x)\,dx = dt$ $1+\frac{1}{2}$

Squaring we get, $\sin 2x = 1 - t^2$ $1\frac{1}{2}$

$$\int \frac{\sin x + \cos x}{\sqrt{\sin 2x}}\, dx = \int \frac{1}{\sqrt{1-t^2}}\, dt = \sin^{-1} t$$

$$= \sin^{-1}(\sin x - \cos x) \qquad 1\frac{1}{2}$$

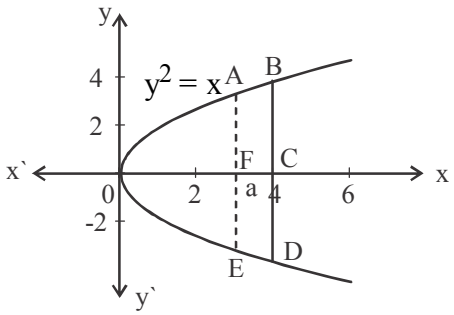
$$\therefore \int_{\pi/6}^{\pi/3} \frac{\sin x + \cos x}{\sqrt{\sin 2x}}\, dx = \sin^{-1}(\sin x - \cos x)\,\Big]_{\pi/6}^{\pi/3}$$

$$= \sin^{-1}\left(\frac{\sqrt{3}}{2} - \frac{1}{2}\right) - \sin^{-1}\left(\frac{1}{2} - \frac{\sqrt{3}}{2}\right) \qquad 1\frac{1}{2}$$

$$= 2 \cdot \sin^{-1}\left(\frac{\sqrt{3}-1}{2}\right)$$

35. If the area between the curves $x = y^2$ and $x = 4$ divided into two equal parts by the line $x = a$, then find the value of a using integration.

Ans. Correct graph $1\frac{1}{2}$



$\text{ar}(OAEO) = \text{ar}(ABDEA)$

$\Rightarrow 2 \cdot \text{ar}(OAFO) = 2 \cdot \text{ar}(ABCFA)$

$$\int_0^a \sqrt{x}\, dx = \int_a^4 \sqrt{x}\, dx \qquad 1\frac{1}{2}$$

$$\frac{2}{3} \cdot a^{3/2} = \frac{2}{3}\left(4^{3/2} - a^{3/2}\right) \qquad 1\frac{1}{2}$$

$$\Rightarrow \frac{2}{3} \cdot a^{3/2} = \frac{2}{3}\left(4^{3/2} - a^{3/2}\right) \qquad 1\frac{1}{2}$$

$$\Rightarrow a^{3/2} = 4, \quad \therefore \; a = 4^{2/3} \qquad 1\frac{1}{2}$$

OR

Find: $\int \frac{x}{(x-1)^2(x+2)}dx$

Ans: $\int \frac{x}{(x-1)^2(x+2)}dx = \frac{2}{9}\int \frac{1}{(x-1)}dx + \frac{1}{3}\int \frac{1}{(x-1)^2}dx - \frac{2}{9}\int \frac{1}{4+2}dx$ **4**

$= \frac{2}{9}\log|x-1| - \frac{1}{3(x-1)} - \frac{2}{9}\log|x+2| + C$ **2**

36. Find the intervals in which the function f defined as $f(x) = \sin x + \cos x$, $0 \le x \le 2\pi$ is strictly increasing or decreasing.

Ans. $f'(x) = \cos x - \sin x$, $0 \le x \le 2\pi$ **1**

$\therefore\ f'(x) = 0 \Rightarrow x = \frac{\pi}{4}, \frac{5\pi}{4}$ **2**

Sign of $f'(x)$: $\underset{0}{\bullet}\ \underset{\frac{\pi}{4}}{\circ}\ \underset{\frac{5\pi}{4}}{\circ}\ \underset{2\pi}{\bullet}$ (+ve, −ve, +ve) **$1\frac{1}{2}$**

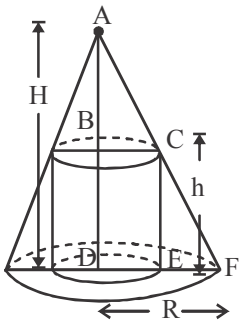
$\therefore$ f(x) is strictly increasing on $\left[0, \frac{\pi}{4}\right) \cup \left(\frac{5\pi}{4}, 2\pi\right]$ **1**

and f(x) is strictly decreasing on $\left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$ **$\frac{1}{2}$**

OR

Prove that the radius of the right circular cylinder of greatest curved surface area which can be inscribed in a given cone is half of that of the cone.

Ans.



Let h = Height of cylinder

r = Radius of cylinder

H = Height of cone

R = Radius of cone

where, H, R are constants

Correct figure **1**

$\frac{H-h}{H} = \frac{r}{R}\ (\because \Delta ABC \sim \Delta ADE)$ ...(i) **$\frac{1}{2}$**

C = curved surface area = $2\pi rh$ **1**

$\therefore\ C = 2\pi rh \cdot H \cdot \left(\frac{R-r}{R}\right)$ (Using (i))

$= \frac{2\pi H}{R}\left(rR - r^2\right)$ **1**

$C'(r) = \frac{2\pi H}{R}(R-2r),\ C''(r) = \frac{-4\pi H}{R} < 0$ **1**

$C'(r) = 0 \Rightarrow r = \frac{R}{2},\ C''\left(r = \frac{R}{2}\right) < 0$ **1**

$\therefore$ Curved surface area of cylinder is Max. iff $r = \frac{R}{2}$ **$\frac{1}{2}$**
